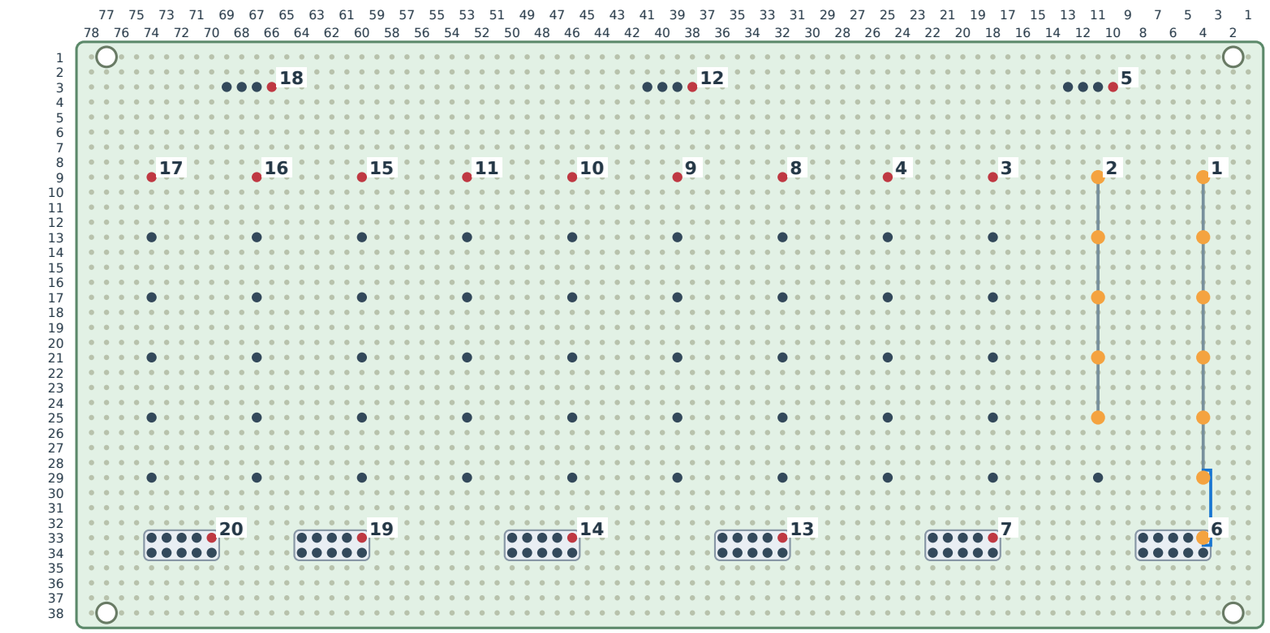
CP00 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

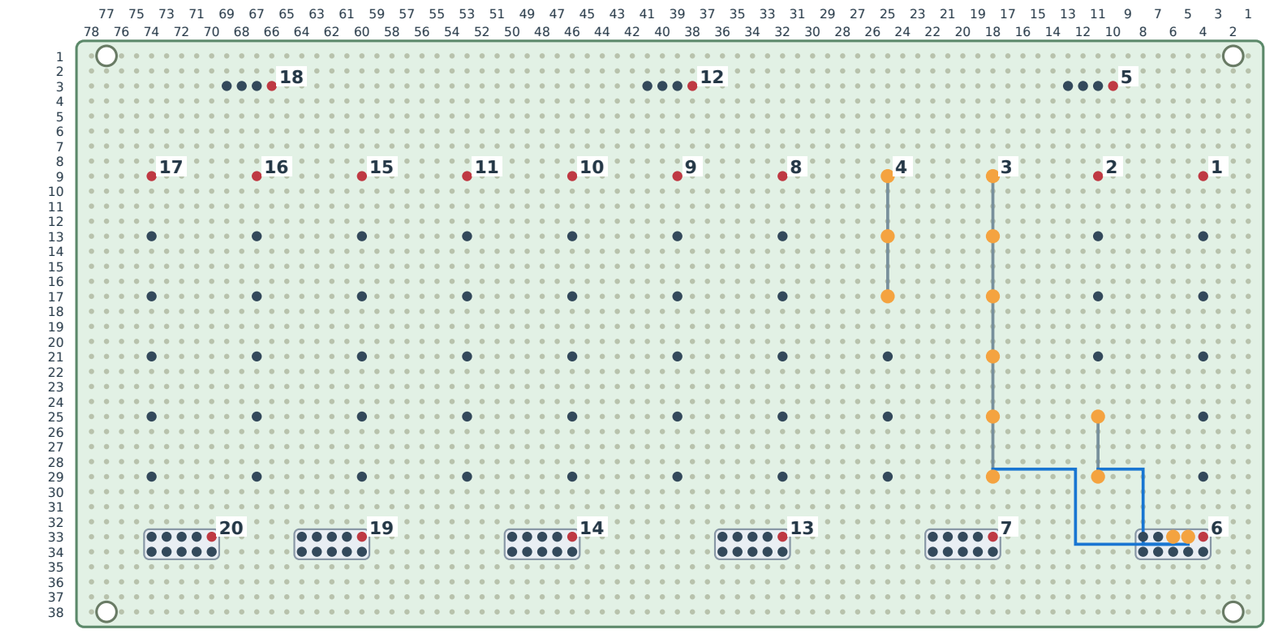
Wire / net	From → to	TOP holes
W1 / NODE00	J_A1_NODE00.1 → J_A1_NODE00.2	4,9 → 4,13
W2 / NODE00	J_A1_NODE00.2 → J_A1_NODE00.3	4,13 → 4,17
W3 / NODE00	J_A1_NODE00.3 → J_A1_NODE00.4	4,17 → 4,21
W4 / NODE00	J_A1_NODE00.4 → J_A1_NODE00.5	4,21 → 4,25
W5 / NODE00	J_A1_NODE00.5 → J_A1_NODE00.6	4,25 → 4,29
W6 / NODE00	J_A1_NODE00.6 → J_A1_NODE.1	4,29 → 4,33
W8 / NODE01	J_A1_NODE01.1 → J_A1_NODE01.2	11,9 → 11,13
W9 / NODE01	J_A1_NODE01.2 → J_A1_NODE01.3	11,13 → 11,17
W10 / NODE01	J_A1_NODE01.3 → J_A1_NODE01.4	11,17 → 11,21
W11 / NODE01	J_A1_NODE01.4 → J_A1_NODE01.5	11,21 → 11,25

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP00 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W12 / NODE01	J_A1_NODE01.5 → J_A1_NODE01.6	11,25 → 11,29
W13 / NODE01	J_A1_NODE01.6 → J_A1_NODE.3	11,29 → 5,33
W15 / NODE02	J_A1_NODE02.1 → J_A1_NODE02.2	18,9 → 18,13
W16 / NODE02	J_A1_NODE02.2 → J_A1_NODE02.3	18,13 → 18,17
W17 / NODE02	J_A1_NODE02.3 → J_A1_NODE02.4	18,17 → 18,21
W18 / NODE02	J_A1_NODE02.4 → J_A1_NODE02.5	18,21 → 18,25
W19 / NODE02	J_A1_NODE02.5 → J_A1_NODE02.6	18,25 → 18,29
W20 / NODE02	J_A1_NODE02.6 → J_A1_NODE.5	18,29 → 6,33
W22 / NODE03	J_A1_NODE03.1 → J_A1_NODE03.2	25,9 → 25,13
W23 / NODE03	J_A1_NODE03.2 → J_A1_NODE03.3	25,13 → 25,17

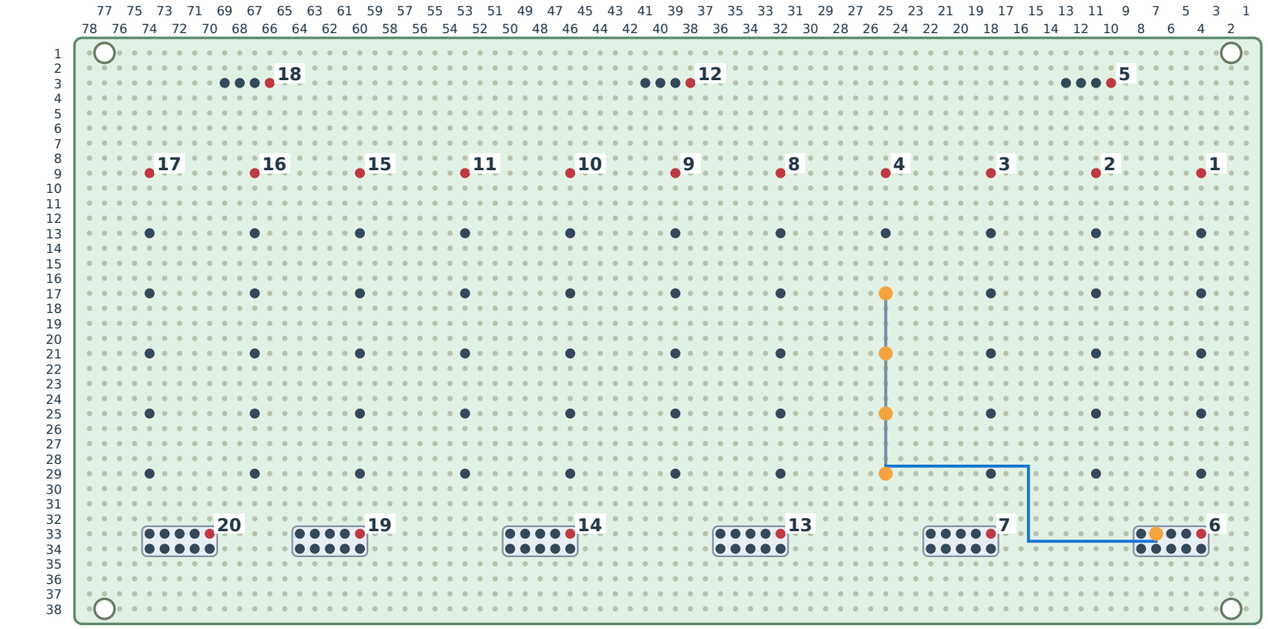
Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP00 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

Wire / net	From → to	TOP holes
W24 / NODE03	J_A1_NODE03.3 → J_A1_NODE03.4	25,17 → 25,21
W25 / NODE03	J_A1_NODE03.4 → J_A1_NODE03.5	25,21 → 25,25
W26 / NODE03	J_A1_NODE03.5 → J_A1_NODE03.6	25,25 → 25,29
W27 / NODE03	J_A1_NODE03.6 → J_A1_NODE.7	25,29 → 7,33

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

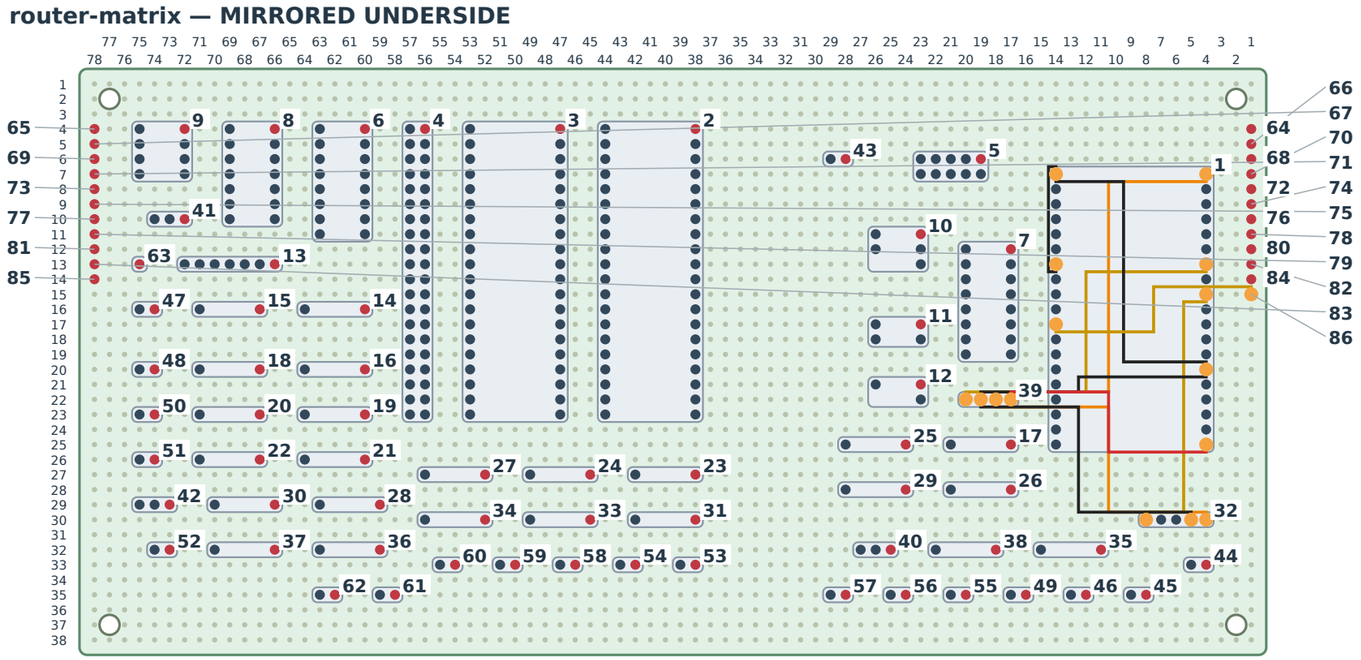
TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP00 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

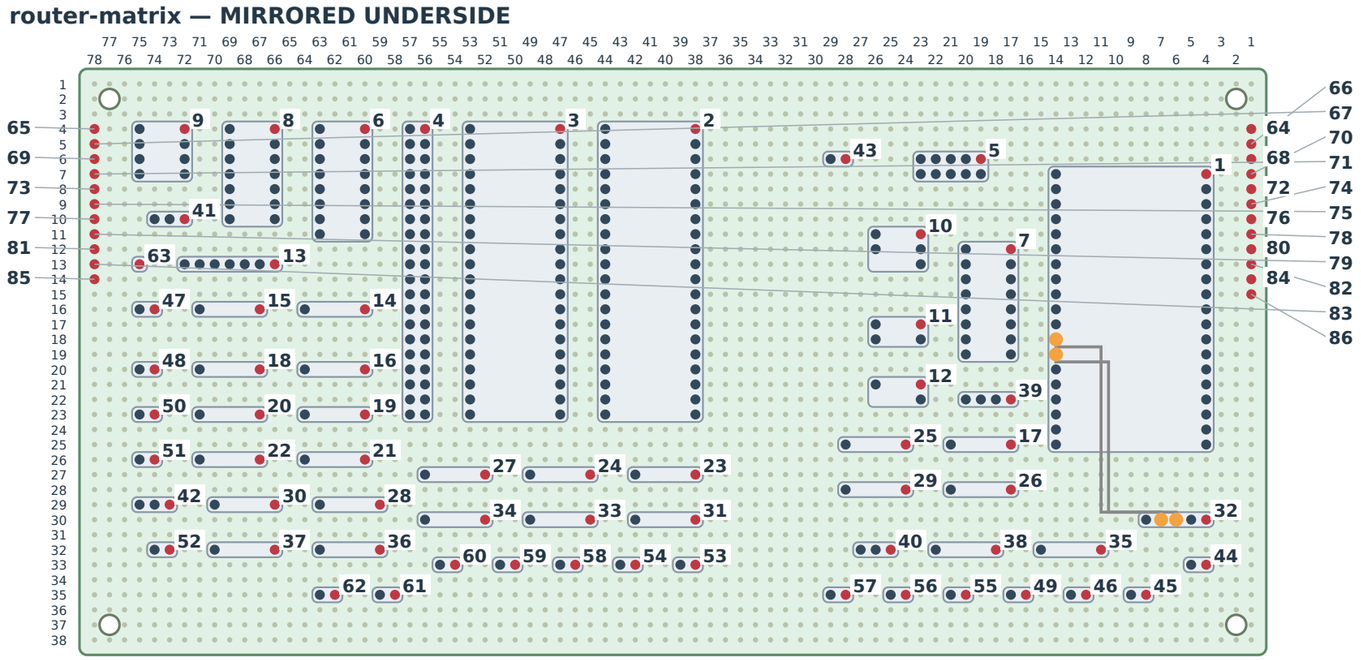
Wire / net	From → to	TOP holes
W1 / USB_ROUTER_3V3	J_ROUTER_CORE_FIXTURE.1 → J_ROUTER_CORE_PROTOTYPE.2	4,30 → 18,22
W2 / USB_ROUTER_3V3	J_ROUTER_CORE_PROTOTYPE.2 → U_ROUTER_CORE_ESP1.1	18,22 → 4,7
W17 / EDU_PWR_EN	J_ROUTER_CORE_PROTOTYPE.4 → U_ROUTER_CORE_ESP1.7	20,22 → 4,13
W21 / FIX_STIM	J_ROUTER_CORE_FIXTURE.5 → U_ROUTER_CORE_ESP1.9	8,30 → 4,15
W24 / GND	J_ROUTER_CORE_FIXTURE.2 → J_ROUTER_CORE_PROTOTYPE.3	5,30 → 19,22
W25 / GND	J_ROUTER_CORE_PROTOTYPE.3 → U_ROUTER_CORE_ESP1.14	19,22 → 4,20
W26 / GND	U_ROUTER_CORE_ESP1.14 → U_ROUTER_CORE_ESP1.20	4,20 → 14,7
W27 / GND	U_ROUTER_CORE_ESP1.20 → U_ROUTER_CORE_ESP1.26	14,7 → 14,13
W87 / USB_5V	J_ROUTER_CORE_PROTOTYPE.1 → U_ROUTER_CORE_ESP1.19	17,22 → 4,25
W98 / EDU_FAULT_RESET	J_BOUNDARY_23.1 → U_ROUTER_CORE_ESP1.30	1,15 → 14,17

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP00 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.

Wire / net	From → to	TOP holes
W100 / FIX_SCL	J_ROUTER_CORE_FIXTURE.4 → U_ROUTER_CORE_ESP1.31	7,30 → 14,18
W101 / FIX_SDA	J_ROUTER_CORE_FIXTURE.3 → U_ROUTER_CORE_ESP1.32	6,30 → 14,19



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

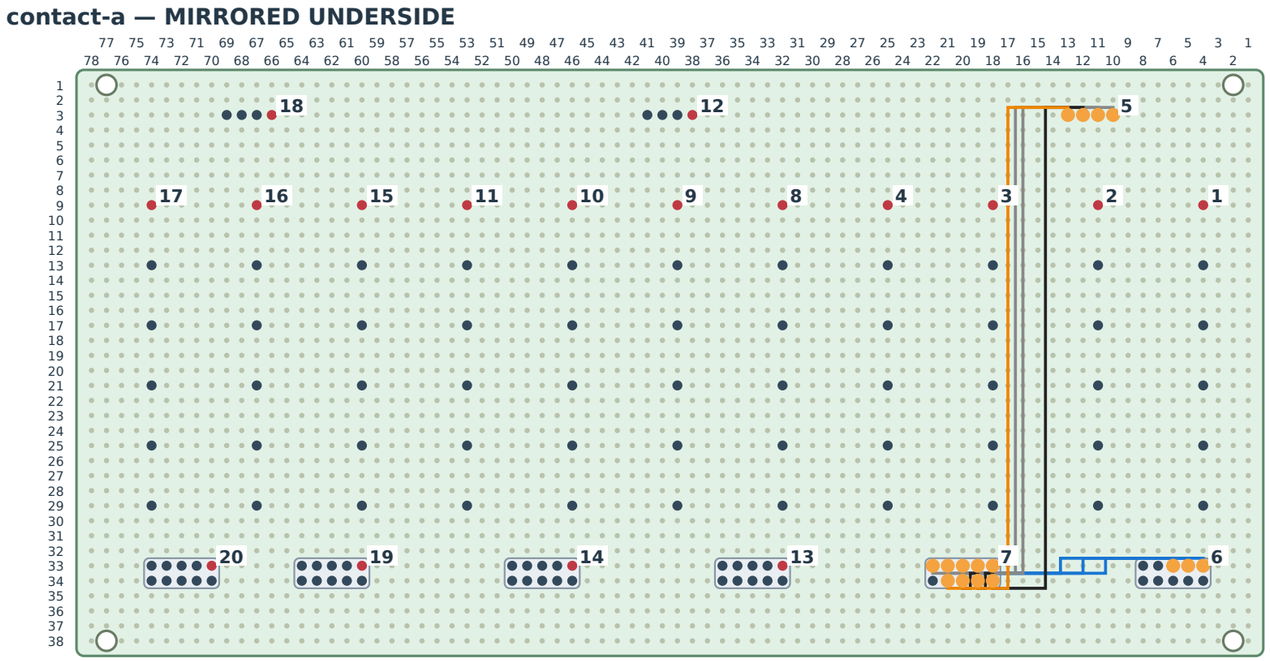
TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP06 — contact-a

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

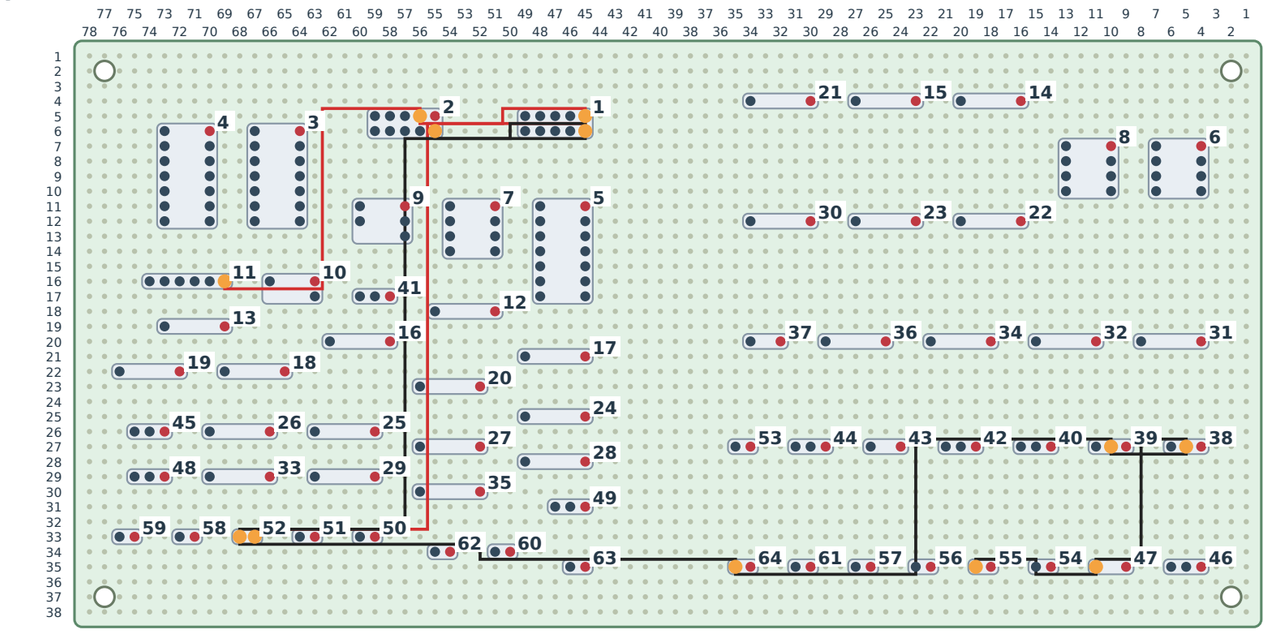
Wire / net	From → to	TOP holes
W7 / NODE00	J_A1_NODE.1 → J_A1_SMART.1	4,33 → 18,33
W14 / NODE01	J_A1_NODE.3 → J_A1_SMART.3	5,33 → 19,33
W21 / NODE02	J_A1_NODE.5 → J_A1_SMART.5	6,33 → 20,33
W28 / SCL_POSITION_0	J_A1_SERVICE.1 → J_A1_SMART.9	10,3 → 22,33
W29 / SDA_POSITION_0	J_A1_SERVICE.2 → J_A1_SMART.7	11,3 → 21,33
W30 / GND	J_A1_SERVICE.3 → J_A1_SMART.2	12,3 → 18,34
W31 / GND	J_A1_SMART.2 → J_A1_SMART.4	18,34 → 19,34
W32 / GND	J_A1_SMART.4 → J_A1_SMART.6	19,34 → 20,34
W41 / 3V3_SYS	J_A1_SERVICE.4 → J_A1_SMART.8	13,3 → 21,34

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP09 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

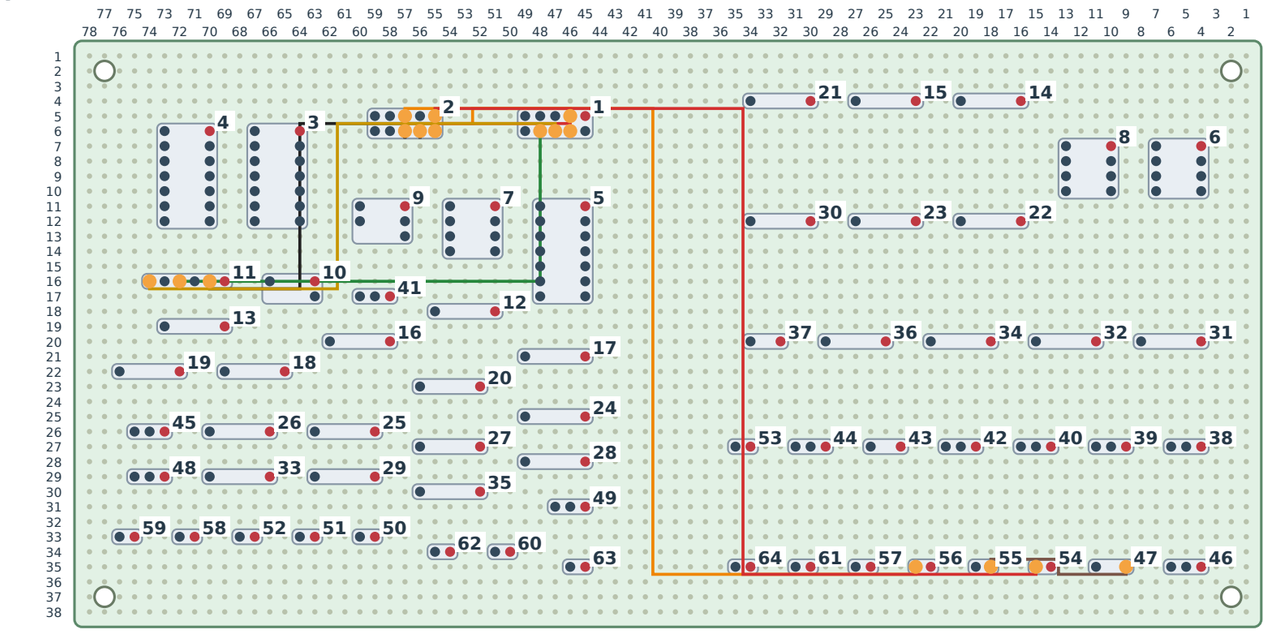
Wire / net	From → to	TOP holes
W1 / 5V_EDU	J_POWER_INTERLOCK_FIXED_BATTERY.1 → J_POWER_INTERLOCK_MATRIX.1	67,33 → 45,5
W2 / 5V_EDU	J_POWER_INTERLOCK_MATRIX.1 → J_POWER_INTERLOCK_SMART.3	45,5 → 56,5
W3 / 5V_EDU	J_POWER_INTERLOCK_SMART.3 → J_POWER_INTERLOCK_TEST.1	56,5 → 69,16
W8 / GND	C_POWER_INPUT_INPUT_HF.2 → C_POWER_INPUT_INPUT.2	19,35 → 11,35
W9 / GND	C_POWER_INPUT_INPUT.2 → J_POWER_INPUT_BUCK_3V3.2	11,35 → 5,27
W10 / GND	J_POWER_INPUT_BUCK_3V3.2 → J_POWER_INPUT_BUCK_5V.2	5,27 → 10,27
W11 / GND	J_POWER_INPUT_BUCK_5V.2 → J_POWER_INPUT_INPUT.2	10,27 → 35,35
W12 / GND	J_POWER_INPUT_INPUT.2 → J_POWER_INTERLOCK_FIXED_BATTERY.2	35,35 → 68,33
W13 / GND	J_POWER_INTERLOCK_FIXED_BATTERY.2 → J_POWER_INTERLOCK_MATRIX.2	68,33 → 45,6
W14 / GND	J_POWER_INTERLOCK_MATRIX.2 → J_POWER_INTERLOCK_SMART.2	45,6 → 55,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP09 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

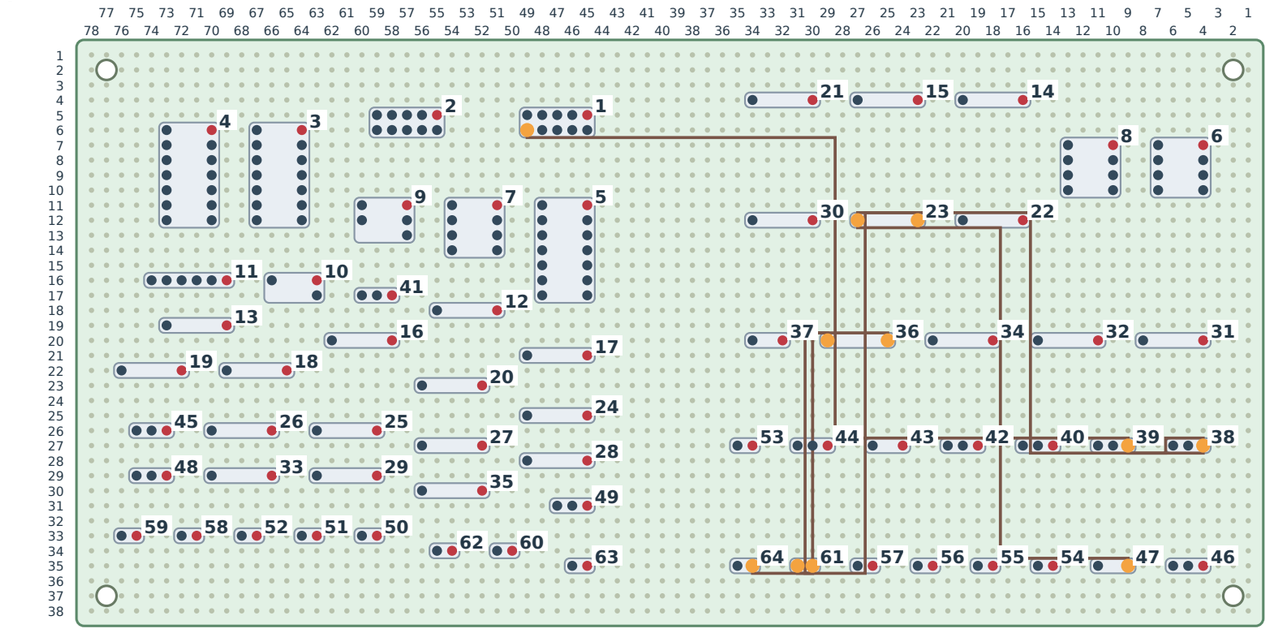
Wire / net	From → to	TOP holes
W15 / GND	J_POWER_INTERLOCK_SMART.2 → J_POWER_INTERLOCK_SMART.4	55,6 → 56,6
W16 / GND	J_POWER_INTERLOCK_SMART.4 → J_POWER_INTERLOCK_SMART.6	56,6 → 57,6
W17 / GND	J_POWER_INTERLOCK_SMART.6 → J_POWER_INTERLOCK_TEST.2	57,6 → 70,16
W63 / 3V3_SYS	J_POWER_INTERLOCK_MATRIX.3 → J_POWER_INTERLOCK_SMART.5	46,5 → 57,5
W64 / 3V3_SYS	J_POWER_INTERLOCK_SMART.5 → JP_POWER_INPUT_3V3.2	57,5 → 23,35
W78 / 5V_SYS	J_POWER_INTERLOCK_MATRIX.4 → J_POWER_INTERLOCK_SMART.1	46,6 → 55,5
W79 / 5V_SYS	J_POWER_INTERLOCK_SMART.1 → JP_POWER_INPUT_LOGIC.2	55,5 → 15,35
W87 / EDU_FAULT	J_POWER_INTERLOCK_MATRIX.6 → J_POWER_INTERLOCK_TEST.6	47,6 → 74,16
W91 / CURRENT_SENSE	J_POWER_INTERLOCK_MATRIX.8 → J_POWER_INTERLOCK_TEST.4	48,6 → 72,16
W96 / 12V_SYS	C_POWER_INPUT_INPUT_HF.1 → C_POWER_INPUT_INPUT.1	18,35 → 9,35

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP09 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

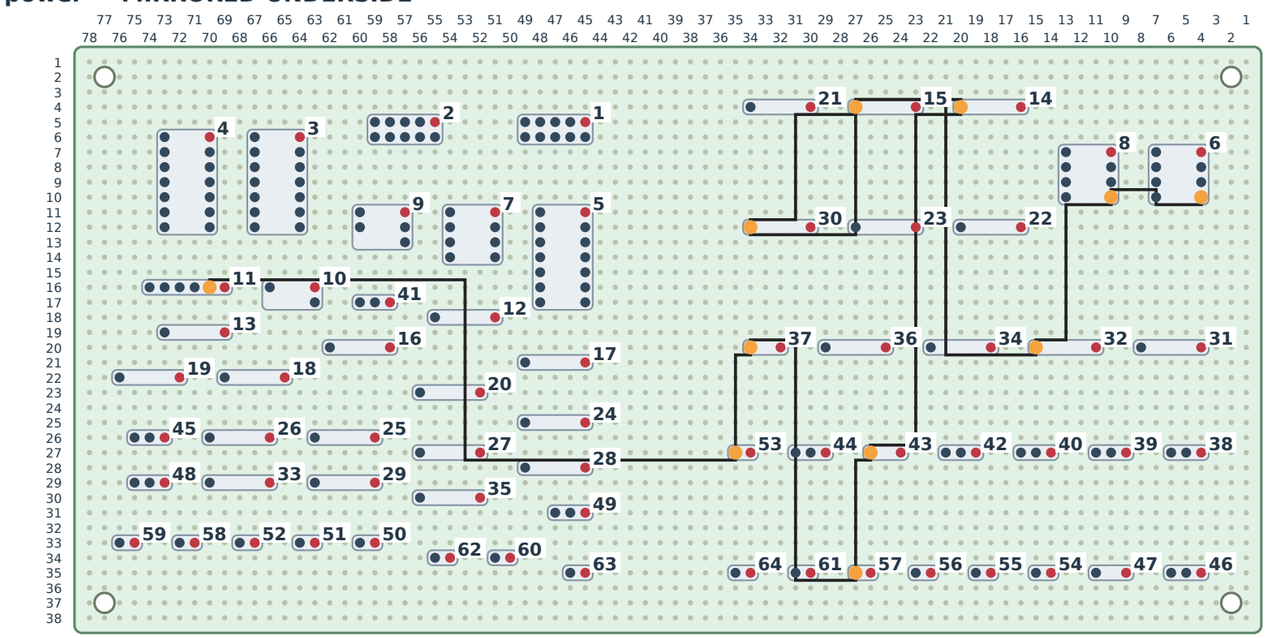
Wire / net	From → to	TOP holes
W97 / 12V_SYS	C_POWER_INPUT_INPUT.1 → D_POWER_INPUT_REVERSE.2	9,35 → 27,12
W98 / 12V_SYS	D_POWER_INPUT_REVERSE.2 → J_POWER_INPUT_BUCK_3V3.1	27,12 → 4,27
W99 / 12V_SYS	J_POWER_INPUT_BUCK_3V3.1 → J_POWER_INPUT_BUCK_5V.1	4,27 → 9,27
W100 / 12V_SYS	J_POWER_INPUT_BUCK_5V.1 → J_POWER_INTERLOCK_MATRIX.10	9,27 → 49,6
W157 / 12V_SWITCHED	D_POWER_INPUT_REVERSE.1 → J_POWER_INPUT_SW_MAIN.2	23,12 → 31,35
W163 / 12V_IN	F_POWER_INPUT_INPUT.1 → J_POWER_INPUT_INPUT.1	25,20 → 34,35
W164 / 12V_FUSED	F_POWER_INPUT_INPUT.2 → J_POWER_INPUT_SW_MAIN.1	29,20 → 30,35

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP10 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

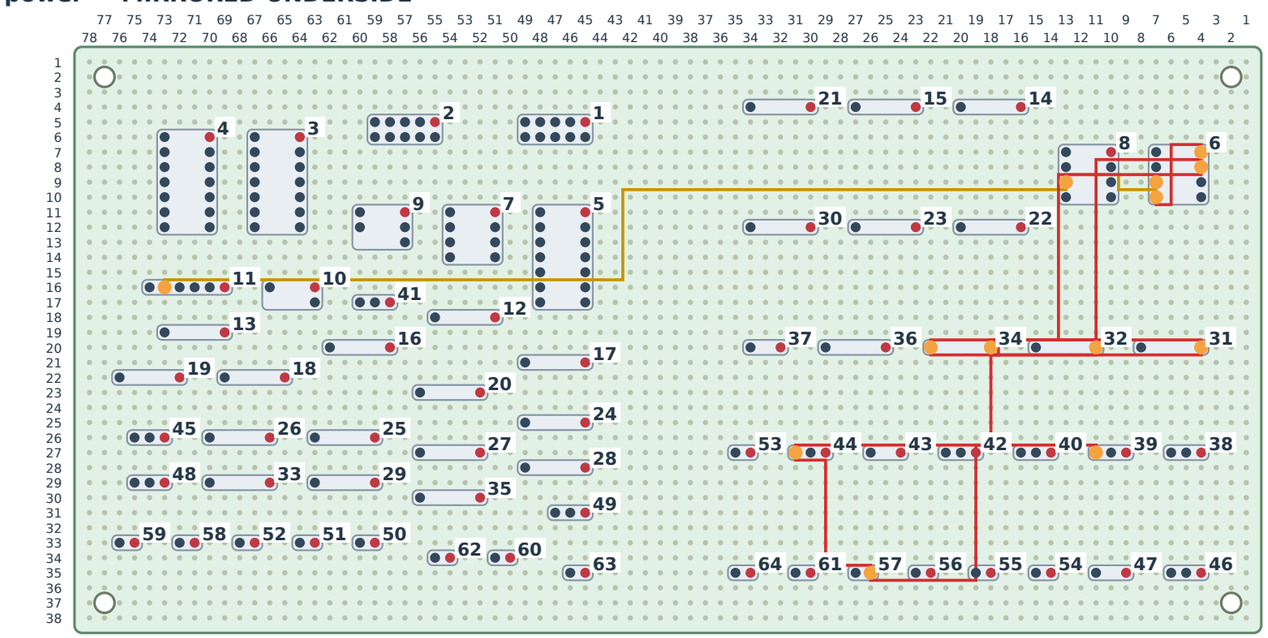
Wire / net	From → to	TOP holes
W18 / GND	J_POWER_INTERLOCK_TEST.2 → C_POWER_GUARDS_OV_3V3_IN.2	70,16 → 35,27
W19 / GND	C_POWER_GUARDS_OV_3V3_IN.2 → C_POWER_GUARDS_OV_3V3_OUT.2	35,27 → 34,20
W20 / GND	C_POWER_GUARDS_OV_3V3_OUT.2 → C_POWER_GUARDS_OV_5V_IN.2	34,20 → 27,35
W21 / GND	C_POWER_GUARDS_OV_5V_IN.2 → C_POWER_GUARDS_OV_5V_OUT.2	27,35 → 26,27
W22 / GND	C_POWER_GUARDS_OV_5V_OUT.2 → R_POWER_GUARDS_OV_3V3_BOTTOM.2	26,27 → 20,4
W23 / GND	R_POWER_GUARDS_OV_3V3_BOTTOM.2 → R_POWER_GUARDS_OV_5V_BOTTOM.2	20,4 → 34,12
W24 / GND	R_POWER_GUARDS_OV_5V_BOTTOM.2 → R_POWER_GUARDS_UV_3V3_BOTTOM.2	34,12 → 27,4
W25 / GND	R_POWER_GUARDS_UV_3V3_BOTTOM.2 → R_POWER_GUARDS_UV_5V_BOTTOM.2	27,4 → 15,20
W26 / GND	R_POWER_GUARDS_UV_5V_BOTTOM.2 → U_POWER_GUARDS_OV_3V3.4	15,20 → 10,10
W27 / GND	U_POWER_GUARDS_OV_3V3.4 → U_POWER_GUARDS_OV_5V.4	10,10 → 4,10

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP10 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

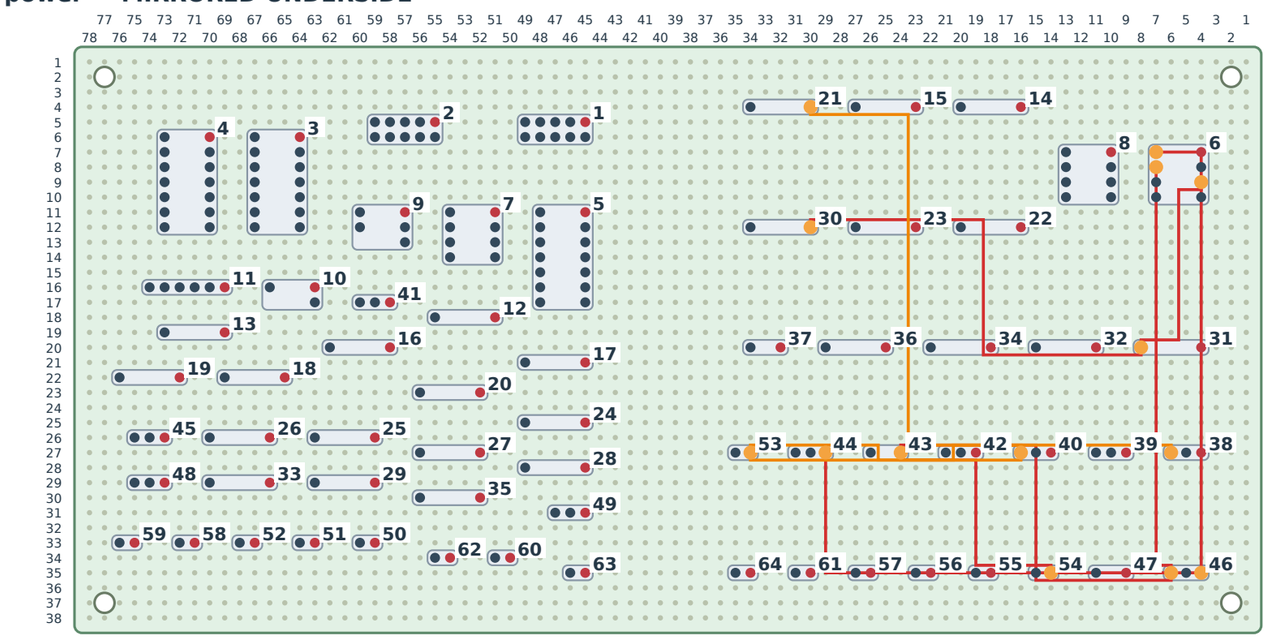
Wire / net	From → to	TOP holes
W106 / PRESET_N	J_POWER_INTERLOCK_TEST.5 → U_POWER_GUARDS_OV_3V3.6	73,16 → 13,9
W107 / PRESET_N	U_POWER_GUARDS_OV_3V3.6 → U_POWER_GUARDS_OV_5V.6	13,9 → 7,9
W116 / BUCK_5V	J_POWER_INPUT_BUCK_5V.3 → C_POWER_GUARDS_OV_5V_IN.1	11,27 → 26,35
W117 / BUCK_5V	C_POWER_GUARDS_OV_5V_IN.1 → Q_POWER_GUARDS_OV_5V_IN.3	26,35 → 31,27
W118 / BUCK_5V	Q_POWER_GUARDS_OV_5V_IN.3 → R_POWER_GUARDS_OV_5V_TOP.1	31,27 → 4,20
W119 / BUCK_5V	R_POWER_GUARDS_OV_5V_TOP.1 → R_POWER_GUARDS_UV_5V_TOP.1	4,20 → 18,20
W120 / BUCK_5V	R_POWER_GUARDS_UV_5V_TOP.1 → U_POWER_GUARDS_OV_5V.1	18,20 → 4,7
W121 / BUCK_5V	U_POWER_GUARDS_OV_5V.1 → U_POWER_GUARDS_OV_5V.5	4,7 → 7,10
W122 / UV_5V	R_POWER_GUARDS_UV_5V_BOTTOM.1 → R_POWER_GUARDS_UV_5V_TOP.2	11,20 → 22,20
W123 / UV_5V	R_POWER_GUARDS_UV_5V_TOP.2 → U_POWER_GUARDS_OV_5V.2	22,20 → 4,8

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP10 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

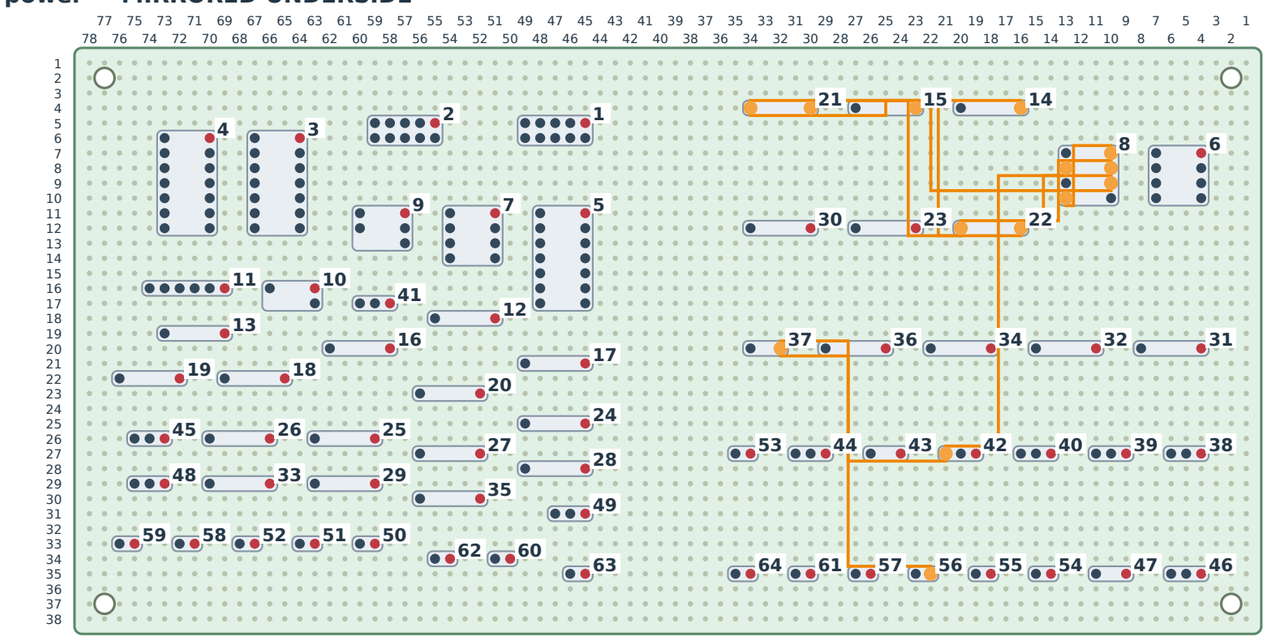
Wire / net	From → to	TOP holes
W124 / OV_5V	R_POWER_GUARDS_OV_5V_BOTTOM.1 → R_POWER_GUARDS_OV_5V_TOP.2	30,12 → 8,20
W125 / OV_5V	R_POWER_GUARDS_OV_5V_TOP.2 → U_POWER_GUARDS_OV_5V.3	8,20 → 4,9
W126 / SAFE_5V	JP_POWER_INPUT_LOGIC.1 → C_POWER_GUARDS_OV_5V_OUT.1	14,35 → 24,27
W127 / SAFE_5V	C_POWER_GUARDS_OV_5V_OUT.1 → Q_POWER_GUARDS_OV_5V_OUT.3	24,27 → 6,35
W128 / SAFE_5V	Q_POWER_GUARDS_OV_5V_OUT.3 → U_POWER_GUARDS_OV_5V.7	6,35 → 7,8
W129 / OV_GATE_5V	Q_POWER_GUARDS_OV_5V_IN.1 → Q_POWER_GUARDS_OV_5V_OUT.1	29,27 → 4,35
W130 / OV_GATE_5V	Q_POWER_GUARDS_OV_5V_OUT.1 → U_POWER_GUARDS_OV_5V.8	4,35 → 7,7
W137 / BUCK_3V3	J_POWER_INPUT_BUCK_3V3.3 → C_POWER_GUARDS_OV_3V3_IN.1	6,27 → 34,27
W138 / BUCK_3V3	C_POWER_GUARDS_OV_3V3_IN.1 → Q_POWER_GUARDS_OV_3V3_IN.3	34,27 → 16,27
W139 / BUCK_3V3	Q_POWER_GUARDS_OV_3V3_IN.3 → R_POWER_GUARDS_OV_3V3_TOP.1	16,27 → 30,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP10 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

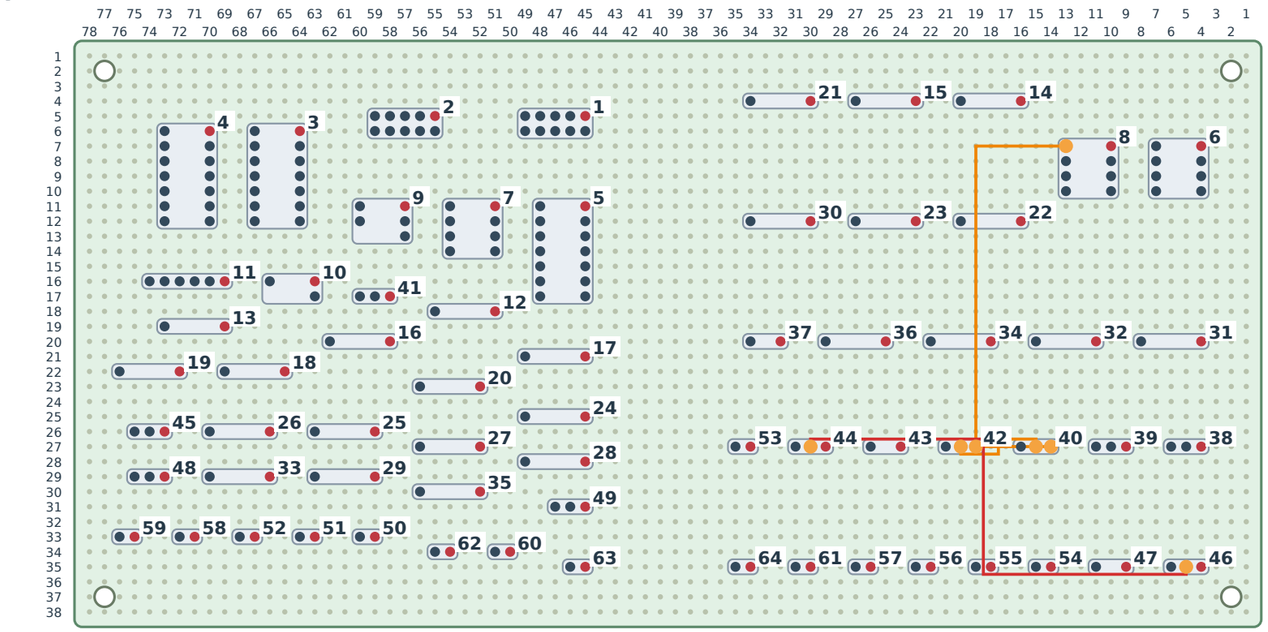
Wire / net	From → to	TOP holes
W140 / BUCK_3V3	R_POWER_GUARDS_OV_3V3_TOP.1 → R_POWER_GUARDS_UV_3V3_TOP.1	30,4 → 16,12
W141 / BUCK_3V3	R_POWER_GUARDS_UV_3V3_TOP.1 → U_POWER_GUARDS_OV_3V3.1	16,12 → 10,7
W142 / BUCK_3V3	U_POWER_GUARDS_OV_3V3.1 → U_POWER_GUARDS_OV_3V3.5	10,7 → 13,10
W143 / UV_3V3	R_POWER_GUARDS_UV_3V3_BOTTOM.1 → R_POWER_GUARDS_UV_3V3_TOP.2	23,4 → 20,12
W144 / UV_3V3	R_POWER_GUARDS_UV_3V3_TOP.2 → U_POWER_GUARDS_OV_3V3.2	20,12 → 10,8
W145 / OV_3V3	R_POWER_GUARDS_OV_3V3_BOTTOM.1 → R_POWER_GUARDS_OV_3V3_TOP.2	16,4 → 34,4
W146 / OV_3V3	R_POWER_GUARDS_OV_3V3_TOP.2 → U_POWER_GUARDS_OV_3V3.3	34,4 → 10,9
W147 / SAFE_3V3	JP_POWER_INPUT_3V3.1 → C_POWER_GUARDS_OV_3V3_OUT.1	22,35 → 32,20
W148 / SAFE_3V3	C_POWER_GUARDS_OV_3V3_OUT.1 → Q_POWER_GUARDS_OV_3V3_OUT.3	32,20 → 21,27
W149 / SAFE_3V3	Q_POWER_GUARDS_OV_3V3_OUT.3 → U_POWER_GUARDS_OV_3V3.7	21,27 → 13,8

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP10 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

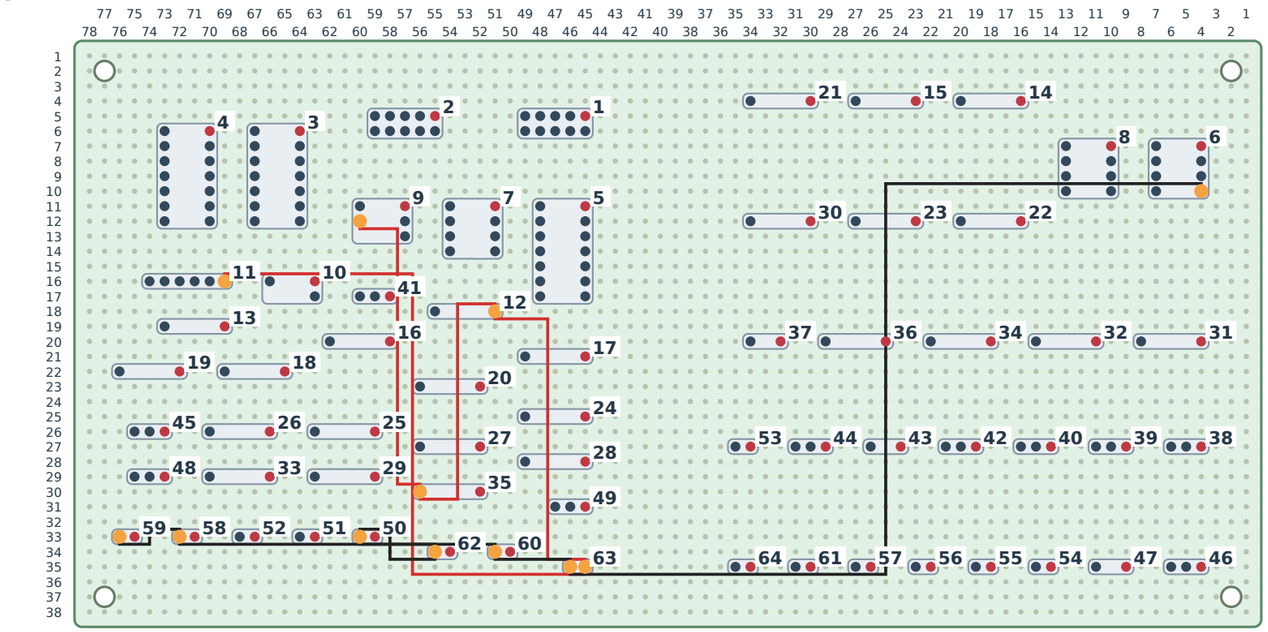
Wire / net	From → to	TOP holes
W150 / OV_GATE_3V3	Q_POWER_GUARDS_OV_3V3_IN.1 → Q_POWER_GUARDS_OV_3V3_OUT.1	14,27 → 19,27
W151 / OV_GATE_3V3	Q_POWER_GUARDS_OV_3V3_OUT.1 → U_POWER_GUARDS_OV_3V3.8	19,27 → 13,7
W165 / OV_MID_3V3	Q_POWER_GUARDS_OV_3V3_IN.2 → Q_POWER_GUARDS_OV_3V3_OUT.2	15,27 → 20,27
W167 / OV_MID_5V	Q_POWER_GUARDS_OV_5V_IN.2 → Q_POWER_GUARDS_OV_5V_OUT.2	30,27 → 5,35

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

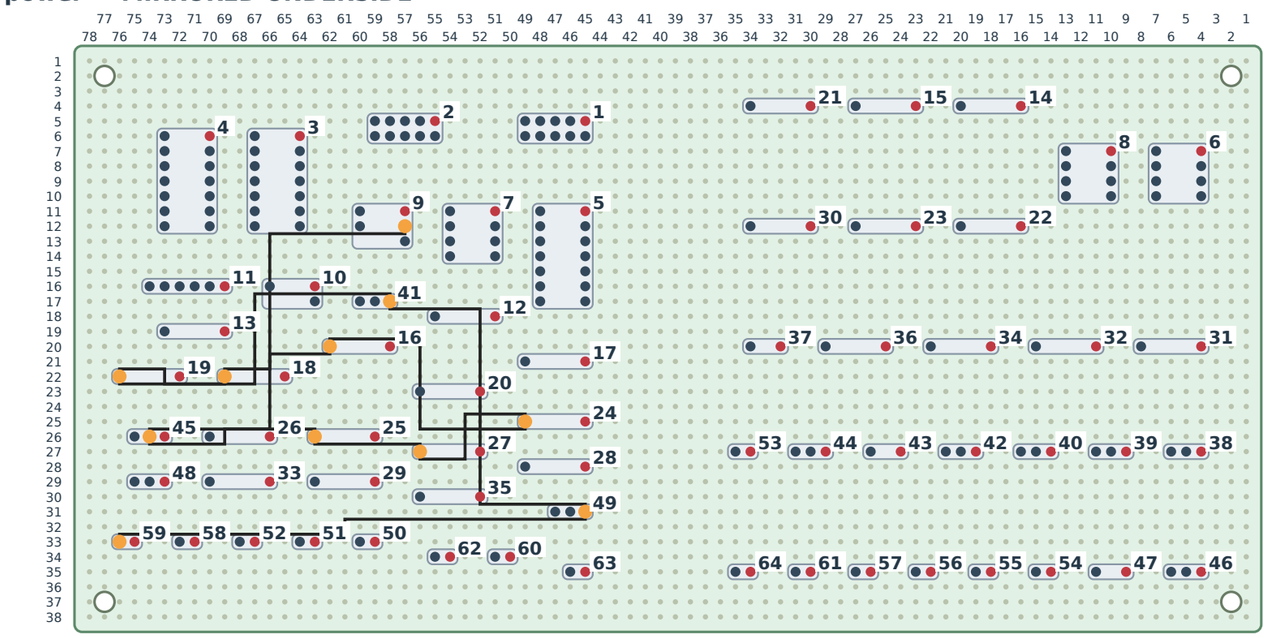
Wire / net	From → to	TOP holes
W4 / 5V_EDU	J_POWER_INTERLOCK_TEST.1 → C_POWER_ANALOG_EDU.1	69,16 → 45,35
W5 / 5V_EDU	C_POWER_ANALOG_EDU.1 → R_POWER_ANALOG_EDU_TOP.1	45,35 → 51,18
W6 / 5V_EDU	R_POWER_ANALOG_EDU_TOP.1 → R_POWER_ANALOG_SHUNT.2	51,18 → 56,30
W7 / 5V_EDU	R_POWER_ANALOG_SHUNT.2 → U_POWER_ANALOG_SENSE.4	56,30 → 60,12
W28 / GND	U_POWER_GUARDS_OV_5V.4 → C_POWER_ANALOG_EDU.2	4,10 → 46,35
W29 / GND	C_POWER_ANALOG_EDU.2 → C_POWER_ANALOG_SENSE.2	46,35 → 51,34
W30 / GND	C_POWER_ANALOG_SENSE.2 → C_POWER_ANALOG_TRIP.2	51,34 → 60,33
W31 / GND	C_POWER_ANALOG_TRIP.2 → C_POWER_INTERLOCK_ALLOW.2	60,33 → 55,34
W32 / GND	C_POWER_INTERLOCK_ALLOW.2 → C_POWER_INTERLOCK_FAULT.2	55,34 → 72,33
W33 / GND	C_POWER_INTERLOCK_FAULT.2 → C_POWER_INTERLOCK_RESET.2	72,33 → 76,33

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

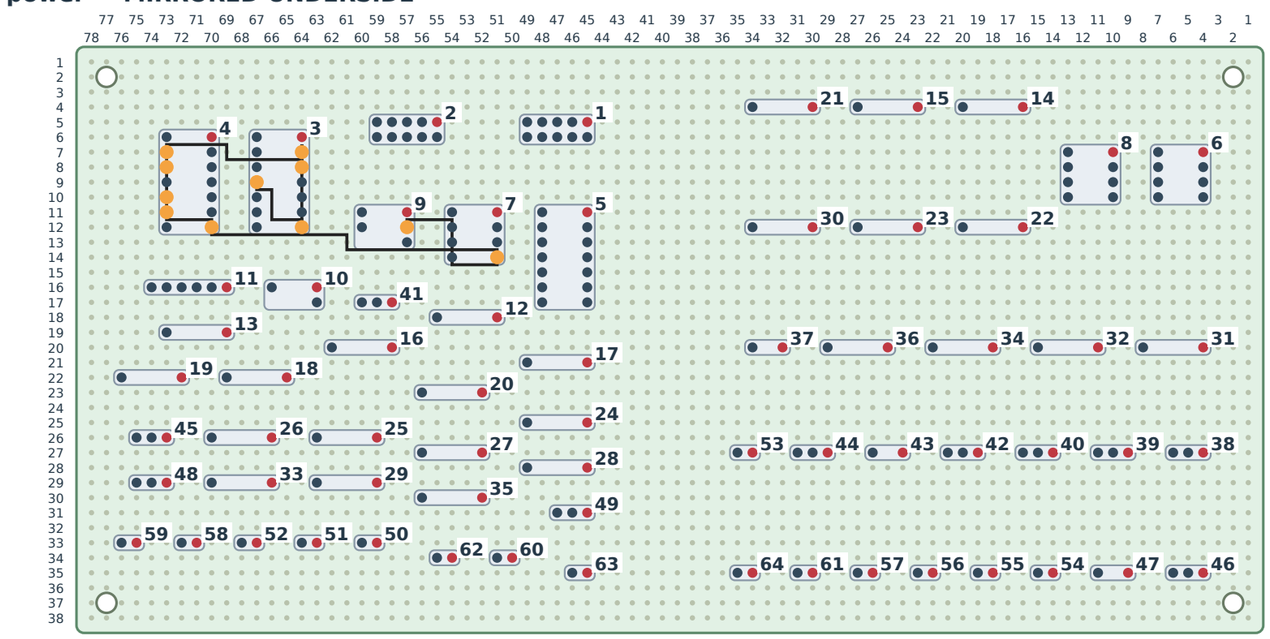
Wire / net	From → to	TOP holes
W34 / GND	C_POWER_INTERLOCK_RESET.2 → Q_POWER_ANALOG_ENABLE.1	76,33 → 45,31
W35 / GND	Q_POWER_ANALOG_ENABLE.1 → Q_POWER_INTERLOCK_BATTERY_LED.1	45,31 → 58,17
W36 / GND	Q_POWER_INTERLOCK_BATTERY_LED.1 → R_POWER_ANALOG_EDU_BOTTOM.2	58,17 → 76,22
W37 / GND	R_POWER_ANALOG_EDU_BOTTOM.2 → R_POWER_ANALOG_TRIP_BOTTOM.2	76,22 → 69,22
W38 / GND	R_POWER_ANALOG_TRIP_BOTTOM.2 → R_POWER_INTERLOCK_EDU_FAULT_RESET_OFF.2	69,22 → 62,20
W39 / GND	R_POWER_INTERLOCK_EDU_FAULT_RESET_OFF.2 → R_POWER_INTERLOCK_EDU_PWR_EN_OFF.2	62,20 → 49,25
W40 / GND	R_POWER_INTERLOCK_EDU_PWR_EN_OFF.2 → R_POWER_INTERLOCK_LED_GATE_OFF.2	49,25 → 56,27
W41 / GND	R_POWER_INTERLOCK_LED_GATE_OFF.2 → R_POWER_INTERLOCK_ROUTER_ALIVE_OFF.2	56,27 → 63,26
W42 / GND	R_POWER_INTERLOCK_ROUTER_ALIVE_OFF.2 → U_POWER_ANALOG_REF.2	63,26 → 74,26
W43 / GND	U_POWER_ANALOG_REF.2 → U_POWER_ANALOG_SENSE.2	74,26 → 57,12

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

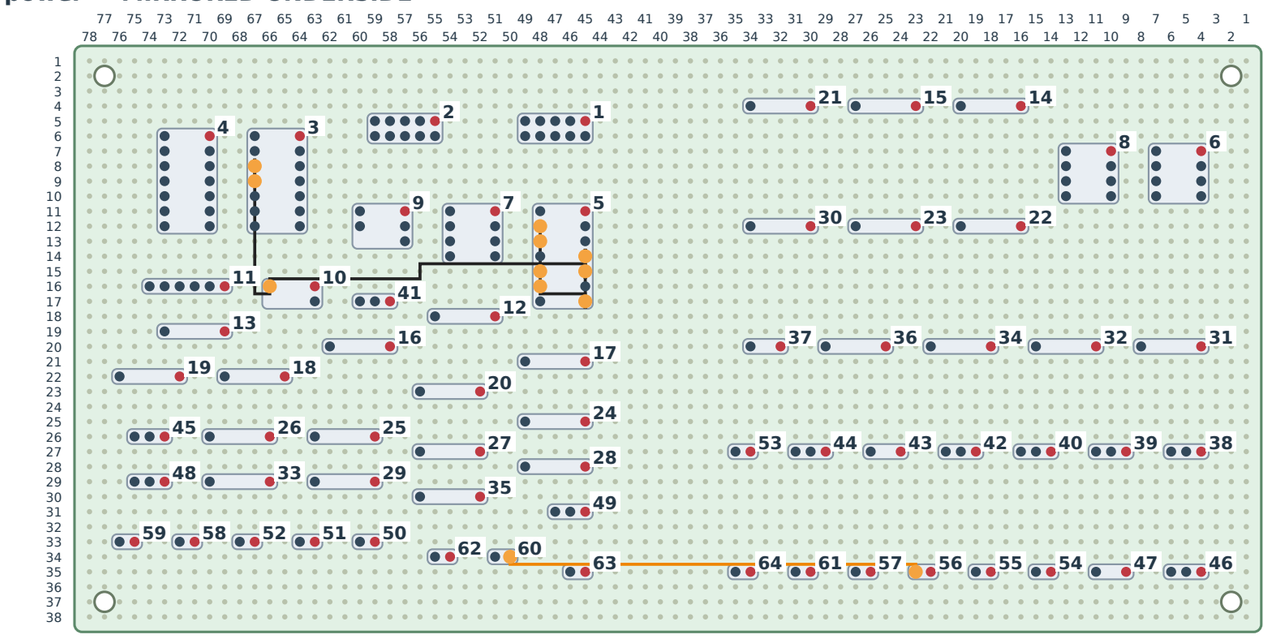
Wire / net	From → to	TOP holes
W44 / GND	U_POWER_ANALOG_SENSE.2 → U_POWER_ANALOG_TRIP.4	57,12 → 51,14
W45 / GND	U_POWER_ANALOG_TRIP.4 → U_POWER_INTERLOCK_ALLOW.7	51,14 → 70,12
W46 / GND	U_POWER_INTERLOCK_ALLOW.7 → U_POWER_INTERLOCK_ALLOW.9	70,12 → 73,11
W47 / GND	U_POWER_INTERLOCK_ALLOW.9 → U_POWER_INTERLOCK_ALLOW.10	73,11 → 73,10
W48 / GND	U_POWER_INTERLOCK_ALLOW.10 → U_POWER_INTERLOCK_ALLOW.12	73,10 → 73,8
W49 / GND	U_POWER_INTERLOCK_ALLOW.12 → U_POWER_INTERLOCK_ALLOW.13	73,8 → 73,7
W50 / GND	U_POWER_INTERLOCK_ALLOW.13 → U_POWER_INTERLOCK_FAULT.2	73,7 → 64,7
W51 / GND	U_POWER_INTERLOCK_FAULT.2 → U_POWER_INTERLOCK_FAULT.3	64,7 → 64,8
W52 / GND	U_POWER_INTERLOCK_FAULT.3 → U_POWER_INTERLOCK_FAULT.7	64,8 → 64,12
W53 / GND	U_POWER_INTERLOCK_FAULT.7 → U_POWER_INTERLOCK_FAULT.11	64,12 → 67,9

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

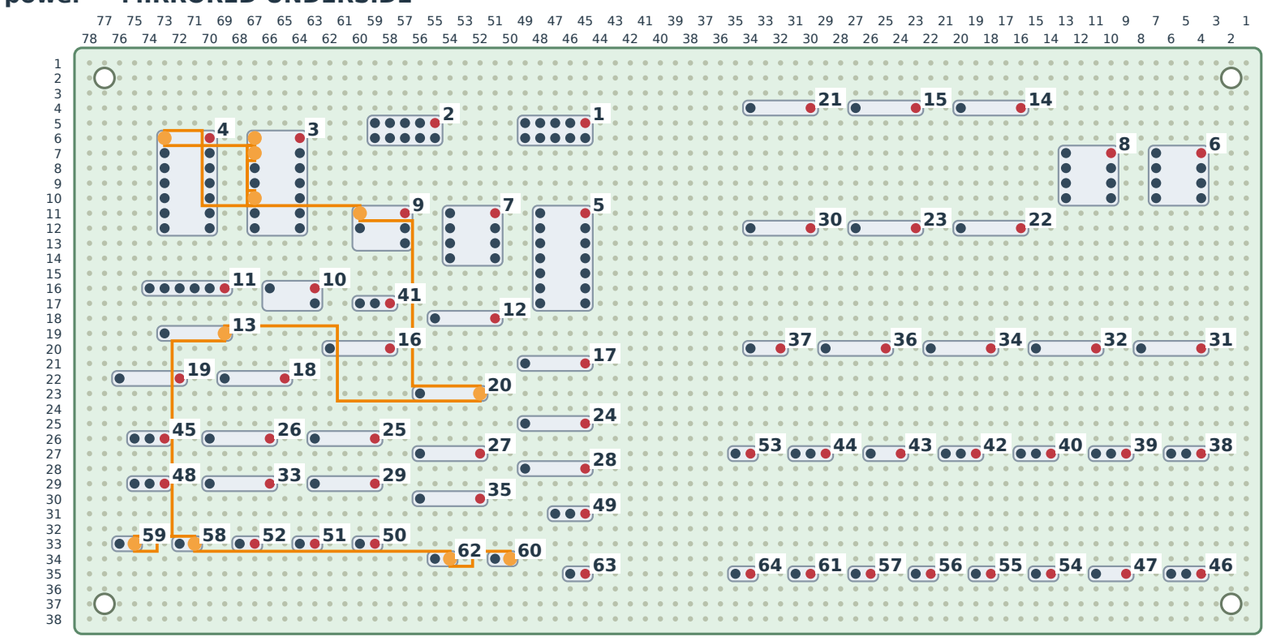
Wire / net	From → to	TOP holes
W54 / GND	U_POWER_INTERLOCK_FAULT.11 → U_POWER_INTERLOCK_FAULT.12	67,9 → 67,8
W55 / GND	U_POWER_INTERLOCK_FAULT.12 → U_POWER_INTERLOCK_POR.3	67,8 → 66,16
W56 / GND	U_POWER_INTERLOCK_POR.3 → U_POWER_INTERLOCK_RESET.4	66,16 → 45,14
W57 / GND	U_POWER_INTERLOCK_RESET.4 → U_POWER_INTERLOCK_RESET.5	45,14 → 45,15
W58 / GND	U_POWER_INTERLOCK_RESET.5 → U_POWER_INTERLOCK_RESET.7	45,15 → 45,17
W59 / GND	U_POWER_INTERLOCK_RESET.7 → U_POWER_INTERLOCK_RESET.9	45,17 → 48,16
W60 / GND	U_POWER_INTERLOCK_RESET.9 → U_POWER_INTERLOCK_RESET.10	48,16 → 48,15
W61 / GND	U_POWER_INTERLOCK_RESET.10 → U_POWER_INTERLOCK_RESET.12	48,15 → 48,13
W62 / GND	U_POWER_INTERLOCK_RESET.12 → U_POWER_INTERLOCK_RESET.13	48,13 → 48,12
W65 / 3V3_SYS	JP_POWER_INPUT_3V3.2 → C_POWER_ANALOG_SENSE.1	23,35 → 50,34

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

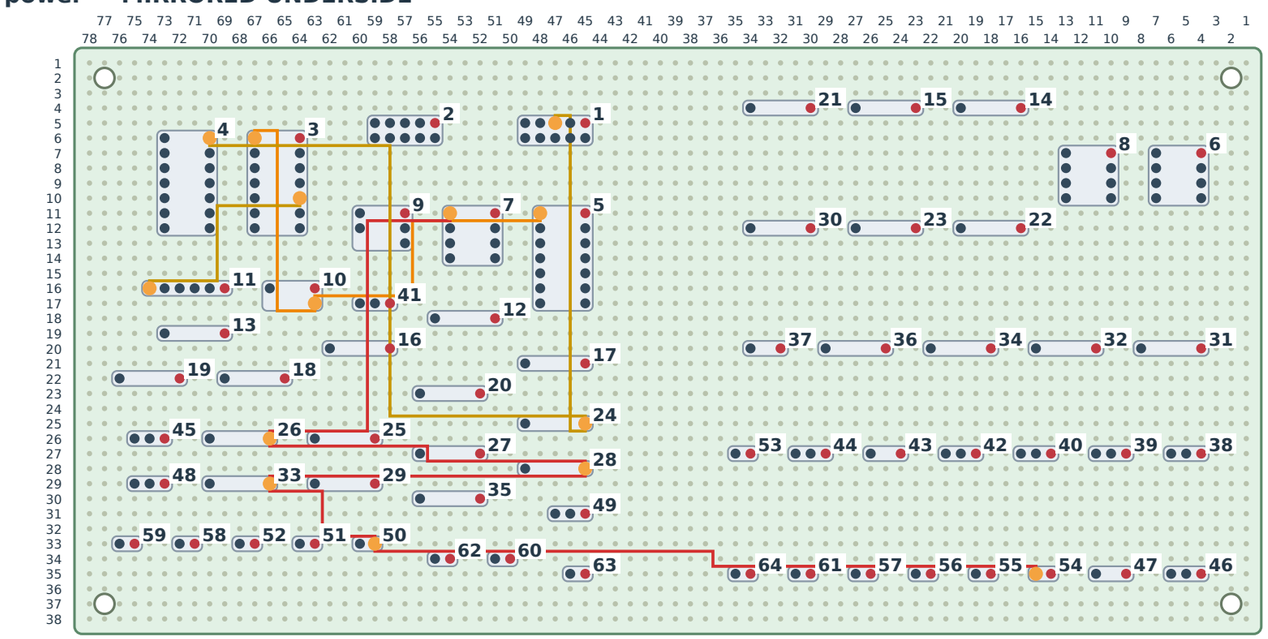
Wire / net	From → to	TOP holes
W66 / 3V3_SYS	C_POWER_ANALOG_SENSE.1 → C_POWER_INTERLOCK_ALLOW.1	50,34 → 54,34
W67 / 3V3_SYS	C_POWER_INTERLOCK_ALLOW.1 → C_POWER_INTERLOCK_FAULT.1	54,34 → 71,33
W68 / 3V3_SYS	C_POWER_INTERLOCK_FAULT.1 → C_POWER_INTERLOCK_RESET.1	71,33 → 75,33
W69 / 3V3_SYS	C_POWER_INTERLOCK_RESET.1 → R_POWER_ANALOG_EDU_ON.1	75,33 → 69,19
W70 / 3V3_SYS	R_POWER_ANALOG_EDU_ON.1 → R_POWER_ANALOG_PRESET.1	69,19 → 52,23
W71 / 3V3_SYS	R_POWER_ANALOG_PRESET.1 → U_POWER_ANALOG_SENSE.5	52,23 → 60,11
W72 / 3V3_SYS	U_POWER_ANALOG_SENSE.5 → U_POWER_INTERLOCK_ALLOW.14	60,11 → 73,6
W73 / 3V3_SYS	U_POWER_INTERLOCK_ALLOW.14 → U_POWER_INTERLOCK_FAULT.10	73,6 → 67,10
W74 / 3V3_SYS	U_POWER_INTERLOCK_FAULT.10 → U_POWER_INTERLOCK_FAULT.13	67,10 → 67,7
W75 / 3V3_SYS	U_POWER_INTERLOCK_FAULT.13 → U_POWER_INTERLOCK_FAULT.14	67,7 → 67,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

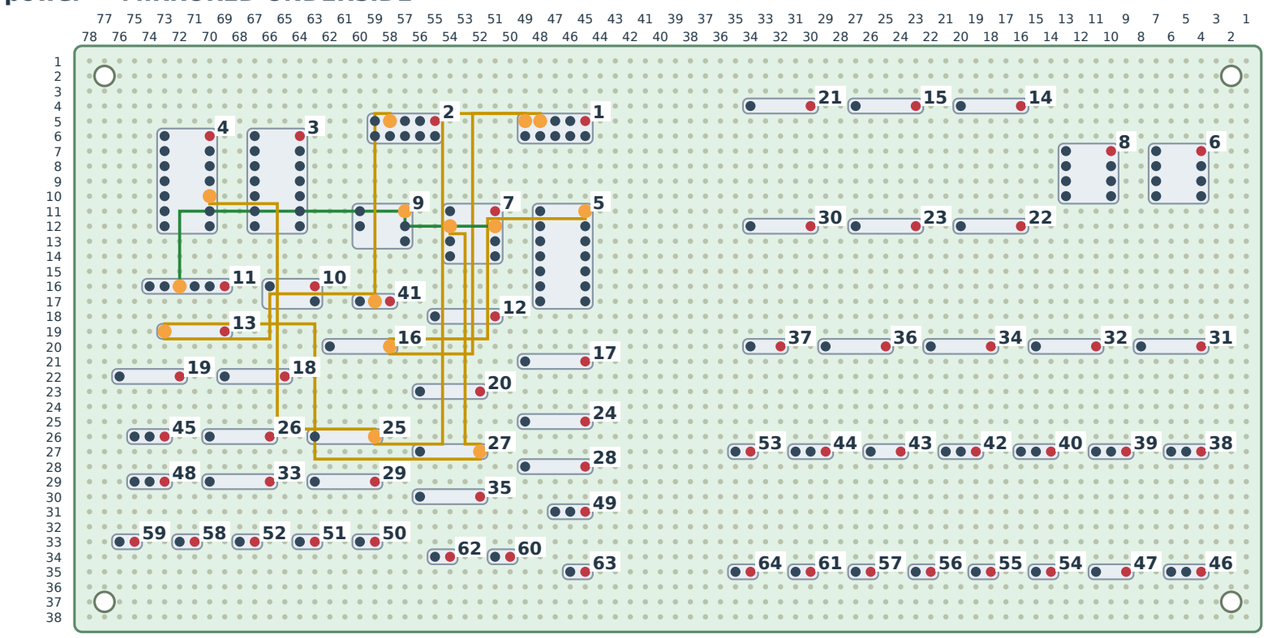
Wire / net	From → to	TOP holes
W76 / 3V3_SYS	U_POWER_INTERLOCK_FAULT.14 → U_POWER_INTERLOCK_POR.2	67,6 → 63,17
W77 / 3V3_SYS	U_POWER_INTERLOCK_POR.2 → U_POWER_INTERLOCK_RESET.14	63,17 → 48,11
W80 / 5V_SYS	JP_POWER_INPUT_LOGIC.2 → C_POWER_ANALOG_TRIP.1	15,35 → 59,33
W81 / 5V_SYS	C_POWER_ANALOG_TRIP.1 → F_POWER_ANALOG_EDU.1	59,33 → 66,29
W82 / 5V_SYS	F_POWER_ANALOG_EDU.1 → R_POWER_ANALOG_REF_FEED.1	66,29 → 45,28
W83 / 5V_SYS	R_POWER_ANALOG_REF_FEED.1 → R_POWER_INTERLOCK_BATTERY_LED.1	45,28 → 66,26
W84 / 5V_SYS	R_POWER_INTERLOCK_BATTERY_LED.1 → U_POWER_ANALOG_TRIP.8	66,26 → 54,11
W85 / EDU_PWR_EN	J_POWER_INTERLOCK_MATRIX.5 → R_POWER_INTERLOCK_EDU_PWR_EN_OFF.1	47,5 → 45,25
W86 / EDU_PWR_EN	R_POWER_INTERLOCK_EDU_PWR_EN_OFF.1 → U_POWER_INTERLOCK_ALLOW.1	45,25 → 70,6
W88 / EDU_FAULT	J_POWER_INTERLOCK_TEST.6 → U_POWER_INTERLOCK_FAULT.5	74,16 → 64,10

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

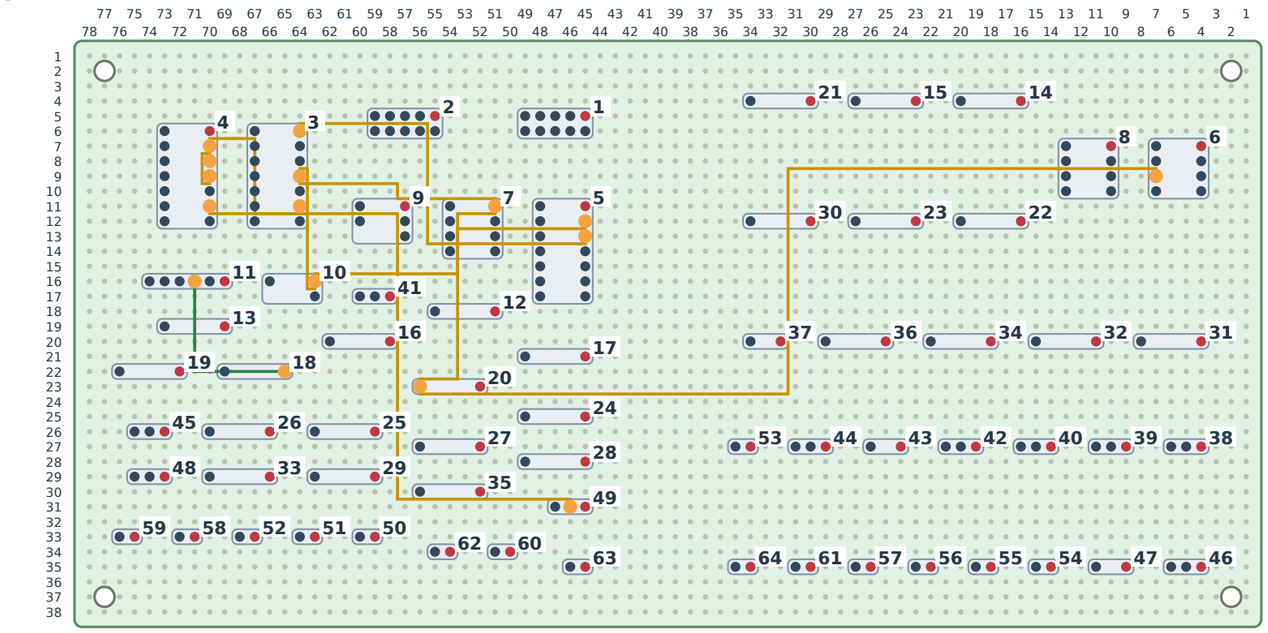
Wire / net	From → to	TOP holes
W89 / EDU_FAULT_RESET	J_POWER_INTERLOCK_MATRIX.7 → R_POWER_INTERLOCK_EDU_FAULT_RESET_OFF.1	48,5 → 58,20
W90 / EDU_FAULT_RESET	R_POWER_INTERLOCK_EDU_FAULT_RESET_OFF.1 → U_POWER_INTERLOCK_RESET.1	58,20 → 45,11
W92 / CURRENT_SENSE	J_POWER_INTERLOCK_TEST.4 → U_POWER_ANALOG_SENSE.1	72,16 → 57,11
W93 / CURRENT_SENSE	U_POWER_ANALOG_SENSE.1 → U_POWER_ANALOG_TRIP.2	57,11 → 51,12
W94 / ROUTER_ALIVE	J_POWER_INTERLOCK_MATRIX.9 → R_POWER_INTERLOCK_ROUTER_ALIVE_OFF.1	49,5 → 59,26
W95 / ROUTER_ALIVE	R_POWER_INTERLOCK_ROUTER_ALIVE_OFF.1 → U_POWER_INTERLOCK_ALLOW.5	59,26 → 70,10
W101 / EDU_ON	J_POWER_INTERLOCK_SMART.7 → Q_POWER_INTERLOCK_BATTERY_LED.2	58,5 → 59,17
W102 / EDU_ON	Q_POWER_INTERLOCK_BATTERY_LED.2 → R_POWER_ANALOG_EDU_ON.2	59,17 → 73,19
W103 / EDU_ON	R_POWER_ANALOG_EDU_ON.2 → R_POWER_INTERLOCK_LED_GATE_OFF.1	73,19 → 52,27
W104 / EDU_ON	R_POWER_INTERLOCK_LED_GATE_OFF.1 → U_POWER_ANALOG_TRIP.7	52,27 → 54,12

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

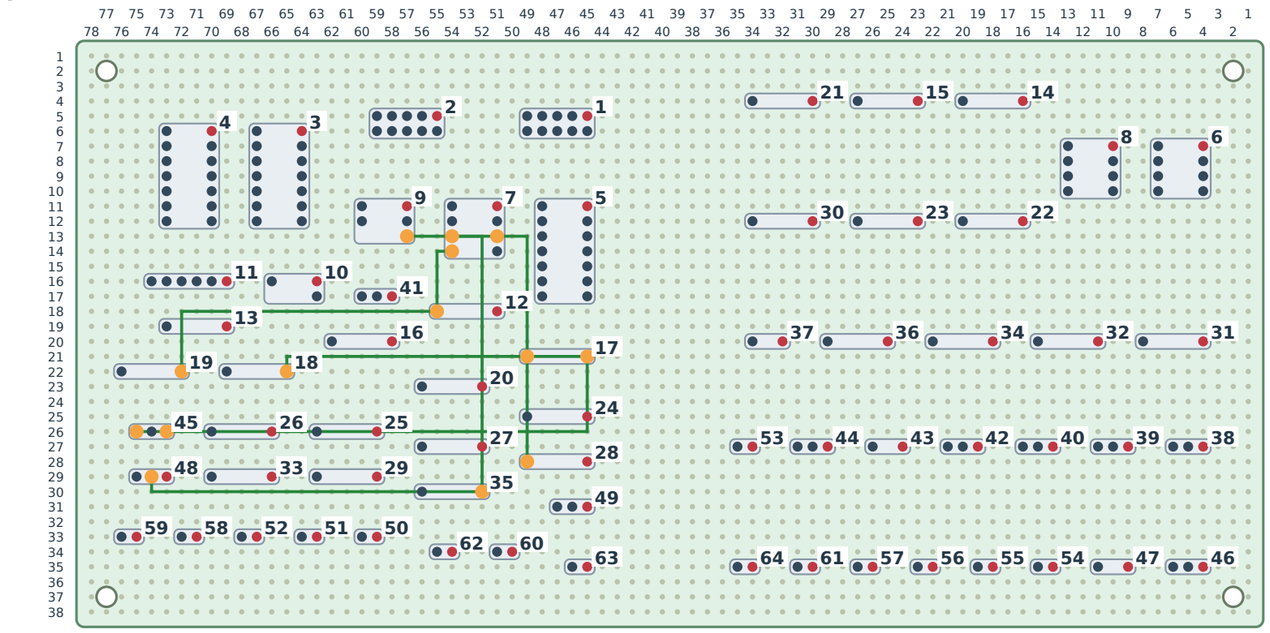
Wire / net	From → to	TOP holes
W105 / CLEAR_N	U_POWER_INTERLOCK_FAULT.1 → U_POWER_INTERLOCK_RESET.3	64,6 → 45,13
W108 / PRESET_N	U_POWER_GUARDS_OV_5V.6 → R_POWER_ANALOG_PRESET.2	7,9 → 56,23
W109 / PRESET_N	R_POWER_ANALOG_PRESET.2 → U_POWER_ANALOG_TRIP.1	56,23 → 51,11
W110 / PRESET_N	U_POWER_ANALOG_TRIP.1 → U_POWER_INTERLOCK_FAULT.4	51,11 → 64,9
W111 / PRESET_N	U_POWER_INTERLOCK_FAULT.4 → U_POWER_INTERLOCK_POR.1	64,9 → 63,16
W112 / PRESET_N	U_POWER_INTERLOCK_POR.1 → U_POWER_INTERLOCK_RESET.2	63,16 → 45,12
W113 / NOT_FAULT	U_POWER_INTERLOCK_ALLOW.2 → U_POWER_INTERLOCK_FAULT.6	70,7 → 64,11
W114 / ENABLE_NO_FAULT	U_POWER_INTERLOCK_ALLOW.3 → U_POWER_INTERLOCK_ALLOW.4	70,8 → 70,9
W115 / ALLOW_RAIL	Q_POWER_ANALOG_ENABLE.2 → U_POWER_INTERLOCK_ALLOW.6	46,31 → 70,11
W131 / REF_TRIP	J_POWER_INTERLOCK_TEST.3 → R_POWER_ANALOG_TRIP_BOTTOM.1	71,16 → 65,22

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

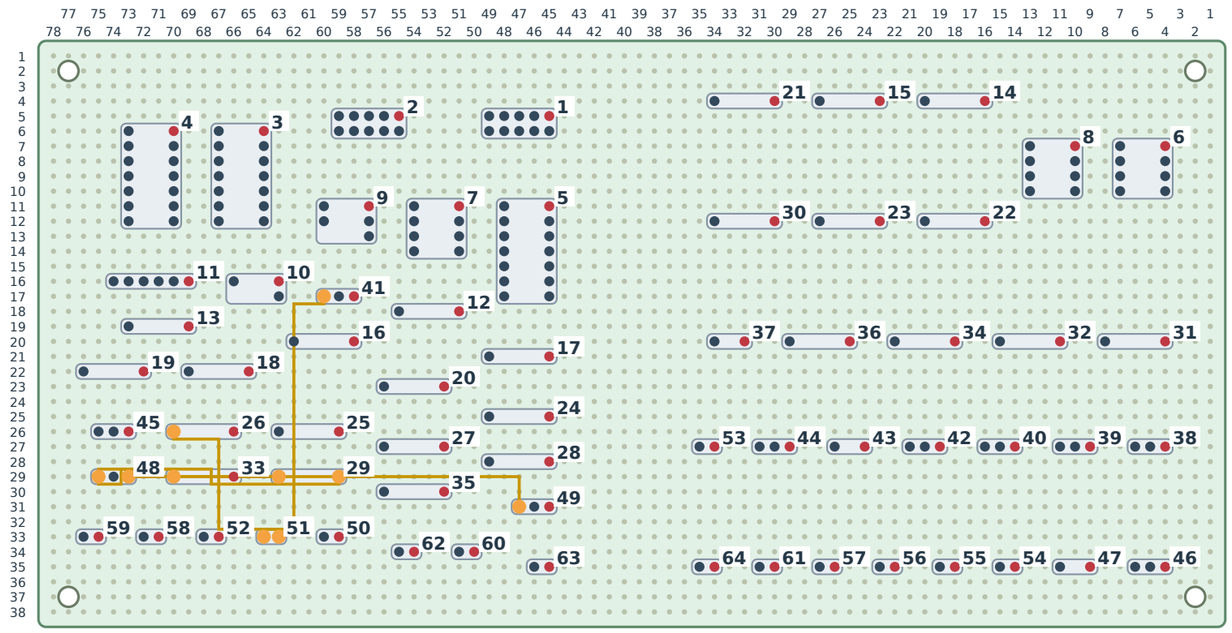
Wire / net	From → to	TOP holes
W132 / REF_TRIP	R_POWER_ANALOG_TRIP_BOTTOM.1 → R_POWER_ANALOG_TRIP_TOP.2	65,22 → 49,21
W133 / REF_TRIP	R_POWER_ANALOG_TRIP_TOP.2 → U_POWER_ANALOG_TRIP.3	49,21 → 51,13
W134 / REF_TRIP	U_POWER_ANALOG_TRIP.3 → U_POWER_ANALOG_TRIP.6	51,13 → 54,13
W135 / EDU_DIV	R_POWER_ANALOG_EDU_BOTTOM.1 → R_POWER_ANALOG_EDU_TOP.2	72,22 → 55,18
W136 / EDU_DIV	R_POWER_ANALOG_EDU_TOP.2 → U_POWER_ANALOG_TRIP.5	55,18 → 54,14
W152 / SHUNT_P	Q_POWER_ANALOG_PASS.2 → R_POWER_ANALOG_SHUNT.1	74,29 → 52,30
W153 / SHUNT_P	R_POWER_ANALOG_SHUNT.1 → U_POWER_ANALOG_SENSE.3	52,30 → 57,13
W154 / REF_2V495	R_POWER_ANALOG_REF_FEED.2 → R_POWER_ANALOG_TRIP_TOP.1	49,28 → 45,21
W155 / REF_2V495	R_POWER_ANALOG_TRIP_TOP.1 → U_POWER_ANALOG_REF.1	45,21 → 73,26
W156 / REF_2V495	U_POWER_ANALOG_REF.1 → U_POWER_ANALOG_REF.3	73,26 → 75,26

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — power

At most ten added wires per view. Inspect each net with power isolated.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

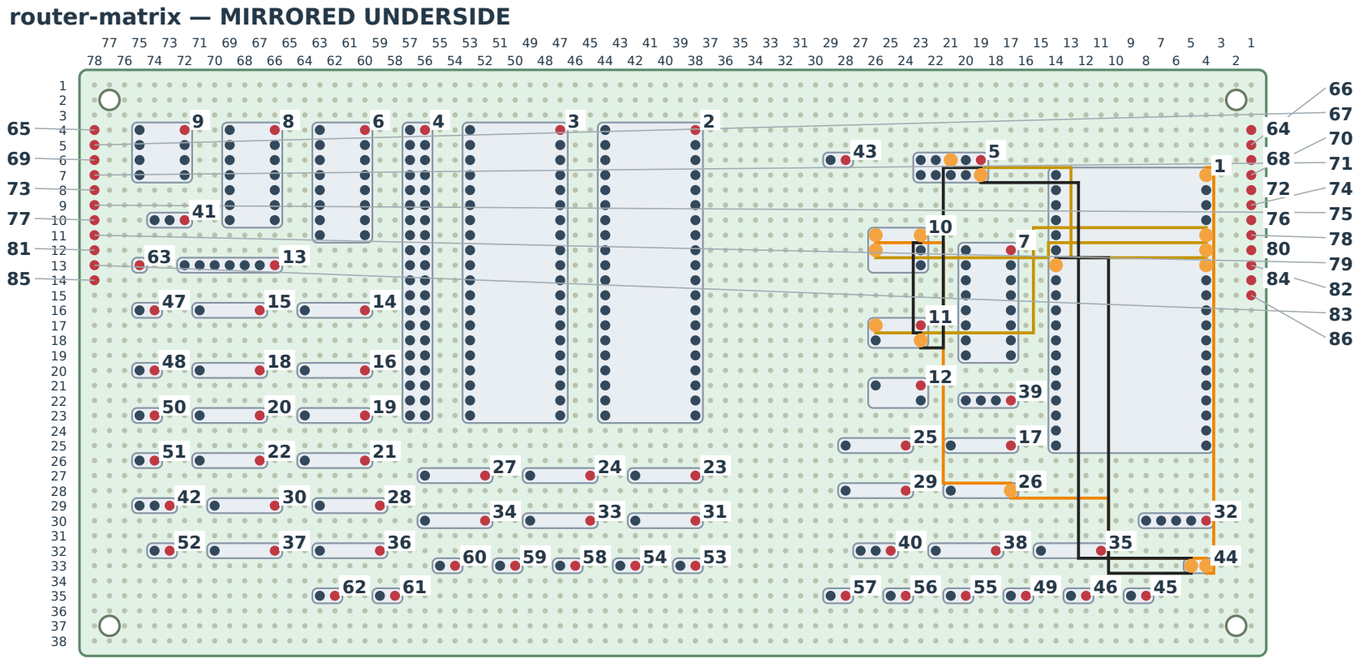
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W158 / BATTERY_LED_A	LED_POWER_INTERLOCK_BATTERY.1 → R_POWER_INTERLOCK_BATTERY_LED.2	63,33 → 70,26
W159 / PTC_OUT	F_POWER_ANALOG_EDU.2 → Q_POWER_ANALOG_PASS.3	70,29 → 75,29
W160 / PTC_OUT	Q_POWER_ANALOG_PASS.3 → R_POWER_ANALOG_GATE_OFF.1	75,29 → 59,29
W161 / PASS_GATE	Q_POWER_ANALOG_ENABLE.3 → Q_POWER_ANALOG_PASS.1	47,31 → 73,29
W162 / PASS_GATE	Q_POWER_ANALOG_PASS.1 → R_POWER_ANALOG_GATE_OFF.2	73,29 → 63,29
W166 / BATTERY_LED_K	LED_POWER_INTERLOCK_BATTERY.2 → Q_POWER_INTERLOCK_BATTERY_LED.3	64,33 → 60,17

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

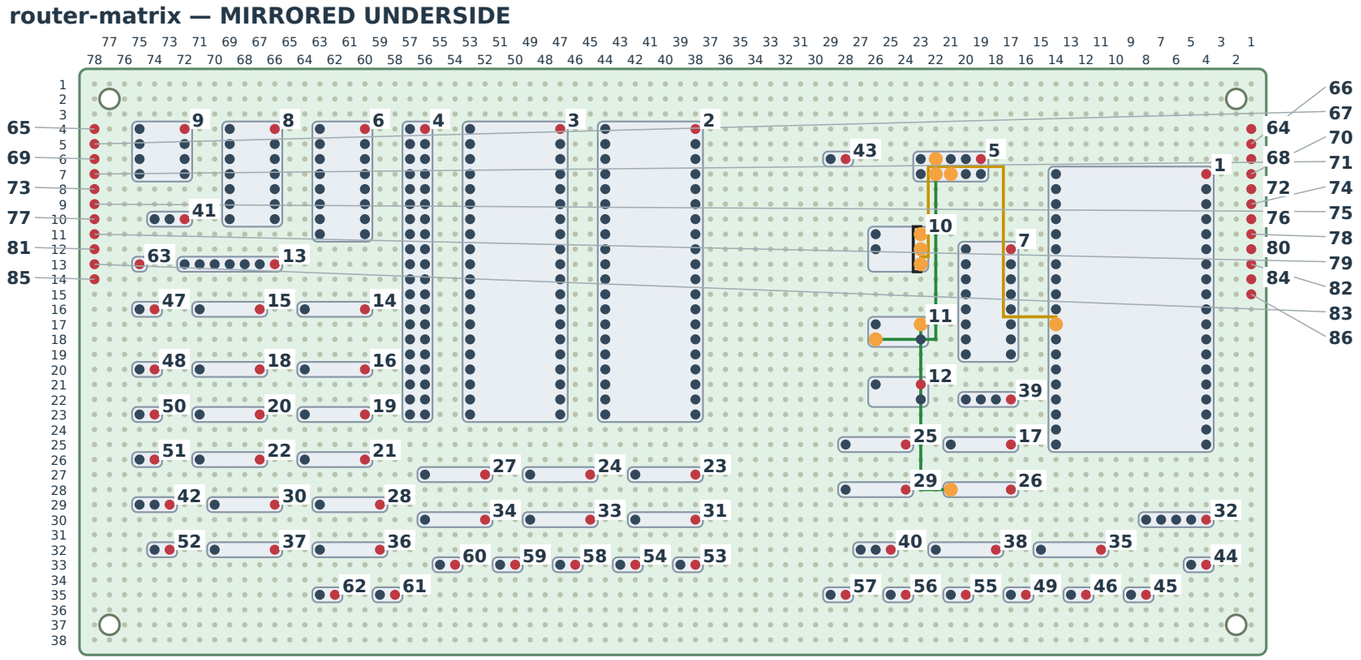
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W3 / USB_ROUTER_3V3	U_ROUTER_CORE_ESP1.1 → C_ROUTER_SAFETY_RETURN.1	4,7 → 4,33
W4 / USB_ROUTER_3V3	C_ROUTER_SAFETY_RETURN.1 → R_ROUTER_SAFETY_KADC_LED.1	4,33 → 17,28
W5 / USB_ROUTER_3V3	R_ROUTER_SAFETY_KADC_LED.1 → U_ROUTER_SAFETY_FAULT_RETURN.5	17,28 → 26,11
W15 / CURRENT_USB	U_ROUTER_CORE_ESP1.5 → K_ROUTER_SAFETY_ADC.4	4,11 → 26,17
W16 / FAULT_USB	U_ROUTER_CORE_ESP1.6 → U_ROUTER_SAFETY_FAULT_RETURN.4	4,12 → 26,12
W18 / EDU_PWR_EN	U_ROUTER_CORE_ESP1.7 → J_ROUTER_CORE_POWER.5	4,13 → 21,6
W28 / GND	U_ROUTER_CORE_ESP1.26 → C_ROUTER_SAFETY_RETURN.2	14,13 → 5,33
W29 / GND	C_ROUTER_SAFETY_RETURN.2 → J_ROUTER_CORE_POWER.2	5,33 → 19,7
W30 / GND	J_ROUTER_CORE_POWER.2 → K_ROUTER_SAFETY_ADC.2	19,7 → 23,18
W31 / GND	K_ROUTER_SAFETY_ADC.2 → U_ROUTER_SAFETY_FAULT_RETURN.1	23,18 → 23,11

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP11 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

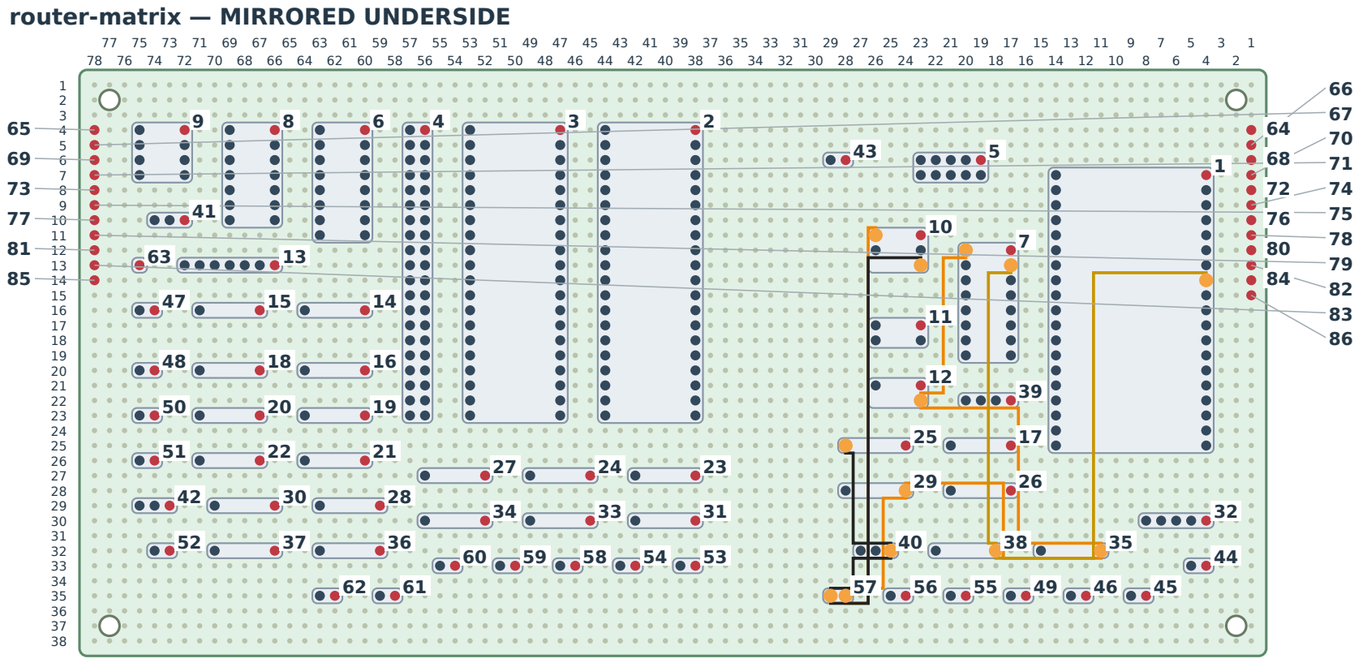
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W32 / GND	U_ROUTER_SAFETY_FAULT_RETURN.1 → U_ROUTER_SAFETY_FAULT_RETURN.3	23,11 → 23,13
W99 / EDU_FAULT_RESET	U_ROUTER_CORE_ESP1.30 → J_ROUTER_CORE_POWER.7	14,17 → 22,6
W197 / EDU_FAULT	J_ROUTER_CORE_POWER.6 → U_ROUTER_SAFETY_FAULT_RETURN.2	21,7 → 23,12
W198 / CURRENT_SENSE	J_ROUTER_CORE_POWER.8 → K_ROUTER_SAFETY_ADC.3	22,7 → 26,18
W216 / K_ADC_LED	K_ROUTER_SAFETY_ADC.1 → R_ROUTER_SAFETY_KADC_LED.2	23,17 → 21,28

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP13 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

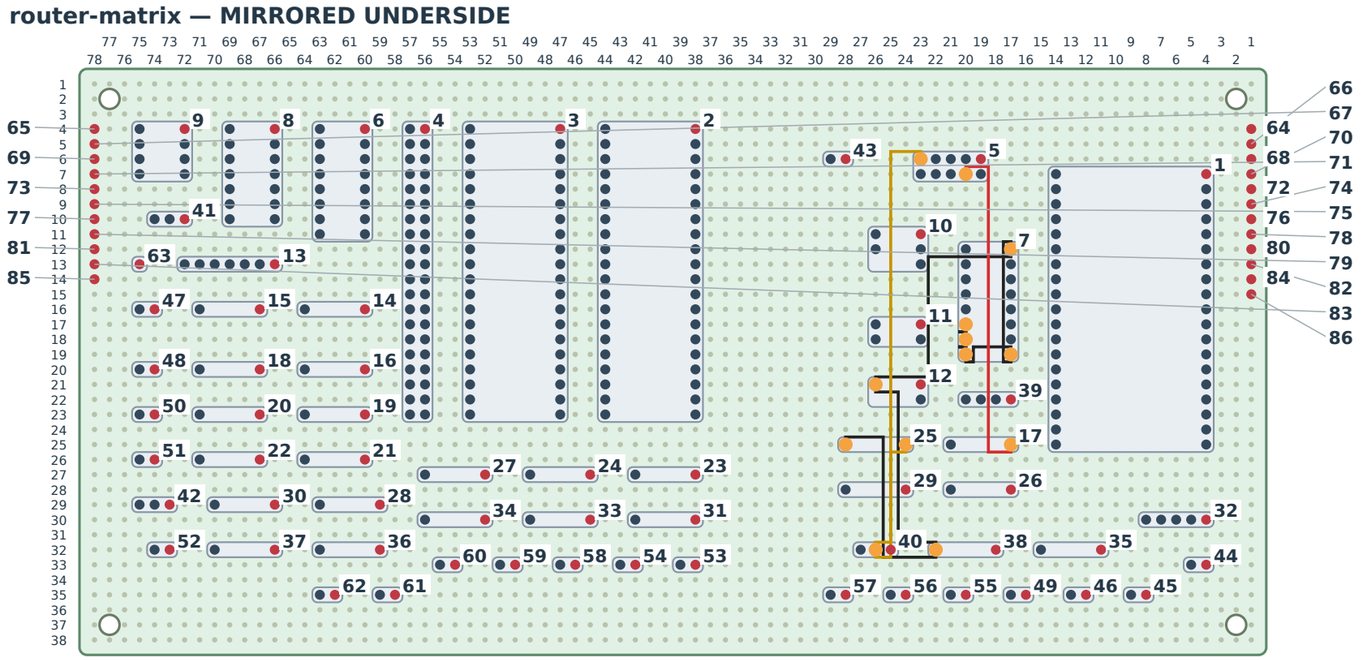
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W6 / USB_ROUTER_3V3	U_ROUTER_SAFETY_FAULT_RETURN.5 → C_ROUTER_SAFETY_WATCH_VCC.1	26,11 → 28,35
W7 / USB_ROUTER_3V3	C_ROUTER_SAFETY_WATCH_VCC.1 → R_ROUTER_SAFETY_USB_POR.1	28,35 → 24,28
W8 / USB_ROUTER_3V3	R_ROUTER_SAFETY_USB_POR.1 → R_ROUTER_SAFETY_WATCH.1	24,28 → 11,32
W9 / USB_ROUTER_3V3	R_ROUTER_SAFETY_WATCH.1 → U_ROUTER_SAFETY_USB_POR.2	11,32 → 23,22
W10 / USB_ROUTER_3V3	U_ROUTER_SAFETY_USB_POR.2 → U_ROUTER_SAFETY_WATCH.16	23,22 → 20,12
W19 / HEARTBEAT	U_ROUTER_CORE_ESP1.8 → R_ROUTER_SAFETY_HEART.1	4,14 → 18,32
W20 / HEARTBEAT	R_ROUTER_SAFETY_HEART.1 → U_ROUTER_SAFETY_WATCH.2	18,32 → 17,13
W33 / GND	U_ROUTER_SAFETY_FAULT_RETURN.3 → C_ROUTER_SAFETY_WATCH_VCC.2	23,13 → 29,35
W34 / GND	C_ROUTER_SAFETY_WATCH_VCC.2 → Q_ROUTER_SAFETY_BUF.1	29,35 → 25,32
W35 / GND	Q_ROUTER_SAFETY_BUF.1 → R_ROUTER_SAFETY_ALIVE_OFF.2	25,32 → 28,25

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP13 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

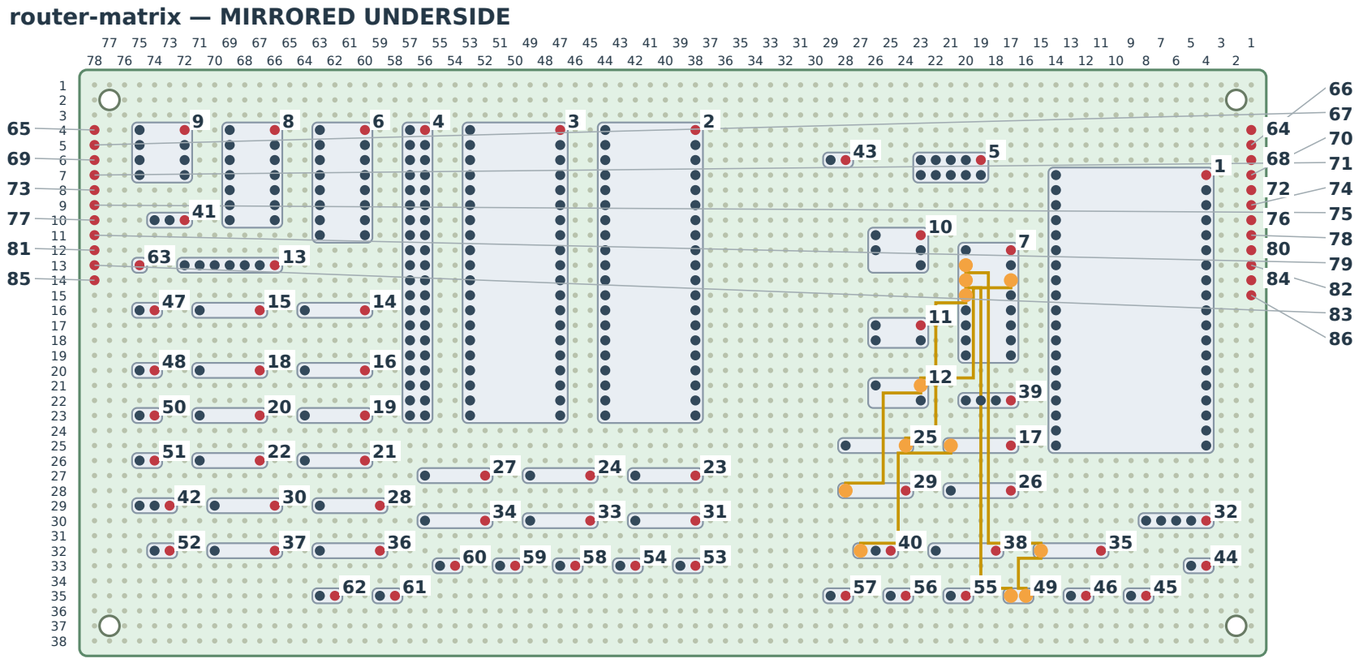
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W36 / GND	R_ROUTER_SAFETY_ALIVE_OFF.2 → R_ROUTER_SAFETY_HEART.2	28,25 → 22,32
W37 / GND	R_ROUTER_SAFETY_HEART.2 → U_ROUTER_SAFETY_USB_POR.3	22,32 → 26,21
W38 / GND	U_ROUTER_SAFETY_USB_POR.3 → U_ROUTER_SAFETY_WATCH.1	26,21 → 17,12
W39 / GND	U_ROUTER_SAFETY_WATCH.1 → U_ROUTER_SAFETY_WATCH.8	17,12 → 17,19
W40 / GND	U_ROUTER_SAFETY_WATCH.8 → U_ROUTER_SAFETY_WATCH.9	17,19 → 20,19
W41 / GND	U_ROUTER_SAFETY_WATCH.9 → U_ROUTER_SAFETY_WATCH.10	20,19 → 20,18
W42 / GND	U_ROUTER_SAFETY_WATCH.10 → U_ROUTER_SAFETY_WATCH.11	20,18 → 20,17
W142 / 5V_SYS	J_ROUTER_CORE_POWER.4 → R_ROUTER_SAFETY_BUF_OE.1	20,7 → 17,25
W199 / ROUTER_ALIVE	J_ROUTER_CORE_POWER.9 → Q_ROUTER_SAFETY_BUF.2	23,6 → 26,32
W200 / ROUTER_ALIVE	Q_ROUTER_SAFETY_BUF.2 → R_ROUTER_SAFETY_ALIVE_OFF.1	26,32 → 24,25

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP13 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

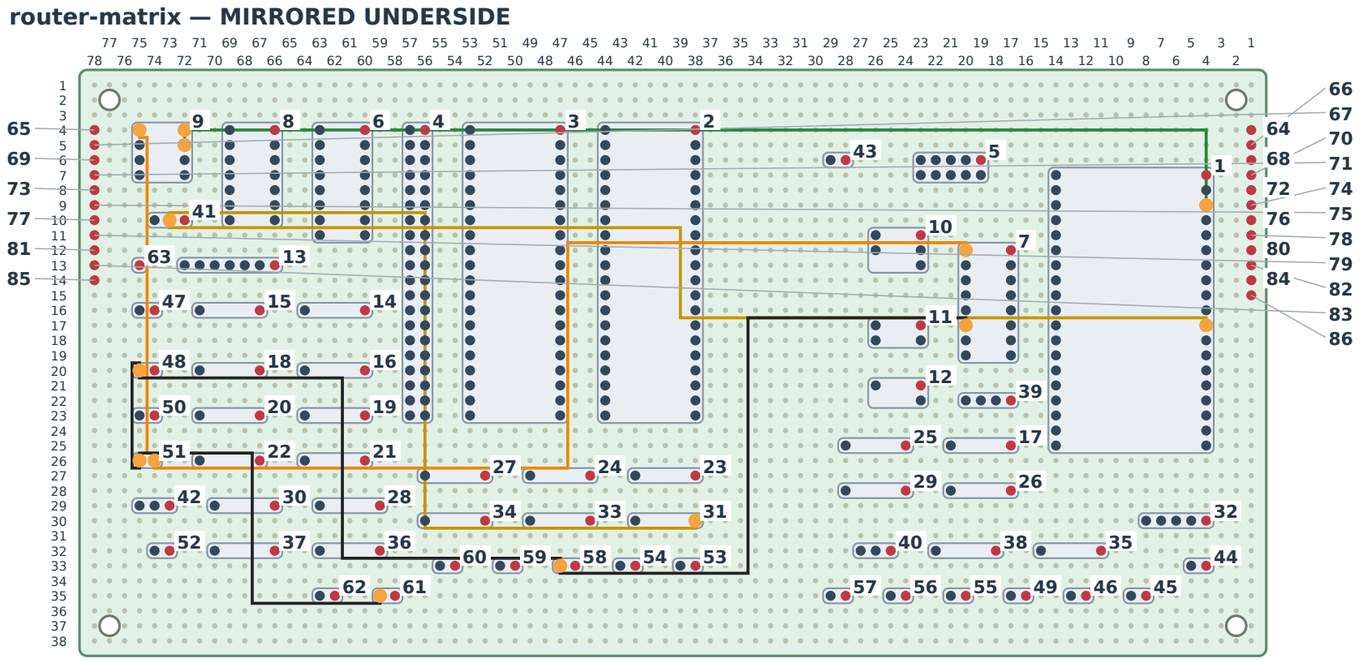
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W201 / ROUTER_ALIVE	R_ROUTER_SAFETY_ALIVE_OFF.1 → U_ROUTER_SAFETY_WATCH.13	24,25 → 20,15
W203 / XBAR_OE_N	Q_ROUTER_SAFETY_BUF.3 → R_ROUTER_SAFETY_BUF_OE.2	27,32 → 21,25
W207 / USB_RESET_N	R_ROUTER_SAFETY_USB_POR.2 → U_ROUTER_SAFETY_USB_POR.1	28,28 → 23,21
W208 / USB_RESET_N	U_ROUTER_SAFETY_USB_POR.1 → U_ROUTER_SAFETY_WATCH.3	23,21 → 17,14
W209 / WATCH_CX	C_ROUTER_SAFETY_WATCH.2 → U_ROUTER_SAFETY_WATCH.14	17,35 → 20,14
W210 / WATCH_RX	C_ROUTER_SAFETY_WATCH.1 → R_ROUTER_SAFETY_WATCH.2	16,35 → 15,32
W211 / WATCH_RX	R_ROUTER_SAFETY_WATCH.2 → U_ROUTER_SAFETY_WATCH.15	15,32 → 20,13

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

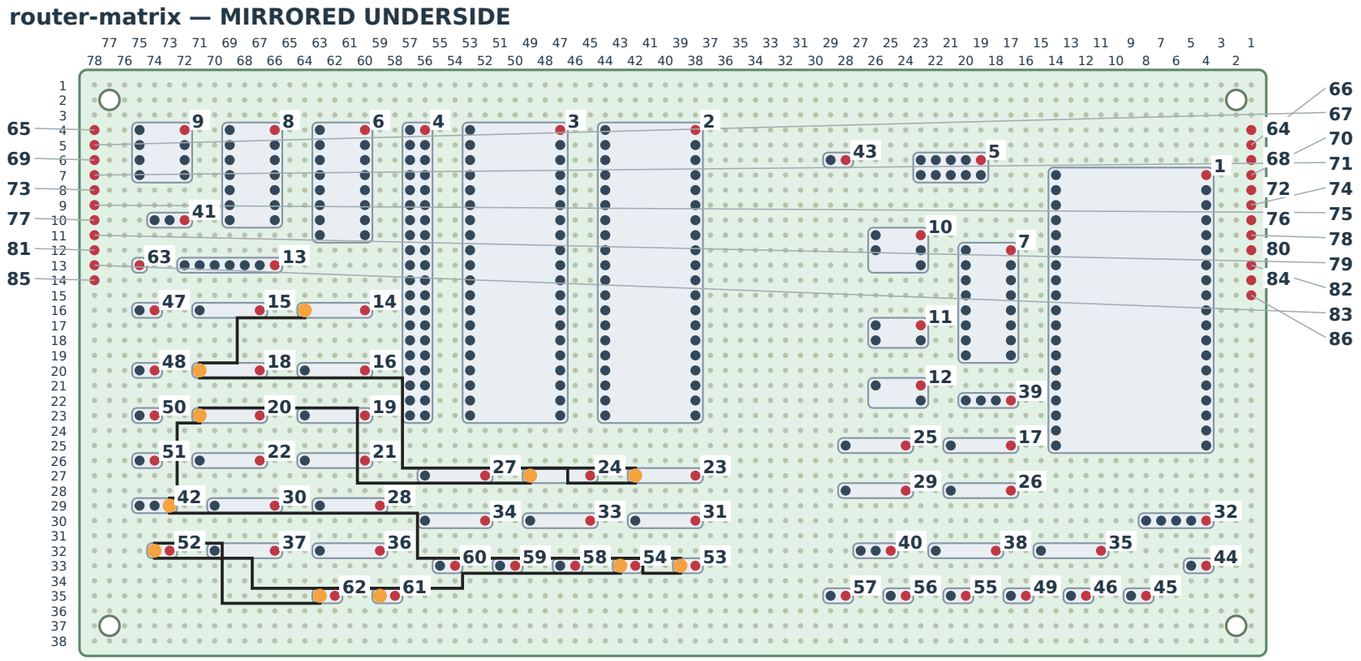
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W11 / USB_ROUTER_3V3	U_ROUTER_SAFETY_WATCH.16 → C_MATRIX_CONTROL_DIAG_VCC.1	20,12 → 74,26
W12 / USB_ROUTER_3V3	C_MATRIX_CONTROL_DIAG_VCC.1 → U_MATRIX_CONTROL_DIAG.8	74,26 → 75,4
W13 / DIAG_ADC	U_ROUTER_CORE_ESP1.3 → U_MATRIX_CONTROL_DIAG.1	4,9 → 72,4
W14 / DIAG_ADC	U_MATRIX_CONTROL_DIAG.1 → U_MATRIX_CONTROL_DIAG.2	72,4 → 72,5
W22 / RST_RELEASE	U_ROUTER_CORE_ESP1.11 → Q_MATRIX_SWITCHES_RESET_RELEASE.2	4,17 → 73,10
W23 / RST_RELEASE	Q_MATRIX_SWITCHES_RESET_RELEASE.2 → R_MATRIX_SWITCHES_RST_RELEASE_OFF.1	73,10 → 38,30
W43 / GND	U_ROUTER_SAFETY_WATCH.11 → C_MATRIX_CONTROL_ADDR.2	20,17 → 47,33
W44 / GND	C_MATRIX_CONTROL_ADDR.2 → C_MATRIX_CONTROL_BUFFER.2	47,33 → 75,20
W45 / GND	C_MATRIX_CONTROL_BUFFER.2 → C_MATRIX_CONTROL_DIAG_VCC.2	75,20 → 75,26
W46 / GND	C_MATRIX_CONTROL_DIAG_VCC.2 → C_MATRIX_CONTROL_DIAG.2	75,26 → 59,35

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

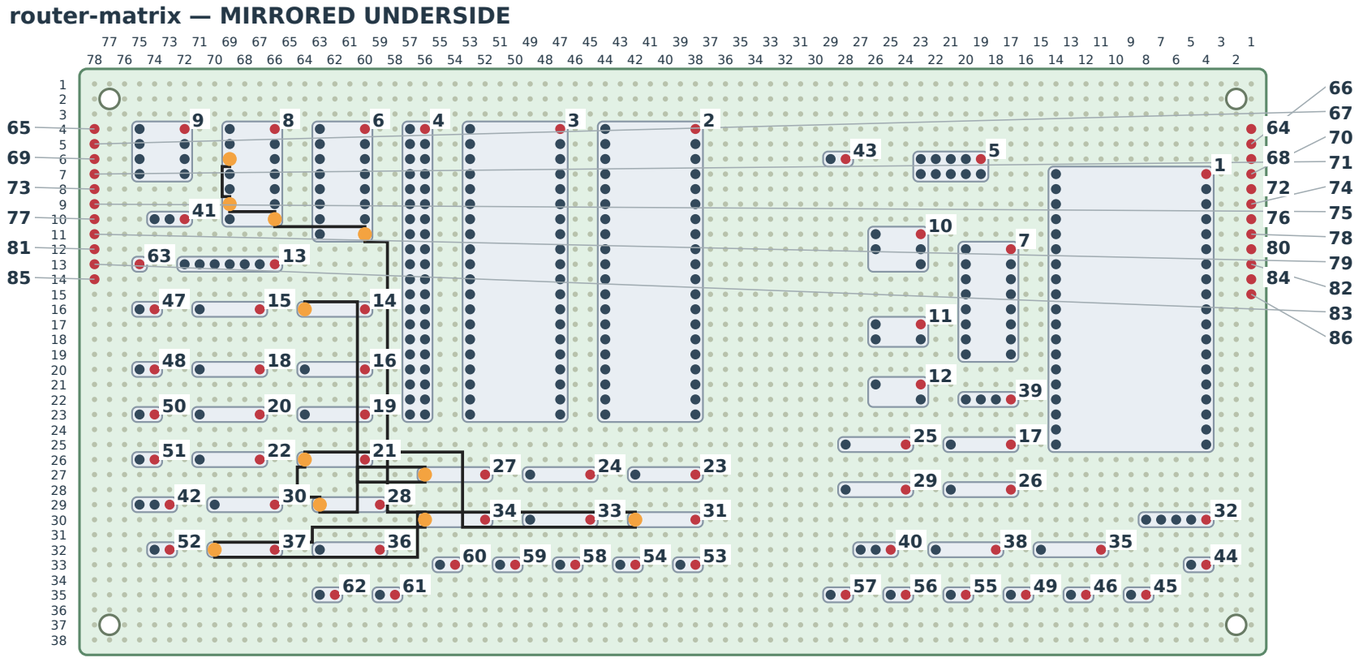
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W47 / GND	C_MATRIX_CONTROL_DIAG.2 → C_MATRIX_SWITCHES_MT0_BULK.2	59,35 → 74,32
W48 / GND	C_MATRIX_SWITCHES_MT0_BULK.2 → C_MATRIX_SWITCHES_MT0.2	74,32 → 63,35
W49 / GND	C_MATRIX_SWITCHES_MT0.2 → C_MATRIX_SWITCHES_MT1_BULK.2	63,35 → 43,33
W50 / GND	C_MATRIX_SWITCHES_MT1_BULK.2 → C_MATRIX_SWITCHES_MT1.2	43,33 → 39,33
W51 / GND	C_MATRIX_SWITCHES_MT1.2 → Q_MATRIX_SWITCHES_RESET_ALIVE.1	39,33 → 73,29
W52 / GND	Q_MATRIX_SWITCHES_RESET_ALIVE.1 → R_MATRIX_CONTROL_CAL.2	73,29 → 71,23
W53 / GND	R_MATRIX_CONTROL_CAL.2 → R_MATRIX_CONTROL_DIAG_BOTTOM.2	71,23 → 49,27
W54 / GND	R_MATRIX_CONTROL_DIAG_BOTTOM.2 → R_MATRIX_CONTROL_DISCHARGE.2	49,27 → 42,27
W55 / GND	R_MATRIX_CONTROL_DISCHARGE.2 → R_MATRIX_CONTROL_STB_A_IN_OFF.2	42,27 → 71,20
W56 / GND	R_MATRIX_CONTROL_STB_A_IN_OFF.2 → R_MATRIX_CONTROL_STB_B_IN_OFF.2	71,20 → 64,16

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

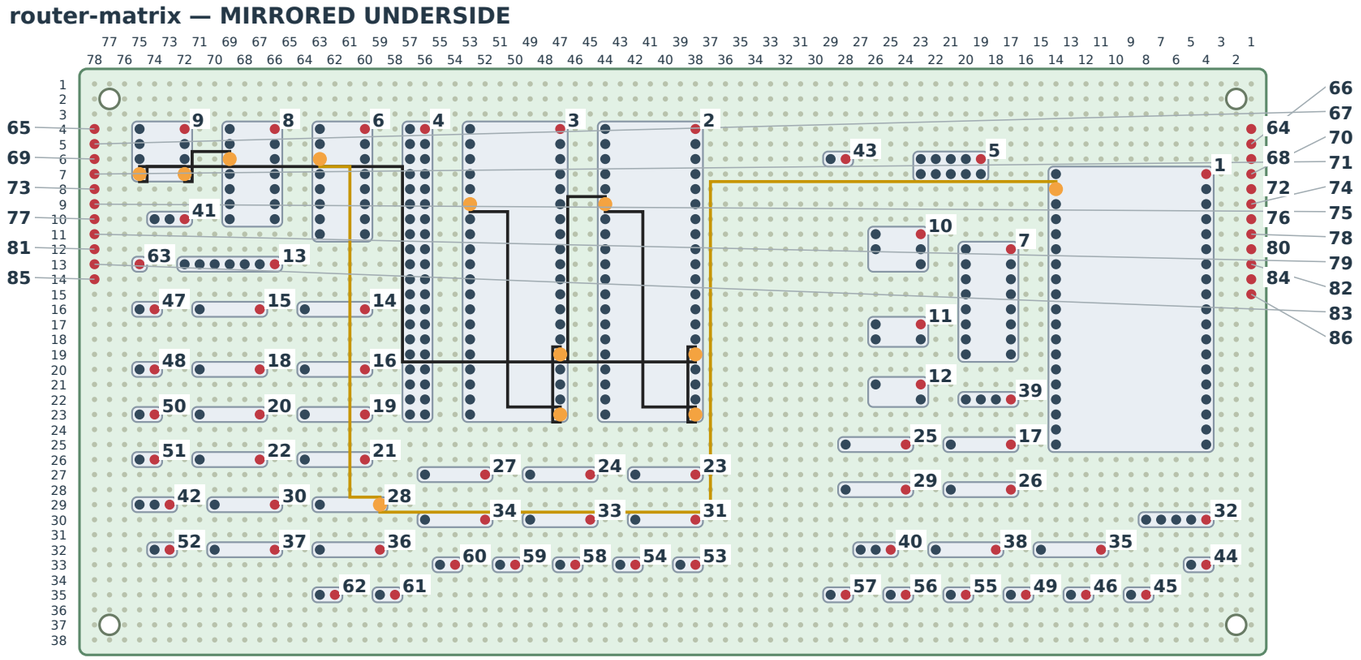
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W57 / GND	R_MATRIX_CONTROL_STB_B_IN_OFF.2 → R_MATRIX_CONTROL_XBAR_CLK_OFF.2	64,16 → 56,27
W58 / GND	R_MATRIX_CONTROL_XBAR_CLK_OFF.2 → R_MATRIX_CONTROL_XBAR_DAT_OFF.2	56,27 → 63,29
W59 / GND	R_MATRIX_CONTROL_XBAR_DAT_OFF.2 → R_MATRIX_CONTROL_XBAR_LATCH_OFF.2	63,29 → 64,26
W60 / GND	R_MATRIX_CONTROL_XBAR_LATCH_OFF.2 → R_MATRIX_SWITCHES_RST_RELEASE_OFF.2	64,26 → 42,30
W61 / GND	R_MATRIX_SWITCHES_RST_RELEASE_OFF.2 → R_MATRIX_SWITCHES_STBA_OFF.2	42,30 → 70,32
W62 / GND	R_MATRIX_SWITCHES_STBA_OFF.2 → R_MATRIX_SWITCHES_STBB_OFF.2	70,32 → 56,30
W63 / GND	R_MATRIX_SWITCHES_STBB_OFF.2 → U_MATRIX_CONTROL_ADDR.8	56,30 → 60,11
W64 / GND	U_MATRIX_CONTROL_ADDR.8 → U_MATRIX_CONTROL_BUFFER.7	60,11 → 66,10
W65 / GND	U_MATRIX_CONTROL_BUFFER.7 → U_MATRIX_CONTROL_BUFFER.9	66,10 → 69,9
W66 / GND	U_MATRIX_CONTROL_BUFFER.9 → U_MATRIX_CONTROL_BUFFER.12	69,9 → 69,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

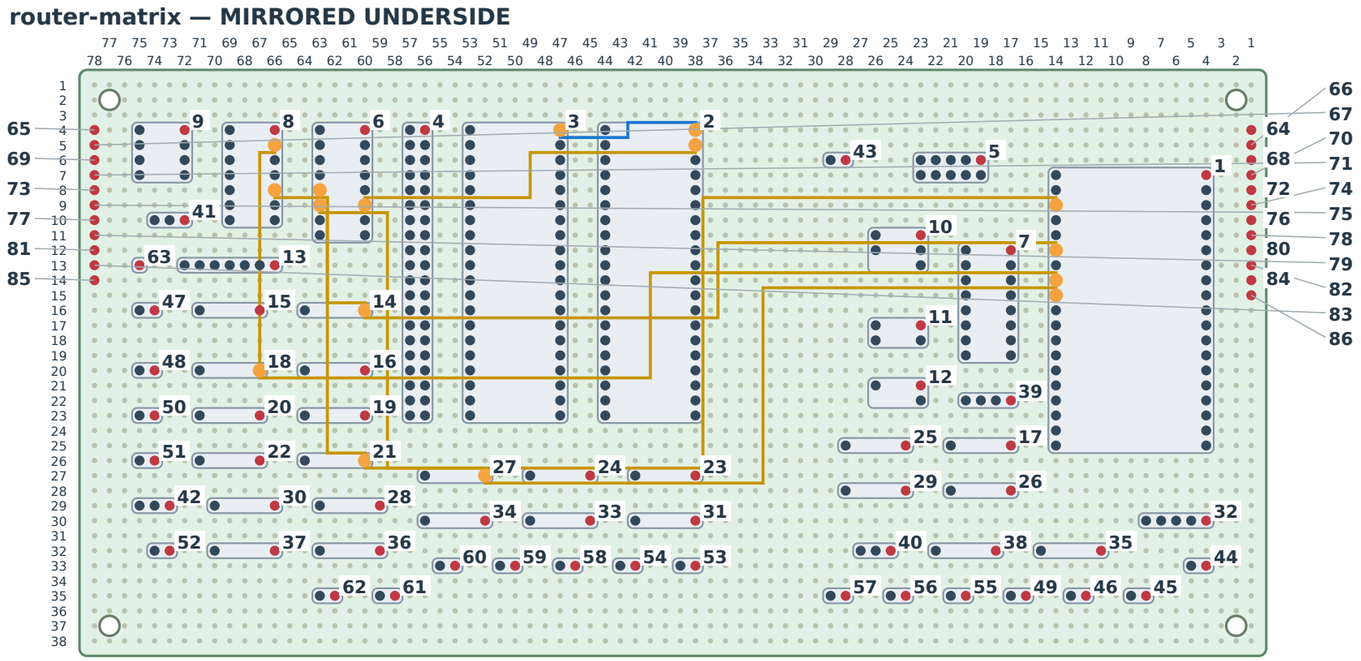
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W67 / GND	U_MATRIX_CONTROL_BUFFER.12 → U_MATRIX_CONTROL_DIAG.4	69,6 → 72,7
W68 / GND	U_MATRIX_CONTROL_DIAG.4 → U_MATRIX_CONTROL_DIAG.5	72,7 → 75,7
W69 / GND	U_MATRIX_CONTROL_DIAG.5 → U_MATRIX_SWITCHES_MT_A.16	75,7 → 38,19
W70 / GND	U_MATRIX_SWITCHES_MT_A.16 → U_MATRIX_SWITCHES_MT_A.20	38,19 → 38,23
W71 / GND	U_MATRIX_SWITCHES_MT_A.20 → U_MATRIX_SWITCHES_MT_A.35	38,23 → 44,9
W72 / GND	U_MATRIX_SWITCHES_MT_A.35 → U_MATRIX_SWITCHES_MT_B.16	44,9 → 47,19
W73 / GND	U_MATRIX_SWITCHES_MT_B.16 → U_MATRIX_SWITCHES_MT_B.20	47,19 → 47,23
W74 / GND	U_MATRIX_SWITCHES_MT_B.20 → U_MATRIX_SWITCHES_MT_B.35	47,23 → 53,9
W88 / XBAR_DAT	U_ROUTER_CORE_ESP1.21 → R_MATRIX_CONTROL_XBAR_DAT_OFF.1	14,8 → 59,29
W89 / XBAR_DAT	R_MATRIX_CONTROL_XBAR_DAT_OFF.1 → U_MATRIX_CONTROL_ADDR.14	59,29 → 63,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

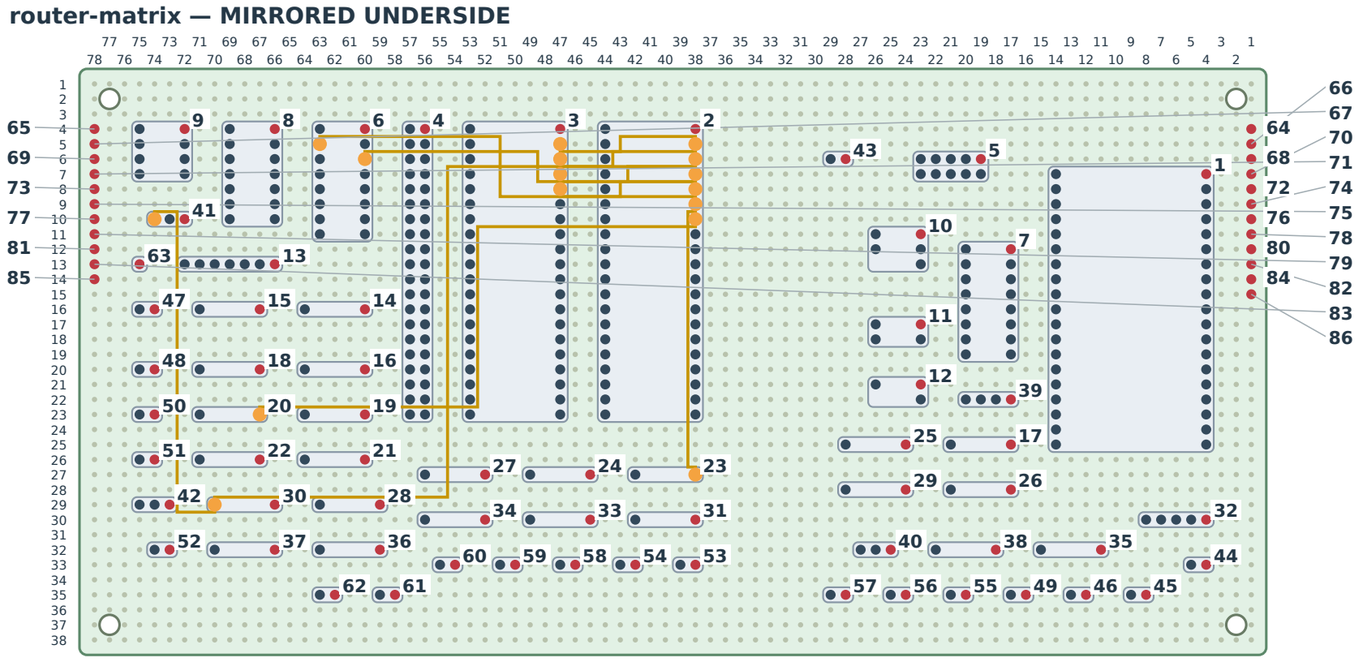
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W90 / XBAR_LATCH	U_ROUTER_CORE_ESP1.22 → R_MATRIX_CONTROL_XBAR_LATCH_OFF.1	14,9 → 60,26
W91 / XBAR_LATCH	R_MATRIX_CONTROL_XBAR_LATCH_OFF.1 → U_MATRIX_CONTROL_ADDR.12	60,26 → 63,8
W92 / STB_B_IN	U_ROUTER_CORE_ESP1.25 → R_MATRIX_CONTROL_STB_B_IN_OFF.1	14,12 → 60,16
W93 / STB_B_IN	R_MATRIX_CONTROL_STB_B_IN_OFF.1 → U_MATRIX_CONTROL_BUFFER.5	60,16 → 66,8
W94 / STB_A_IN	U_ROUTER_CORE_ESP1.27 → R_MATRIX_CONTROL_STB_A_IN_OFF.1	14,14 → 67,20
W95 / STB_A_IN	R_MATRIX_CONTROL_STB_A_IN_OFF.1 → U_MATRIX_CONTROL_BUFFER.2	67,20 → 66,5
W96 / XBAR_CLK	U_ROUTER_CORE_ESP1.28 → R_MATRIX_CONTROL_XBAR_CLK_OFF.1	14,15 → 52,27
W97 / XBAR_CLK	R_MATRIX_CONTROL_XBAR_CLK_OFF.1 → U_MATRIX_CONTROL_ADDR.11	52,27 → 63,9
W102 / BUS3	U_MATRIX_SWITCHES_MT_A.1 → U_MATRIX_SWITCHES_MT_B.1	38,4 → 47,4
W103 / AY2	U_MATRIX_CONTROL_ADDR.6 → U_MATRIX_SWITCHES_MT_A.2	60,9 → 38,5

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

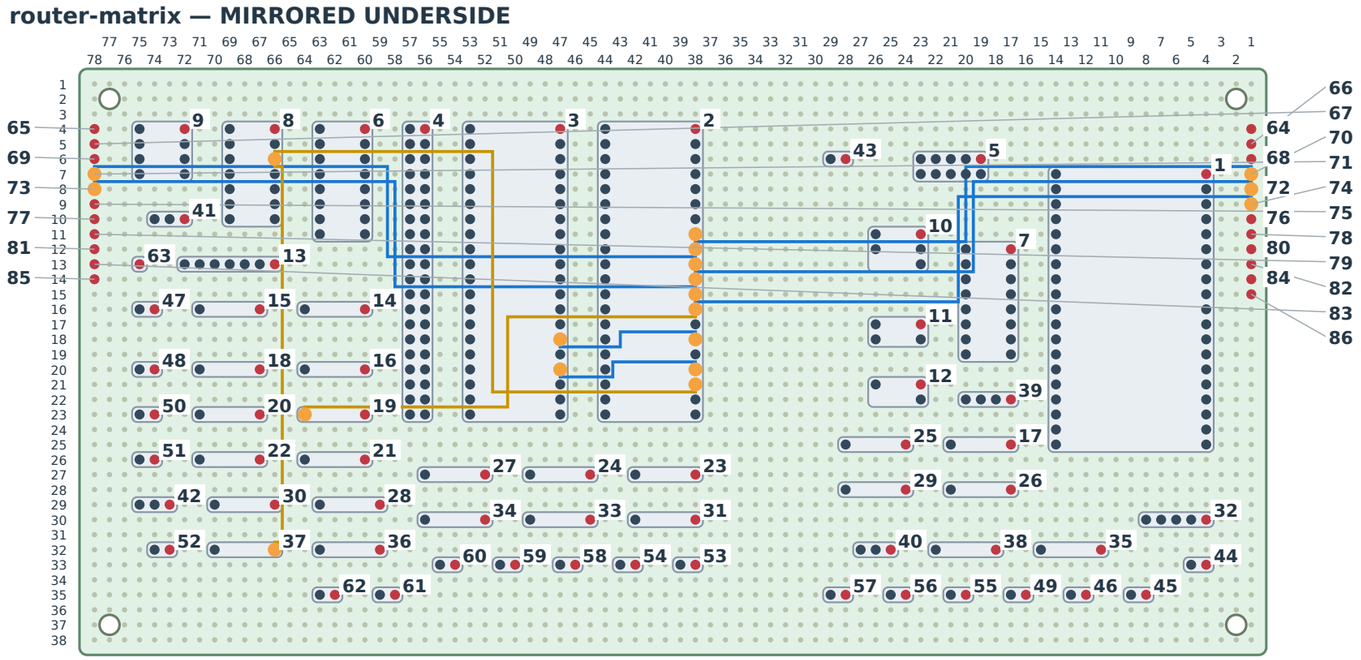
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W104 / AY2	U_MATRIX_SWITCHES_MT_A.2 → U_MATRIX_SWITCHES_MT_B.2	38,5 → 47,5
W105 / XBAR_RESET	Q_MATRIX_SWITCHES_RESET_RELEASE.3 → R_MATRIX_SWITCHES_RESET_ON.2	74,10 → 70,29
W106 / XBAR_RESET	R_MATRIX_SWITCHES_RESET_ON.2 → U_MATRIX_SWITCHES_MT_A.3	70,29 → 38,6
W107 / XBAR_RESET	U_MATRIX_SWITCHES_MT_A.3 → U_MATRIX_SWITCHES_MT_B.3	38,6 → 47,6
W108 / AX3	U_MATRIX_CONTROL_ADDR.3 → U_MATRIX_SWITCHES_MT_A.4	60,6 → 38,7
W109 / AX3	U_MATRIX_SWITCHES_MT_A.4 → U_MATRIX_SWITCHES_MT_B.4	38,7 → 47,7
W110 / AX0	U_MATRIX_CONTROL_ADDR.15 → U_MATRIX_SWITCHES_MT_A.5	63,5 → 38,8
W111 / AX0	U_MATRIX_SWITCHES_MT_A.5 → U_MATRIX_SWITCHES_MT_B.5	38,8 → 47,8
W112 / DISCHARGE	R_MATRIX_CONTROL_DISCHARGE.1 → U_MATRIX_SWITCHES_MT_A.6	38,27 → 38,9
W113 / CAL_LOAD	R_MATRIX_CONTROL_CAL.1 → U_MATRIX_SWITCHES_MT_A.7	67,23 → 38,10

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

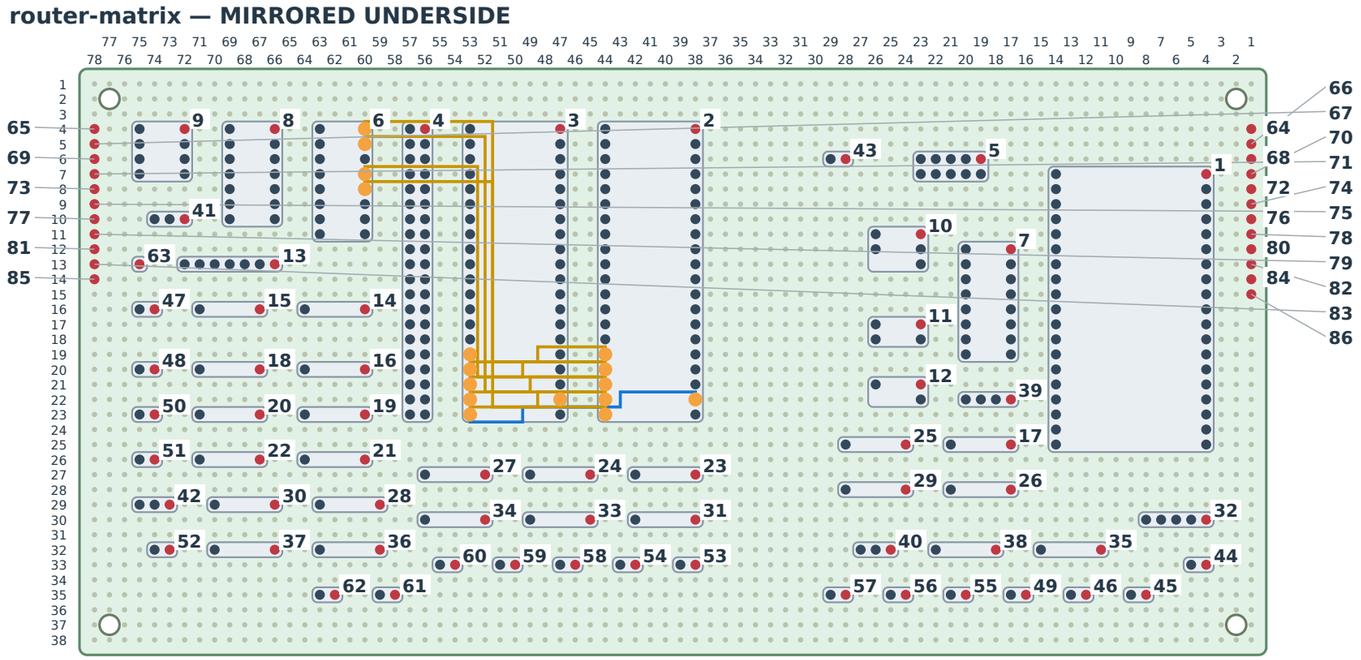
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W114 / NODE06	J_BOUNDARY_7.1 → U_MATRIX_SWITCHES_MT_A.8	1,7 → 38,11
W115 / NODE07	J_BOUNDARY_8.1 → U_MATRIX_SWITCHES_MT_A.9	78,7 → 38,12
W116 / NODE08	J_BOUNDARY_9.1 → U_MATRIX_SWITCHES_MT_A.10	1,8 → 38,13
W117 / NODE09	J_BOUNDARY_10.1 → U_MATRIX_SWITCHES_MT_A.11	78,8 → 38,14
W118 / NODE10	J_BOUNDARY_11.1 → U_MATRIX_SWITCHES_MT_A.12	1,9 → 38,15
W119 / SOURCE_HI	R_MATRIX_CONTROL_SOURCE_HI.2 → U_MATRIX_SWITCHES_MT_A.13	64,23 → 38,16
W120 / BUS7	U_MATRIX_SWITCHES_MT_A.15 → U_MATRIX_SWITCHES_MT_B.15	38,18 → 47,18
W121 / BUS6	U_MATRIX_SWITCHES_MT_A.17 → U_MATRIX_SWITCHES_MT_B.17	38,20 → 47,20
W122 / STROBE_A	R_MATRIX_SWITCHES_STBA_OFF.1 → U_MATRIX_CONTROL_BUFFER.3	66,32 → 66,6
W123 / STROBE_A	U_MATRIX_CONTROL_BUFFER.3 → U_MATRIX_SWITCHES_MT_A.18	66,6 → 38,21

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

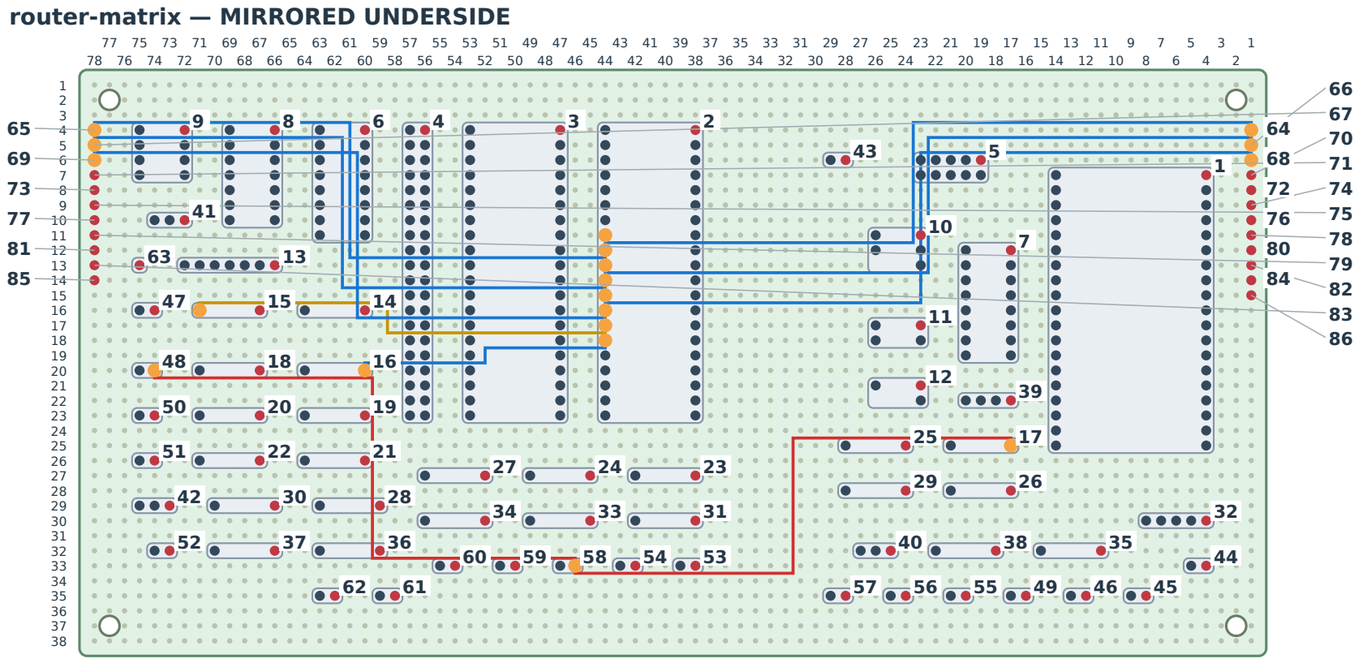
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W124 / BUS5	U_MATRIX_SWITCHES_MT_A.19 → U_MATRIX_SWITCHES_MT_B.19	38,22 → 47,22
W125 / BUS4	U_MATRIX_SWITCHES_MT_A.21 → U_MATRIX_SWITCHES_MT_B.21	44,23 → 53,23
W126 / AX1	U_MATRIX_CONTROL_ADDR.1 → U_MATRIX_SWITCHES_MT_A.22	60,4 → 44,22
W127 / AX1	U_MATRIX_SWITCHES_MT_A.22 → U_MATRIX_SWITCHES_MT_B.22	44,22 → 53,22
W128 / AX2	U_MATRIX_CONTROL_ADDR.2 → U_MATRIX_SWITCHES_MT_A.23	60,5 → 44,21
W129 / AX2	U_MATRIX_SWITCHES_MT_A.23 → U_MATRIX_SWITCHES_MT_B.23	44,21 → 53,21
W130 / AY0	U_MATRIX_CONTROL_ADDR.4 → U_MATRIX_SWITCHES_MT_A.24	60,7 → 44,20
W131 / AY0	U_MATRIX_SWITCHES_MT_A.24 → U_MATRIX_SWITCHES_MT_B.24	44,20 → 53,20
W132 / AY1	U_MATRIX_CONTROL_ADDR.5 → U_MATRIX_SWITCHES_MT_A.25	60,8 → 44,19
W133 / AY1	U_MATRIX_SWITCHES_MT_A.25 → U_MATRIX_SWITCHES_MT_B.25	44,19 → 53,19

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



Wire / net	From → to	TOP holes
W134 / DIAG_NODE	R_MATRIX_CONTROL_DIAG_TOP.1 → U_MATRIX_SWITCHES_MT_A.26	60,20 → 44,18
W135 / SOURCE_LO	R_MATRIX_CONTROL_SOURCE_LO.2 → U_MATRIX_SWITCHES_MT_A.27	71,16 → 44,17
W136 / NODE05	J_BOUNDARY_6.1 → U_MATRIX_SWITCHES_MT_A.28	78,6 → 44,16
W137 / NODE04	J_BOUNDARY_5.1 → U_MATRIX_SWITCHES_MT_A.29	1,6 → 44,15
W138 / NODE03	J_BOUNDARY_4.1 → U_MATRIX_SWITCHES_MT_A.30	78,5 → 44,14
W139 / NODE02	J_BOUNDARY_3.1 → U_MATRIX_SWITCHES_MT_A.31	1,5 → 44,13
W140 / NODE01	J_BOUNDARY_2.1 → U_MATRIX_SWITCHES_MT_A.32	78,4 → 44,12
W141 / NODE00	J_BOUNDARY_1.1 → U_MATRIX_SWITCHES_MT_A.33	1,4 → 44,11
W143 / 5V_SYS	R_ROUTER_SAFETY_BUF_OE.1 → C_MATRIX_CONTROL_ADDR.1	17,25 → 46,33
W144 / 5V_SYS	C_MATRIX_CONTROL_ADDR.1 → C_MATRIX_CONTROL_BUFFER.1	46,33 → 74,20

200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

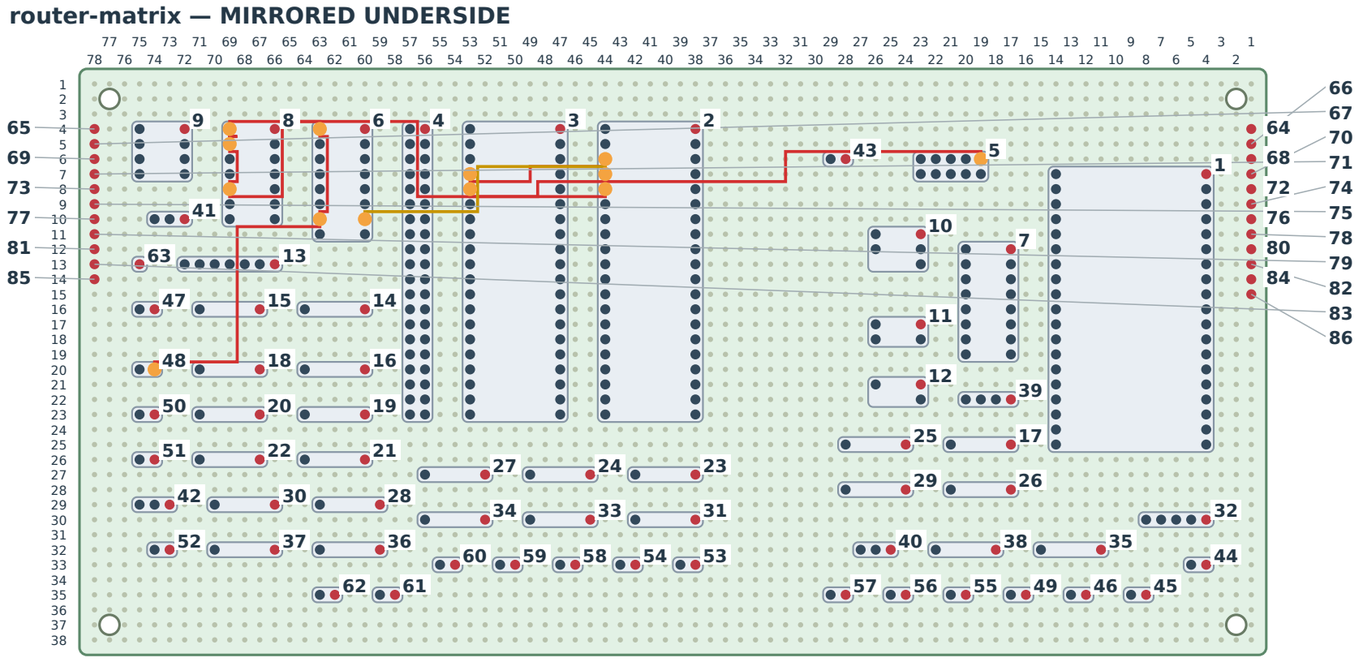
TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

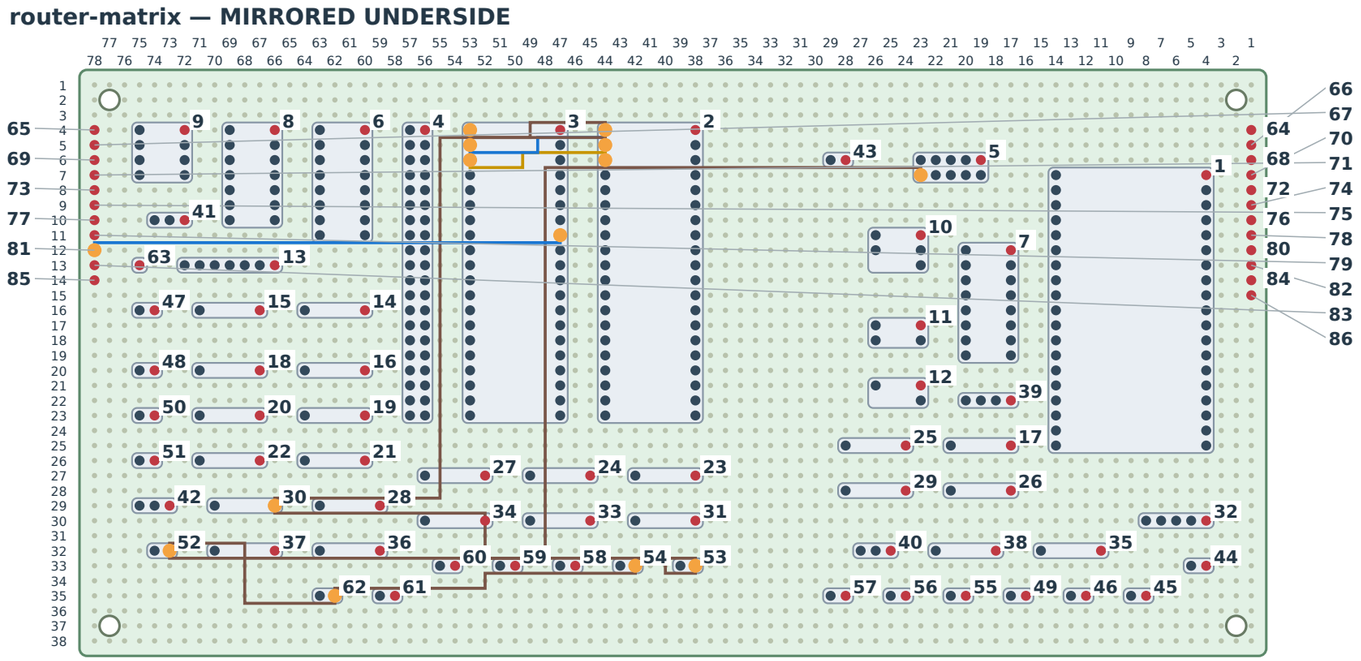
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W145 / 5V_SYS	C_MATRIX_CONTROL_BUFFER.1 → U_MATRIX_CONTROL_ADDR.10	74,20 → 63,10
W146 / 5V_SYS	U_MATRIX_CONTROL_ADDR.10 → U_MATRIX_CONTROL_ADDR.16	63,10 → 63,4
W147 / 5V_SYS	U_MATRIX_CONTROL_ADDR.16 → U_MATRIX_CONTROL_BUFFER.10	63,4 → 69,8
W148 / 5V_SYS	U_MATRIX_CONTROL_BUFFER.10 → U_MATRIX_CONTROL_BUFFER.13	69,8 → 69,5
W149 / 5V_SYS	U_MATRIX_CONTROL_BUFFER.13 → U_MATRIX_CONTROL_BUFFER.14	69,5 → 69,4
W150 / 5V_SYS	U_MATRIX_CONTROL_BUFFER.14 → U_MATRIX_SWITCHES_MT_A.36	69,4 → 44,8
W151 / 5V_SYS	U_MATRIX_SWITCHES_MT_A.36 → U_MATRIX_SWITCHES_MT_B.36	44,8 → 53,8
W154 / 5V_EDU	J_ROUTER_CORE_POWER.1 → U_MATRIX_SWITCHES_MT_A.37	19,6 → 44,7
W155 / 5V_EDU	U_MATRIX_SWITCHES_MT_A.37 → U_MATRIX_SWITCHES_MT_B.37	44,7 → 53,7
W156 / XBAR_DATA	U_MATRIX_CONTROL_ADDR.7 → U_MATRIX_SWITCHES_MT_A.38	60,10 → 44,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

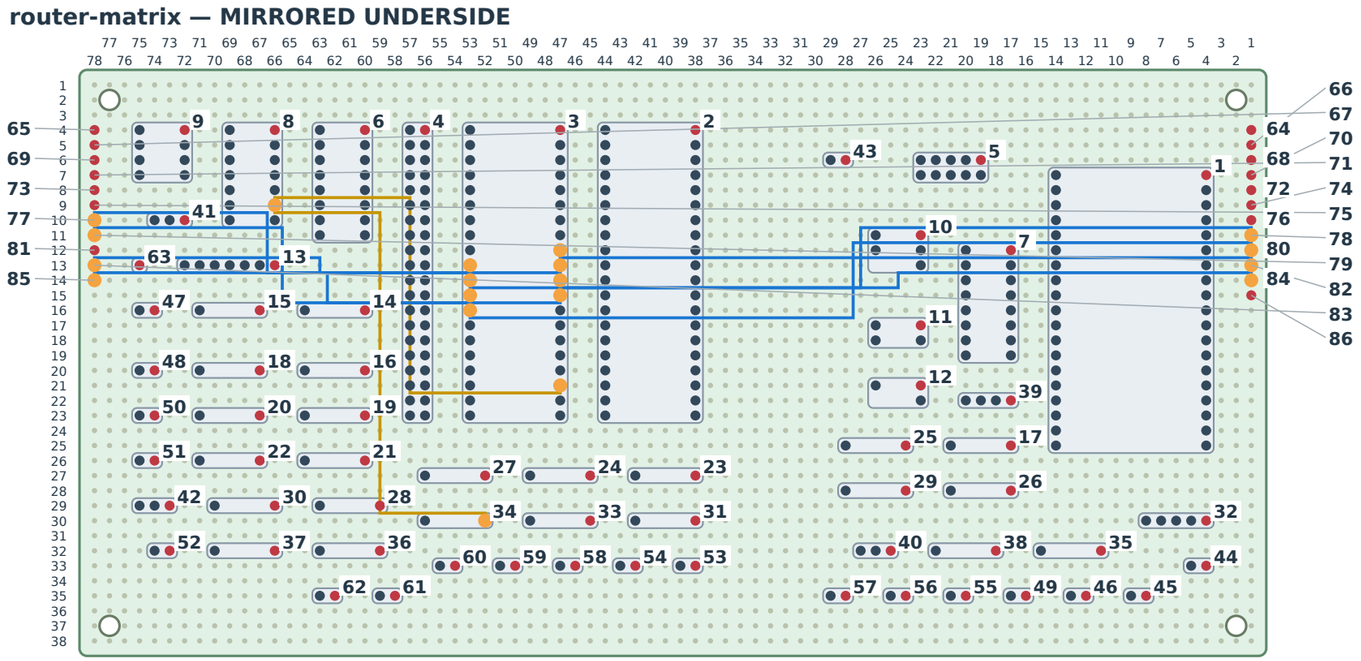
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W157 / XBAR_DATA	U_MATRIX_SWITCHES_MT_A.38 → U_MATRIX_SWITCHES_MT_B.38	44,6 → 53,6
W158 / BUS2	U_MATRIX_SWITCHES_MT_A.39 → U_MATRIX_SWITCHES_MT_B.39	44,5 → 53,5
W159 / 12V_SYS	J_ROUTER_CORE_POWER.10 → C_MATRIX_SWITCHES_MT0_BULK.1	23,7 → 73,32
W160 / 12V_SYS	C_MATRIX_SWITCHES_MT0_BULK.1 → C_MATRIX_SWITCHES_MT0.1	73,32 → 62,35
W161 / 12V_SYS	C_MATRIX_SWITCHES_MT0.1 → C_MATRIX_SWITCHES_MT1_BULK.1	62,35 → 42,33
W162 / 12V_SYS	C_MATRIX_SWITCHES_MT1_BULK.1 → C_MATRIX_SWITCHES_MT1.1	42,33 → 38,33
W163 / 12V_SYS	C_MATRIX_SWITCHES_MT1.1 → R_MATRIX_SWITCHES_RESET_ON.1	38,33 → 66,29
W164 / 12V_SYS	R_MATRIX_SWITCHES_RESET_ON.1 → U_MATRIX_SWITCHES_MT_A.40	66,29 → 44,4
W165 / 12V_SYS	U_MATRIX_SWITCHES_MT_A.40 → U_MATRIX_SWITCHES_MT_B.40	44,4 → 53,4
W169 / NODE17	J_BOUNDARY_18.1 → U_MATRIX_SWITCHES_MT_B.8	78,12 → 47,11

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

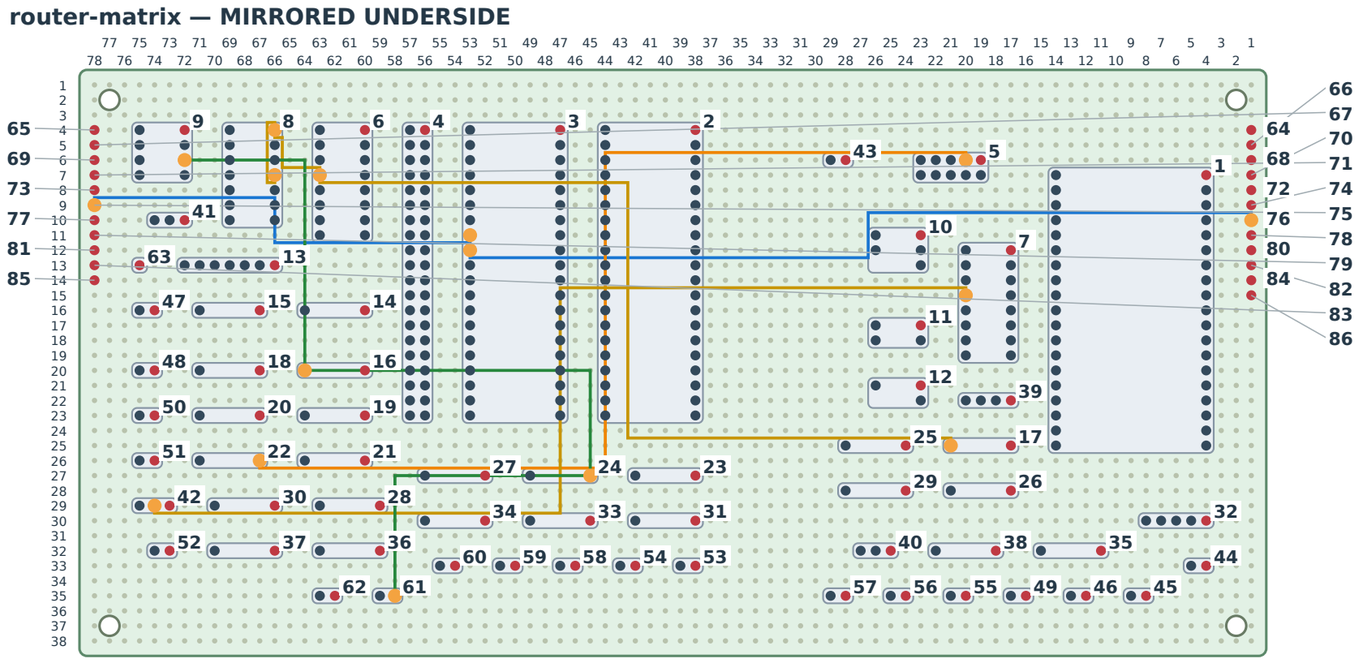
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W170 / NODE18	J_BOUNDARY_19.1 → U_MATRIX_SWITCHES_MT_B.9	1,13 → 47,12
W171 / NODE19	J_BOUNDARY_20.1 → U_MATRIX_SWITCHES_MT_B.10	78,13 → 47,13
W172 / NODE20	J_BOUNDARY_21.1 → U_MATRIX_SWITCHES_MT_B.11	1,14 → 47,14
W173 / NODE21	J_BOUNDARY_22.1 → U_MATRIX_SWITCHES_MT_B.12	78,14 → 47,15
W174 / STROBE_B	R_MATRIX_SWITCHES_STBB_OFF.1 → U_MATRIX_CONTROL_BUFFER.6	52,30 → 66,9
W175 / STROBE_B	U_MATRIX_CONTROL_BUFFER.6 → U_MATRIX_SWITCHES_MT_B.18	66,9 → 47,21
W176 / NODE16	J_BOUNDARY_17.1 → U_MATRIX_SWITCHES_MT_B.28	1,12 → 53,16
W177 / NODE15	J_BOUNDARY_16.1 → U_MATRIX_SWITCHES_MT_B.29	78,11 → 53,15
W178 / NODE14	J_BOUNDARY_15.1 → U_MATRIX_SWITCHES_MT_B.30	1,11 → 53,14
W179 / NODE13	J_BOUNDARY_14.1 → U_MATRIX_SWITCHES_MT_B.31	78,10 → 53,13

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

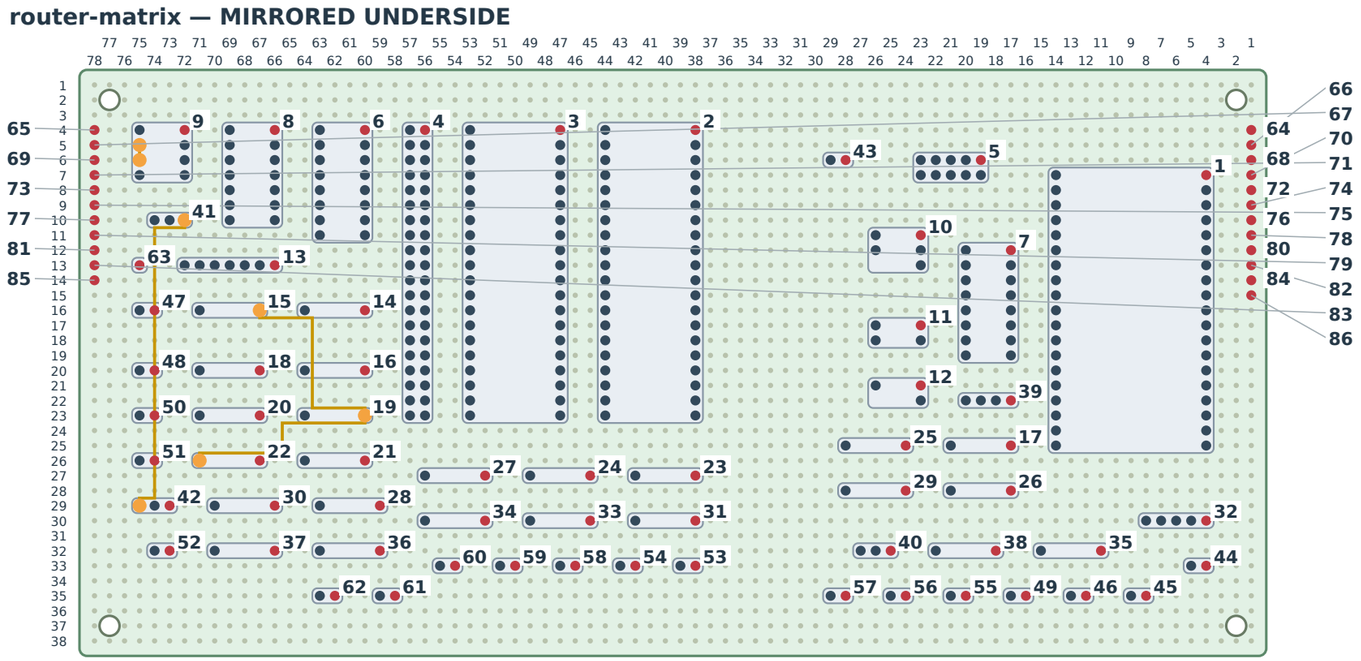
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W180 / NODE12	J_BOUNDARY_13.1 → U_MATRIX_SWITCHES_MT_B.32	1,10 → 53,12
W181 / NODE11	J_BOUNDARY_12.1 → U_MATRIX_SWITCHES_MT_B.33	78,9 → 53,11
W196 / 3V3_SYS	J_ROUTER_CORE_POWER.3 → D_MATRIX_CONTROL_SOURCE.1	20,6 → 67,26
W202 / ROUTER_ALIVE	U_ROUTER_SAFETY_WATCH.13 → Q_MATRIX_SWITCHES_RESET_ALIVE.2	20,15 → 74,29
W204 / XBAR_OE_N	R_ROUTER_SAFETY_BUF_OE.2 → U_MATRIX_CONTROL_ADDR.13	21,25 → 63,7
W205 / XBAR_OE_N	U_MATRIX_CONTROL_ADDR.13 → U_MATRIX_CONTROL_BUFFER.1	63,7 → 66,4
W206 / XBAR_OE_N	U_MATRIX_CONTROL_BUFFER.1 → U_MATRIX_CONTROL_BUFFER.4	66,4 → 66,7
W212 / DIAG_DIV	C_MATRIX_CONTROL_DIAG.1 → R_MATRIX_CONTROL_DIAG_BOTTOM.1	58,35 → 45,27
W213 / DIAG_DIV	R_MATRIX_CONTROL_DIAG_BOTTOM.1 → R_MATRIX_CONTROL_DIAG_TOP.2	45,27 → 64,20
W214 / DIAG_DIV	R_MATRIX_CONTROL_DIAG_TOP.2 → U_MATRIX_CONTROL_DIAG.3	64,20 → 72,6

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP24 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

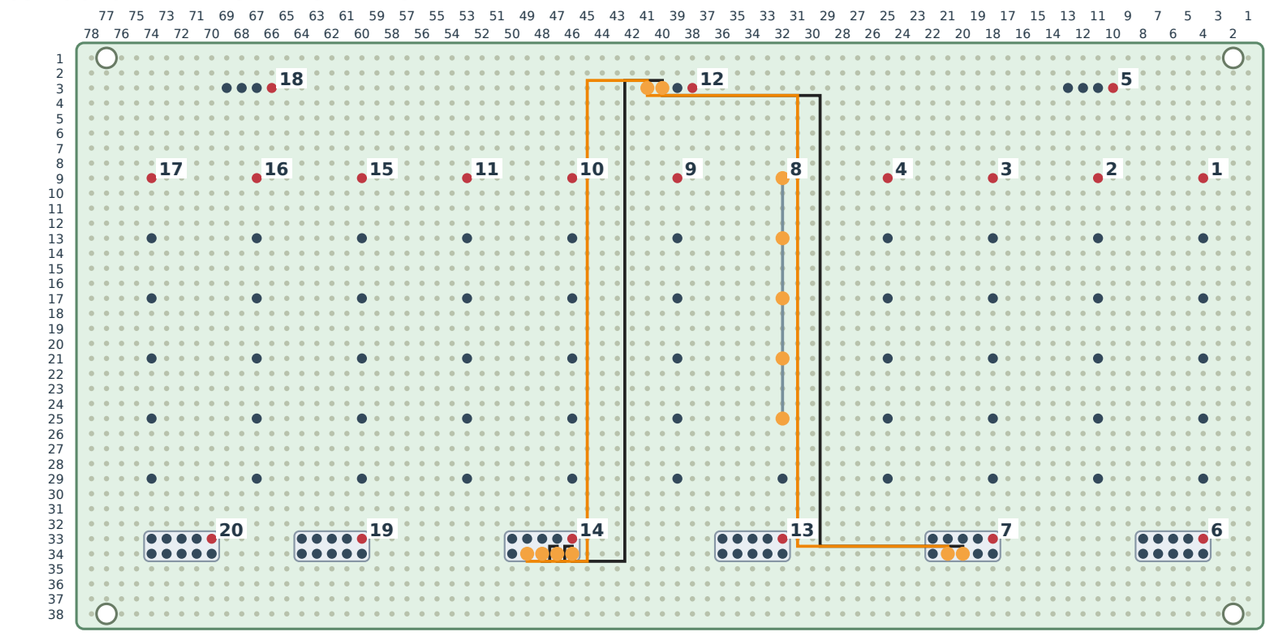
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W215 / DIAG_UNUSED	U_MATRIX_CONTROL_DIAG.6 → U_MATRIX_CONTROL_DIAG.7	75,6 → 75,5
W217 / SOURCE_FEED	D_MATRIX_CONTROL_SOURCE.2 → R_MATRIX_CONTROL_SOURCE_HI.1	71,26 → 60,23
W218 / SOURCE_FEED	R_MATRIX_CONTROL_SOURCE_HI.1 → R_MATRIX_CONTROL_SOURCE_LO.1	60,23 → 67,16
W219 / RESET_MID	Q_MATRIX_SWITCHES_RESET_ALIVE.3 → Q_MATRIX_SWITCHES_RESET_RELEASE.1	75,29 → 72,10

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP27 — contact-a

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

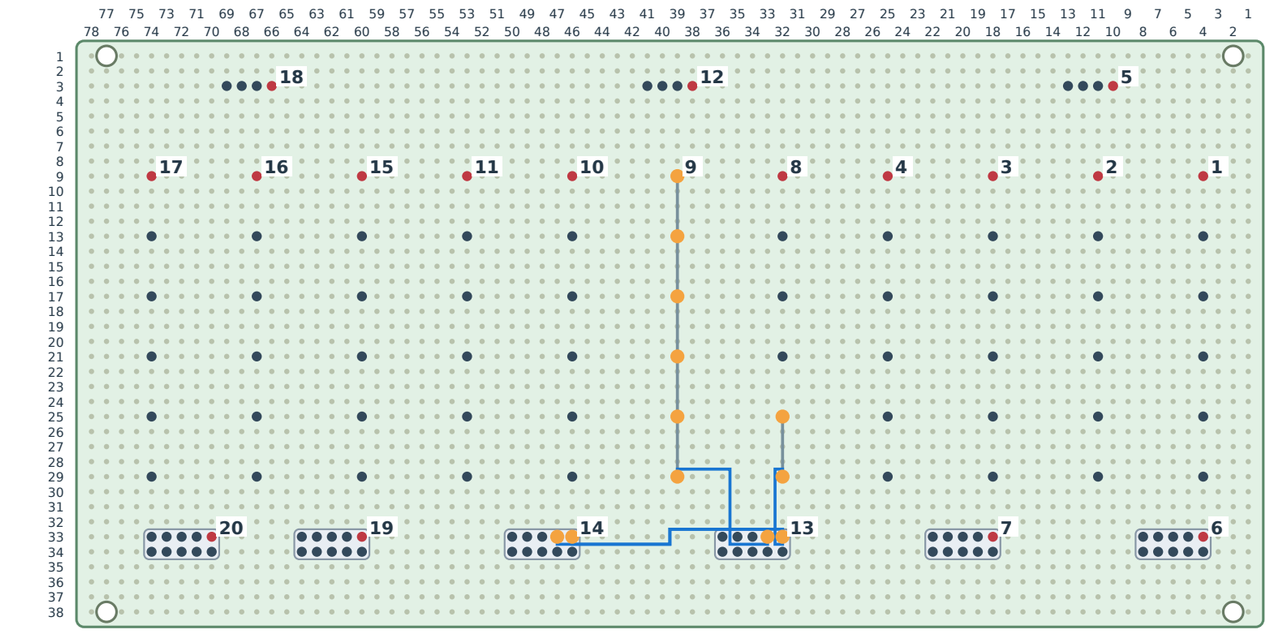
Wire / net	From → to	TOP holes
W33 / GND	J_A1_SMART.6 → J_A2_SERVICE.3	20,34 → 40,3
W34 / GND	J_A2_SERVICE.3 → J_A2_SMART.2	40,3 → 46,34
W35 / GND	J_A2_SMART.2 → J_A2_SMART.4	46,34 → 47,34
W36 / GND	J_A2_SMART.4 → J_A2_SMART.6	47,34 → 48,34
W42 / 3V3_SYS	J_A1_SMART.8 → J_A2_SERVICE.4	21,34 → 41,3
W43 / 3V3_SYS	J_A2_SERVICE.4 → J_A2_SMART.8	41,3 → 49,34
W46 / NODE04	J_A2_NODE04.1 → J_A2_NODE04.2	32,9 → 32,13
W47 / NODE04	J_A2_NODE04.2 → J_A2_NODE04.3	32,13 → 32,17
W48 / NODE04	J_A2_NODE04.3 → J_A2_NODE04.4	32,17 → 32,21
W49 / NODE04	J_A2_NODE04.4 → J_A2_NODE04.5	32,21 → 32,25

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP27 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

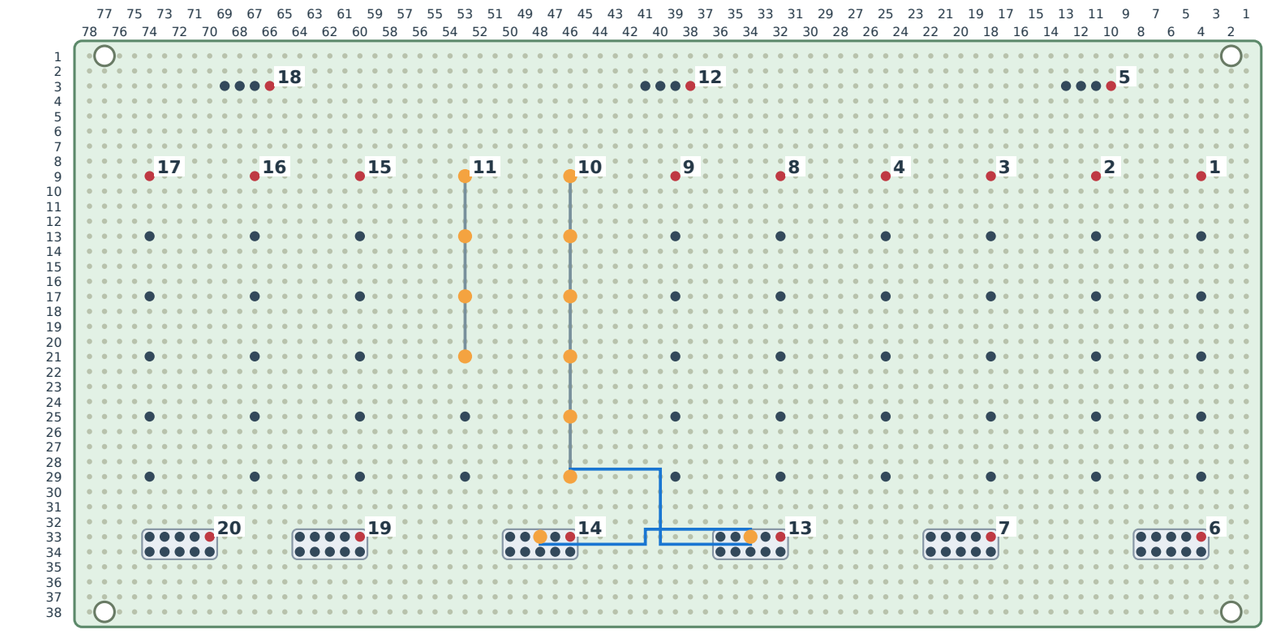
Wire / net	From → to	TOP holes
W50 / NODE04	J_A2_NODE04.5 → J_A2_NODE04.6	32,25 → 32,29
W51 / NODE04	J_A2_NODE04.6 → J_A2_NODE.1	32,29 → 32,33
W52 / NODE04	J_A2_NODE.1 → J_A2_SMART.1	32,33 → 46,33
W53 / NODE05	J_A2_NODE05.1 → J_A2_NODE05.2	39,9 → 39,13
W54 / NODE05	J_A2_NODE05.2 → J_A2_NODE05.3	39,13 → 39,17
W55 / NODE05	J_A2_NODE05.3 → J_A2_NODE05.4	39,17 → 39,21
W56 / NODE05	J_A2_NODE05.4 → J_A2_NODE05.5	39,21 → 39,25
W57 / NODE05	J_A2_NODE05.5 → J_A2_NODE05.6	39,25 → 39,29
W58 / NODE05	J_A2_NODE05.6 → J_A2_NODE.3	39,29 → 33,33
W59 / NODE05	J_A2_NODE.3 → J_A2_SMART.3	33,33 → 47,33

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP27 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

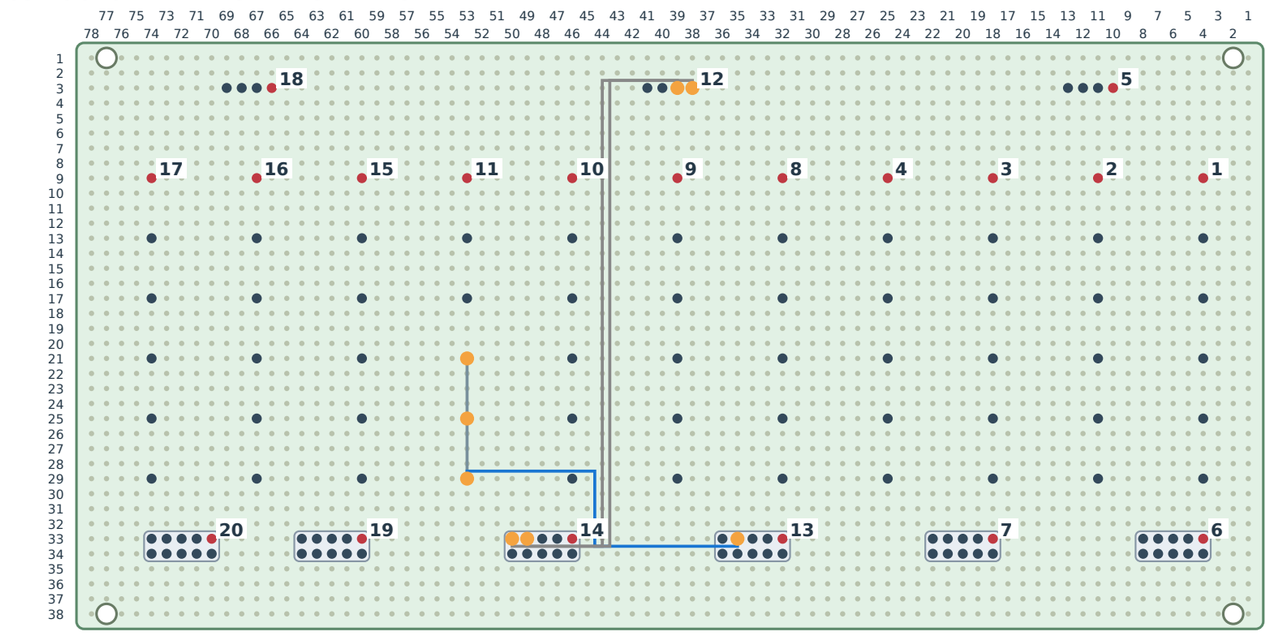
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W60 / NODE06	J_A2_NODE06.1 → J_A2_NODE06.2	46,9 → 46,13
W61 / NODE06	J_A2_NODE06.2 → J_A2_NODE06.3	46,13 → 46,17
W62 / NODE06	J_A2_NODE06.3 → J_A2_NODE06.4	46,17 → 46,21
W63 / NODE06	J_A2_NODE06.4 → J_A2_NODE06.5	46,21 → 46,25
W64 / NODE06	J_A2_NODE06.5 → J_A2_NODE06.6	46,25 → 46,29
W65 / NODE06	J_A2_NODE06.6 → J_A2_NODE.5	46,29 → 34,33
W66 / NODE06	J_A2_NODE.5 → J_A2_SMART.5	34,33 → 48,33
W67 / NODE07	J_A2_NODE07.1 → J_A2_NODE07.2	53,9 → 53,13
W68 / NODE07	J_A2_NODE07.2 → J_A2_NODE07.3	53,13 → 53,17
W69 / NODE07	J_A2_NODE07.3 → J_A2_NODE07.4	53,17 → 53,21

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP27 — contact-a

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

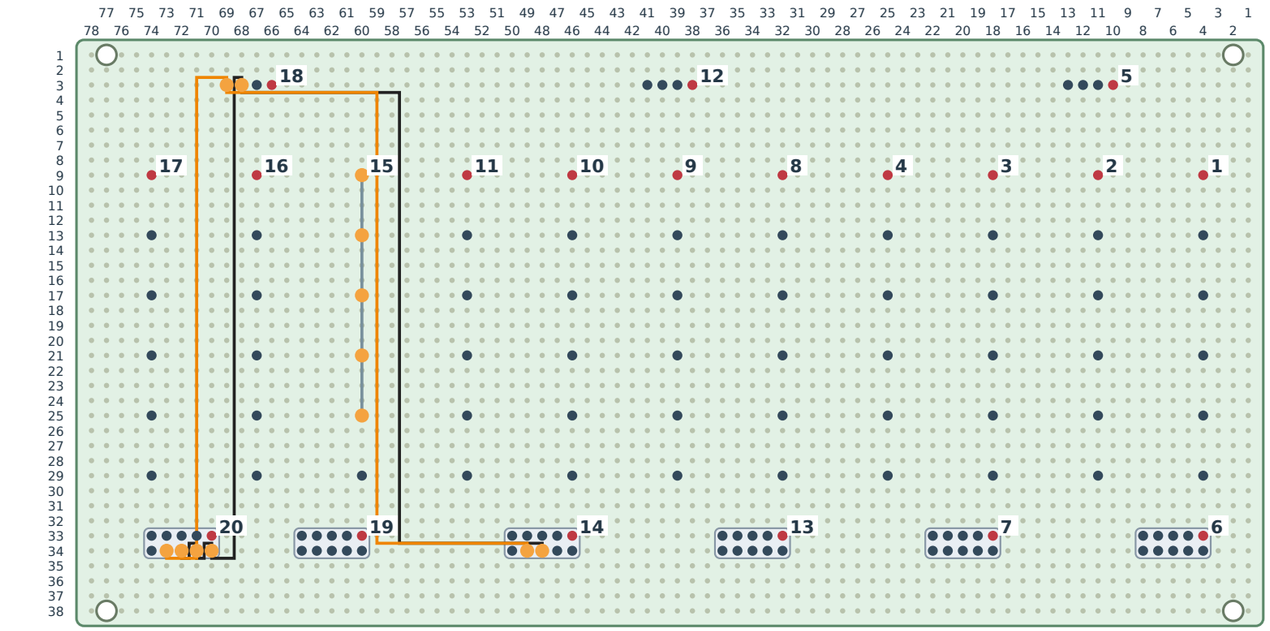
Wire / net	From → to	TOP holes
W70 / NODE07	J_A2_NODE07.4 → J_A2_NODE07.5	53,21 → 53,25
W71 / NODE07	J_A2_NODE07.5 → J_A2_NODE07.6	53,25 → 53,29
W72 / NODE07	J_A2_NODE07.6 → J_A2_NODE.7	53,29 → 35,33
W73 / SCL_POSITION_1	J_A2_SERVICE.1 → J_A2_SMART.9	38,3 → 50,33
W74 / SDA_POSITION_1	J_A2_SERVICE.2 → J_A2_SMART.7	39,3 → 49,33

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP28 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

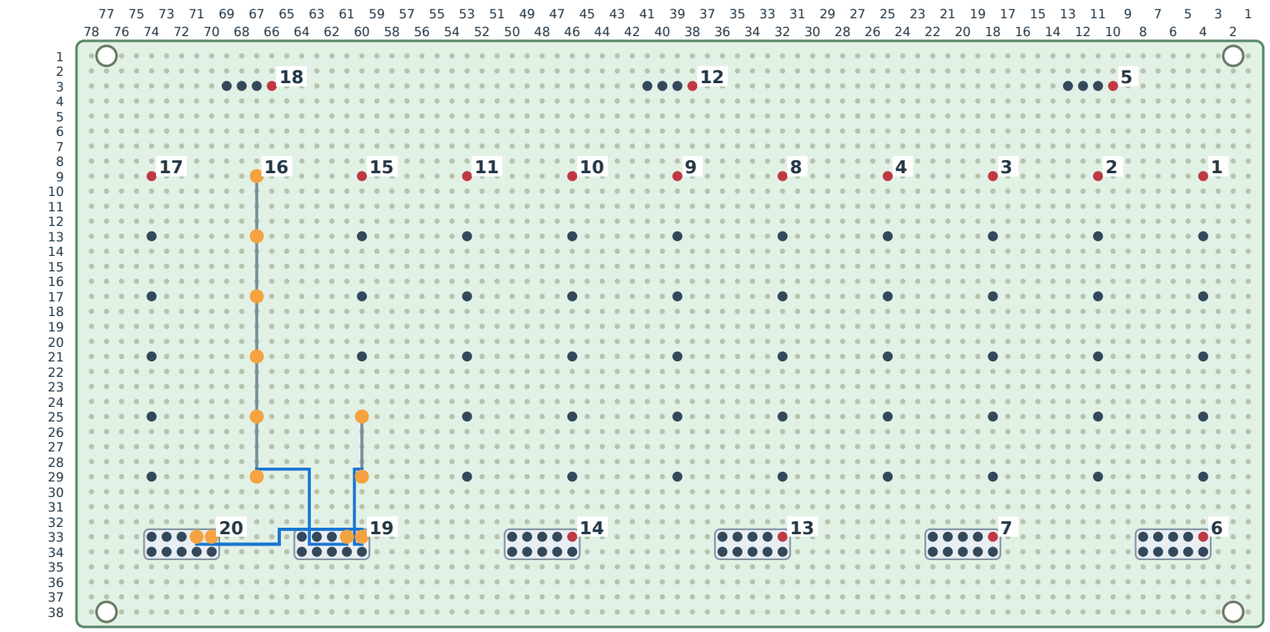
Wire / net	From → to	TOP holes
W37 / GND	J_A2_SMART.6 → J_A3_SERVICE.3	48,34 → 68,3
W38 / GND	J_A3_SERVICE.3 → J_A3_SMART.2	68,3 → 70,34
W39 / GND	J_A3_SMART.2 → J_A3_SMART.4	70,34 → 71,34
W40 / GND	J_A3_SMART.4 → J_A3_SMART.6	71,34 → 72,34
W44 / 3V3_SYS	J_A2_SMART.8 → J_A3_SERVICE.4	49,34 → 69,3
W45 / 3V3_SYS	J_A3_SERVICE.4 → J_A3_SMART.8	69,3 → 73,34
W75 / NODE08	J_A3_NODE08.1 → J_A3_NODE08.2	60,9 → 60,13
W76 / NODE08	J_A3_NODE08.2 → J_A3_NODE08.3	60,13 → 60,17
W77 / NODE08	J_A3_NODE08.3 → J_A3_NODE08.4	60,17 → 60,21
W78 / NODE08	J_A3_NODE08.4 → J_A3_NODE08.5	60,21 → 60,25

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP28 — contact-a

At most ten added wires per view. Inspect each net with power isolated.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

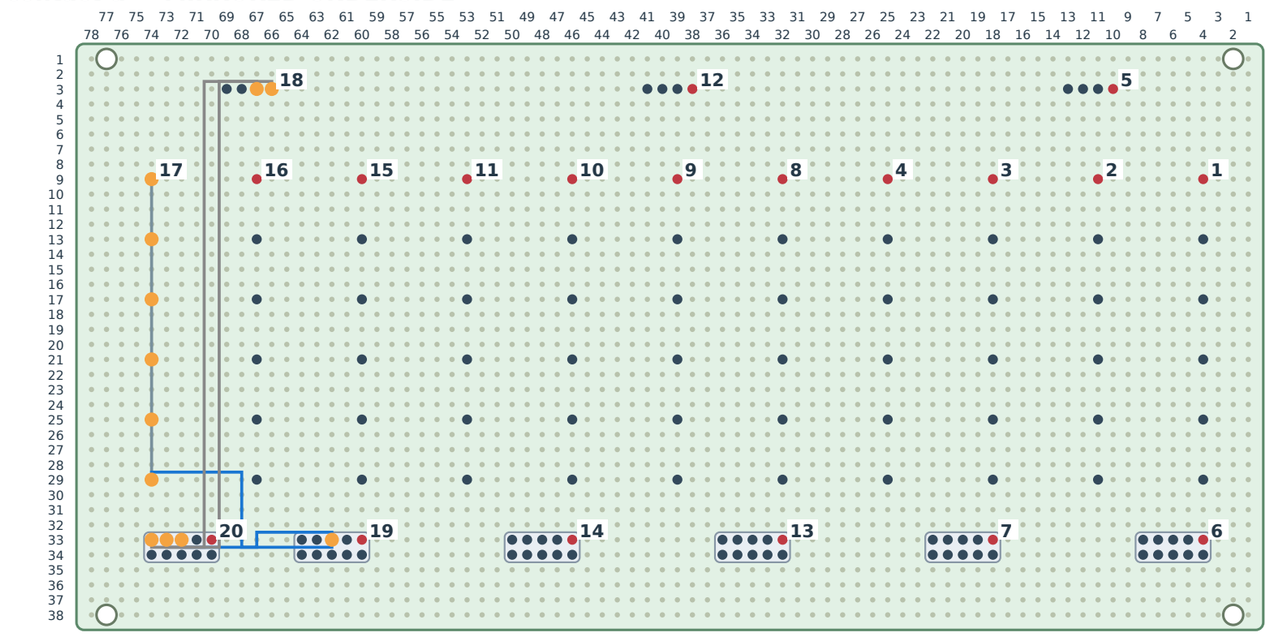
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W79 / NODE08	J_A3_NODE08.5 → J_A3_NODE08.6	60,25 → 60,29
W80 / NODE08	J_A3_NODE08.6 → J_A3_NODE.1	60,29 → 60,33
W81 / NODE08	J_A3_NODE.1 → J_A3_SMART.1	60,33 → 70,33
W82 / NODE09	J_A3_NODE09.1 → J_A3_NODE09.2	67,9 → 67,13
W83 / NODE09	J_A3_NODE09.2 → J_A3_NODE09.3	67,13 → 67,17
W84 / NODE09	J_A3_NODE09.3 → J_A3_NODE09.4	67,17 → 67,21
W85 / NODE09	J_A3_NODE09.4 → J_A3_NODE09.5	67,21 → 67,25
W86 / NODE09	J_A3_NODE09.5 → J_A3_NODE09.6	67,25 → 67,29
W87 / NODE09	J_A3_NODE09.6 → J_A3_NODE.3	67,29 → 61,33
W88 / NODE09	J_A3_NODE.3 → J_A3_SMART.3	61,33 → 71,33

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP28 — contact-a

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

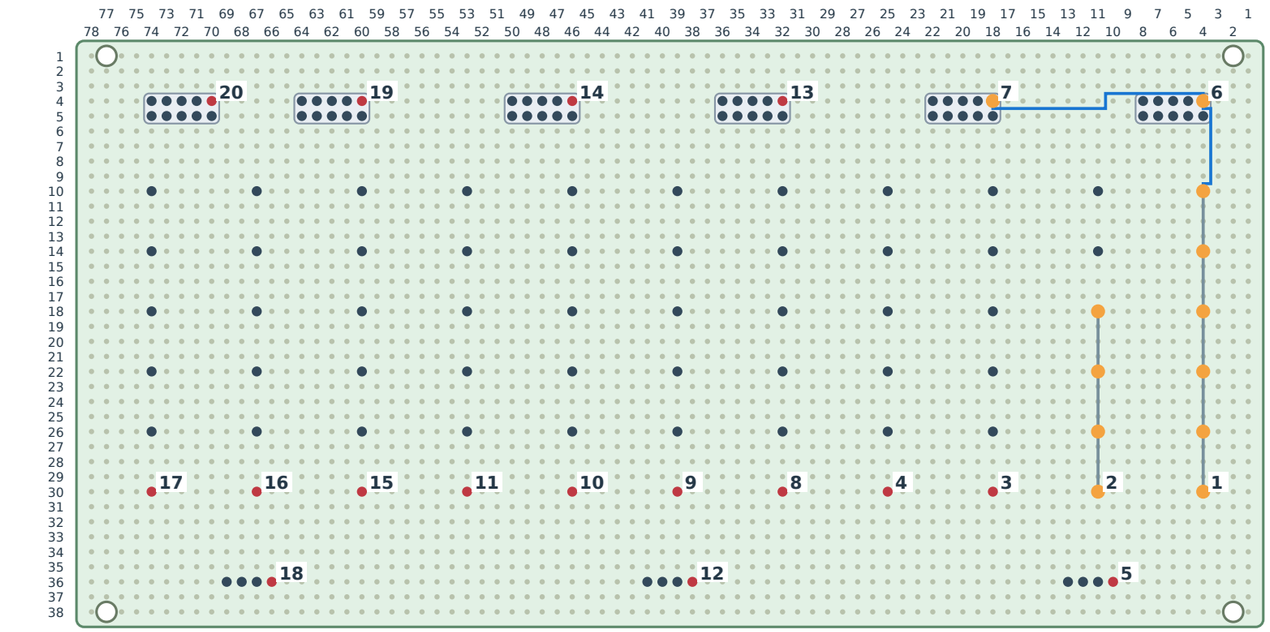
Wire / net	From → to	TOP holes
W89 / NODE10	J_A3_NODE10.1 → J_A3_NODE10.2	74,9 → 74,13
W90 / NODE10	J_A3_NODE10.2 → J_A3_NODE10.3	74,13 → 74,17
W91 / NODE10	J_A3_NODE10.3 → J_A3_NODE10.4	74,17 → 74,21
W92 / NODE10	J_A3_NODE10.4 → J_A3_NODE10.5	74,21 → 74,25
W93 / NODE10	J_A3_NODE10.5 → J_A3_NODE10.6	74,25 → 74,29
W94 / NODE10	J_A3_NODE10.6 → J_A3_NODE.5	74,29 → 62,33
W95 / NODE10	J_A3_NODE.5 → J_A3_SMART.5	62,33 → 72,33
W96 / SCL_POSITION_2	J_A3_SERVICE.1 → J_A3_SMART.9	66,3 → 74,33
W97 / SDA_POSITION_2	J_A3_SERVICE.2 → J_A3_SMART.7	67,3 → 73,33

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP29 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

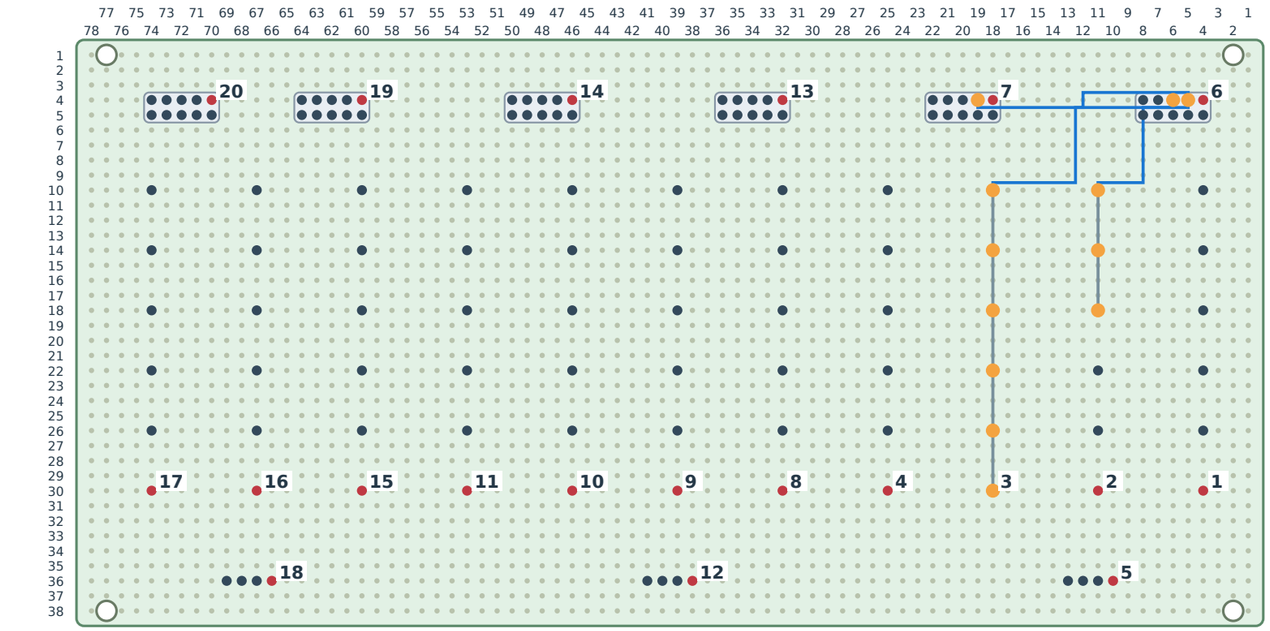
Wire / net	From → to	TOP holes
W1 / NODE11	J_B1_NODE11.1 → J_B1_NODE11.2	4,30 → 4,26
W2 / NODE11	J_B1_NODE11.2 → J_B1_NODE11.3	4,26 → 4,22
W3 / NODE11	J_B1_NODE11.3 → J_B1_NODE11.4	4,22 → 4,18
W4 / NODE11	J_B1_NODE11.4 → J_B1_NODE11.5	4,18 → 4,14
W5 / NODE11	J_B1_NODE11.5 → J_B1_NODE11.6	4,14 → 4,10
W6 / NODE11	J_B1_NODE11.6 → J_B1_NODE.1	4,10 → 4,4
W7 / NODE11	J_B1_NODE.1 → J_B1_SMART.1	4,4 → 18,4
W8 / NODE12	J_B1_NODE12.1 → J_B1_NODE12.2	11,30 → 11,26
W9 / NODE12	J_B1_NODE12.2 → J_B1_NODE12.3	11,26 → 11,22
W10 / NODE12	J_B1_NODE12.3 → J_B1_NODE12.4	11,22 → 11,18

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP29 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

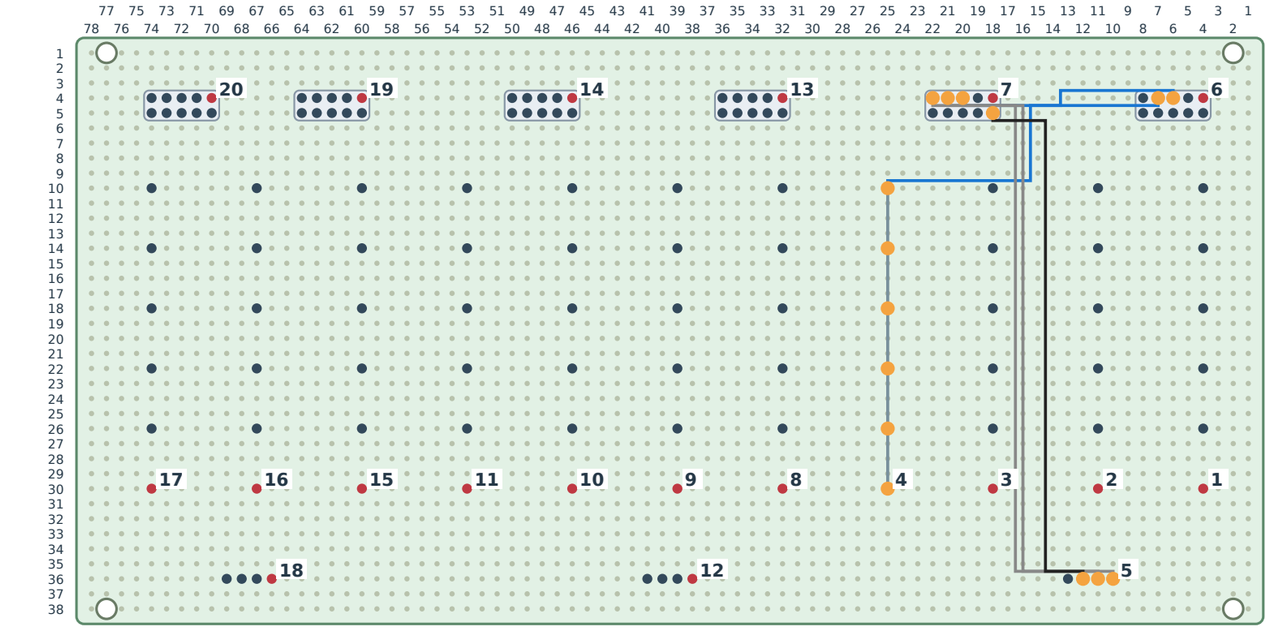
Wire / net	From → to	TOP holes
W11 / NODE12	J_B1_NODE12.4 → J_B1_NODE12.5	11,18 → 11,14
W12 / NODE12	J_B1_NODE12.5 → J_B1_NODE12.6	11,14 → 11,10
W13 / NODE12	J_B1_NODE12.6 → J_B1_NODE.3	11,10 → 5,4
W14 / NODE12	J_B1_NODE.3 → J_B1_SMART.3	5,4 → 19,4
W15 / NODE13	J_B1_NODE13.1 → J_B1_NODE13.2	18,30 → 18,26
W16 / NODE13	J_B1_NODE13.2 → J_B1_NODE13.3	18,26 → 18,22
W17 / NODE13	J_B1_NODE13.3 → J_B1_NODE13.4	18,22 → 18,18
W18 / NODE13	J_B1_NODE13.4 → J_B1_NODE13.5	18,18 → 18,14
W19 / NODE13	J_B1_NODE13.5 → J_B1_NODE13.6	18,14 → 18,10
W20 / NODE13	J_B1_NODE13.6 → J_B1_NODE.5	18,10 → 6,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP29 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W21 / NODE13	J_B1_NODE.5 → J_B1_SMART.5	6,4 → 20,4
W22 / NODE14	J_B1_NODE14.1 → J_B1_NODE14.2	25,30 → 25,26
W23 / NODE14	J_B1_NODE14.2 → J_B1_NODE14.3	25,26 → 25,22
W24 / NODE14	J_B1_NODE14.3 → J_B1_NODE14.4	25,22 → 25,18
W25 / NODE14	J_B1_NODE14.4 → J_B1_NODE14.5	25,18 → 25,14
W26 / NODE14	J_B1_NODE14.5 → J_B1_NODE14.6	25,14 → 25,10
W27 / NODE14	J_B1_NODE14.6 → J_B1_NODE.7	25,10 → 7,4
W28 / SCL_POSITION_3	J_B1_SERVICE.1 → J_B1_SMART.9	10,36 → 22,4
W29 / SDA_POSITION_3	J_B1_SERVICE.2 → J_B1_SMART.7	11,36 → 21,4
W30 / GND	J_B1_SERVICE.3 → J_B1_SMART.2	12,36 → 18,5

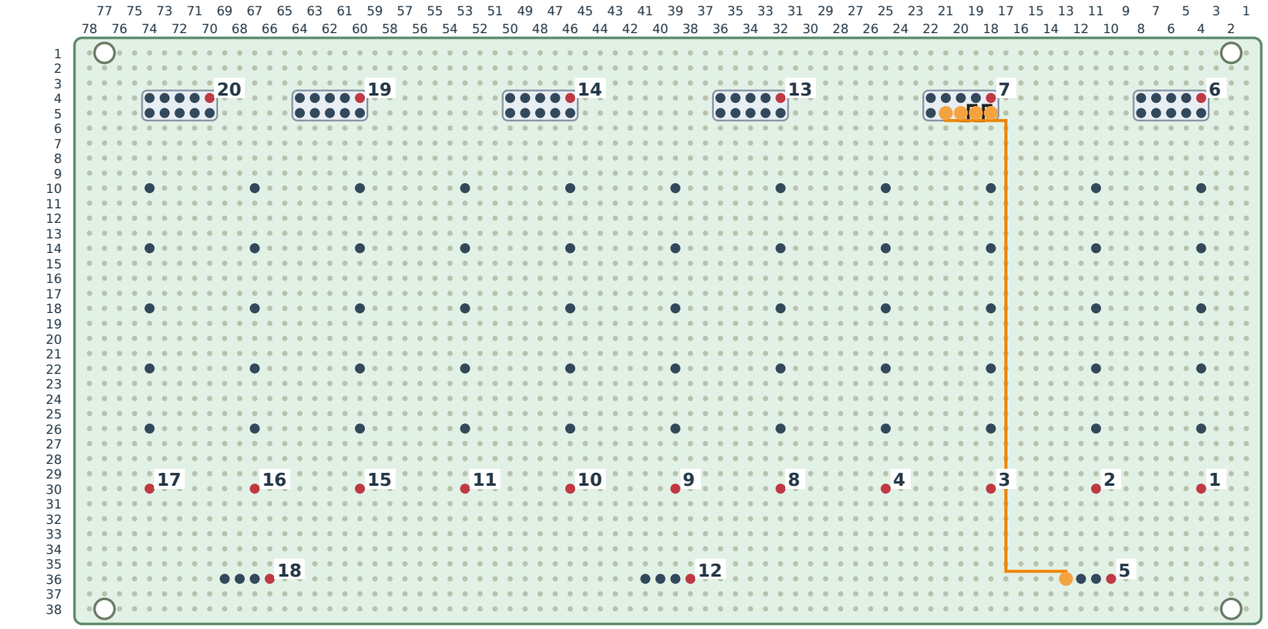
Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP29 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

Wire / net	From → to	TOP holes
W31 / GND	J_B1_SMART.2 → J_B1_SMART.4	18,5 → 19,5
W32 / GND	J_B1_SMART.4 → J_B1_SMART.6	19,5 → 20,5
W41 / 3V3_SYS	J_B1_SERVICE.4 → J_B1_SMART.8	13,36 → 21,5

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

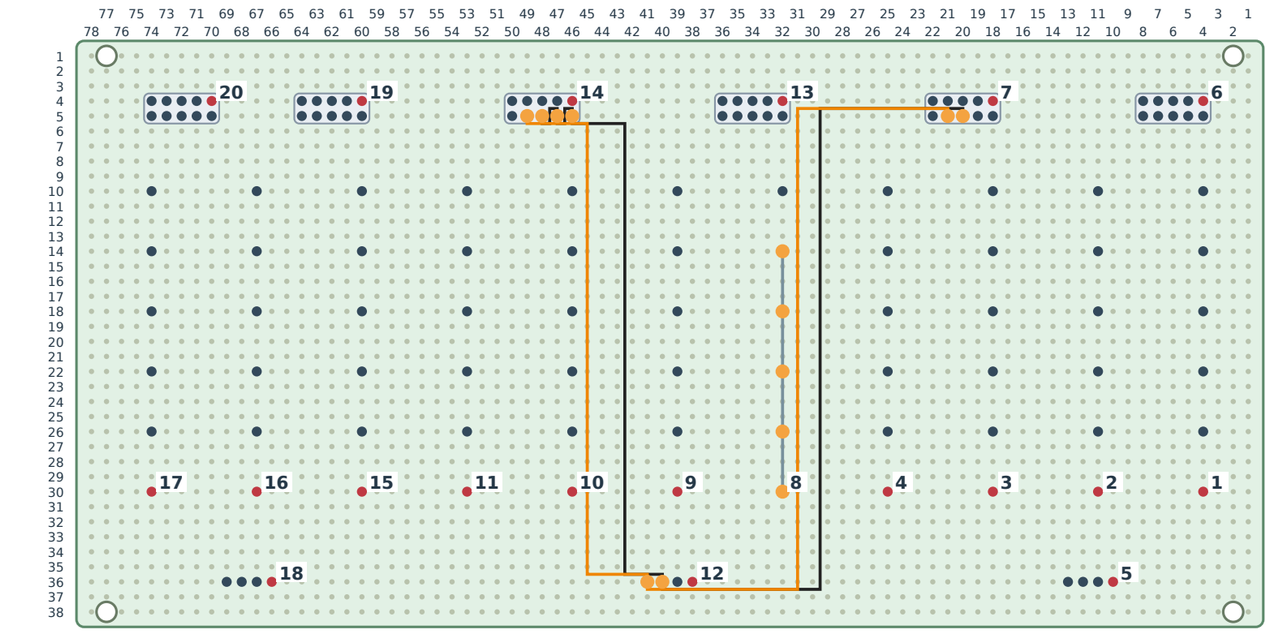
PROTOTYPE: physical fit and hardware tests pending.

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP30 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

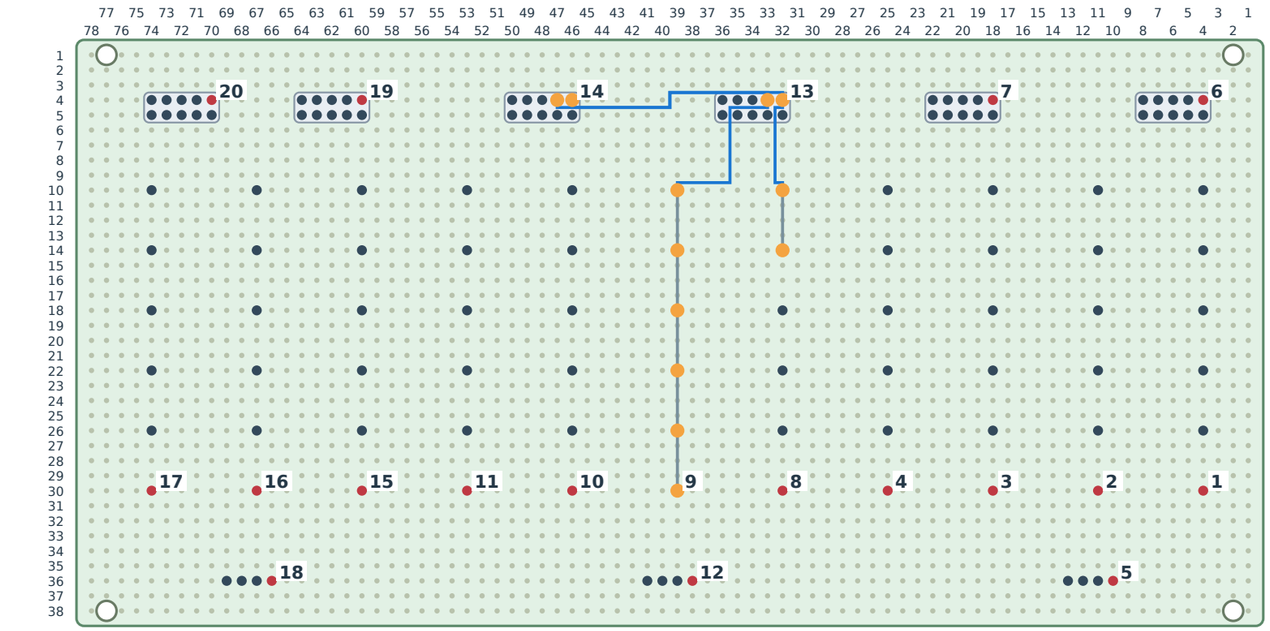
Wire / net	From → to	TOP holes
W33 / GND	J_B1_SMART.6 → J_B2_SERVICE.3	20,5 → 40,36
W34 / GND	J_B2_SERVICE.3 → J_B2_SMART.2	40,36 → 46,5
W35 / GND	J_B2_SMART.2 → J_B2_SMART.4	46,5 → 47,5
W36 / GND	J_B2_SMART.4 → J_B2_SMART.6	47,5 → 48,5
W42 / 3V3_SYS	J_B1_SMART.8 → J_B2_SERVICE.4	21,5 → 41,36
W43 / 3V3_SYS	J_B2_SERVICE.4 → J_B2_SMART.8	41,36 → 49,5
W46 / NODE15	J_B2_NODE15.1 → J_B2_NODE15.2	32,30 → 32,26
W47 / NODE15	J_B2_NODE15.2 → J_B2_NODE15.3	32,26 → 32,22
W48 / NODE15	J_B2_NODE15.3 → J_B2_NODE15.4	32,22 → 32,18
W49 / NODE15	J_B2_NODE15.4 → J_B2_NODE15.5	32,18 → 32,14

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP30 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

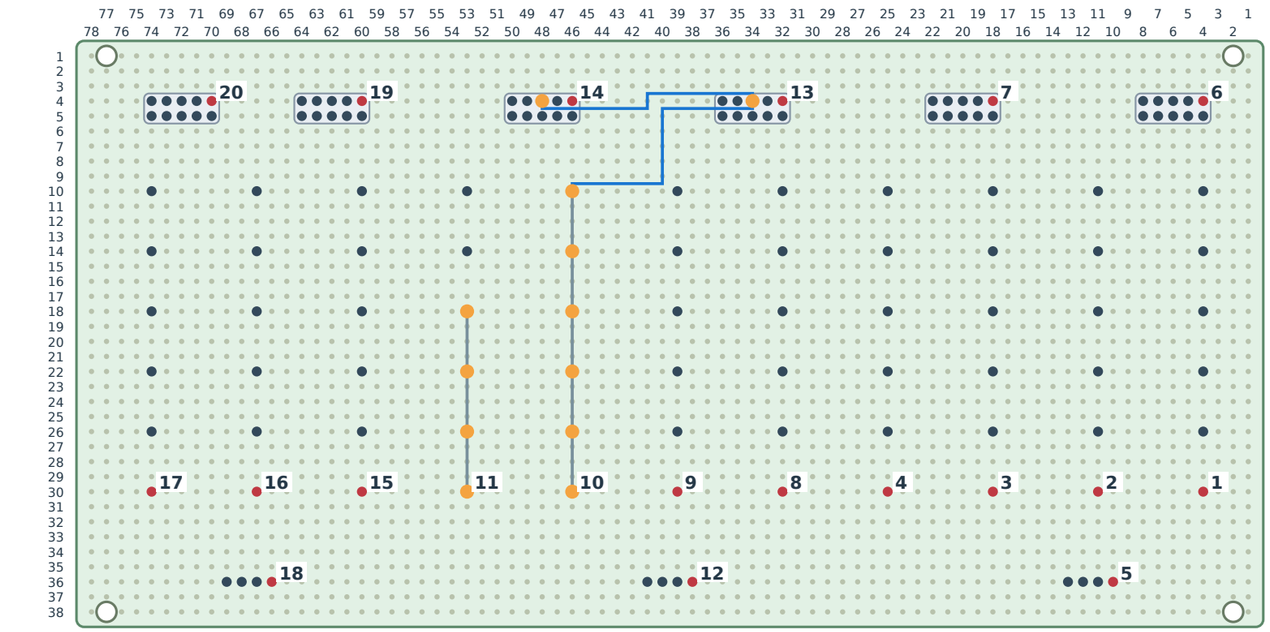
Wire / net	From → to	TOP holes
W50 / NODE15	J_B2_NODE15.5 → J_B2_NODE15.6	32,14 → 32,10
W51 / NODE15	J_B2_NODE15.6 → J_B2_NODE.1	32,10 → 32,4
W52 / NODE15	J_B2_NODE.1 → J_B2_SMART.1	32,4 → 46,4
W53 / NODE16	J_B2_NODE16.1 → J_B2_NODE16.2	39,30 → 39,26
W54 / NODE16	J_B2_NODE16.2 → J_B2_NODE16.3	39,26 → 39,22
W55 / NODE16	J_B2_NODE16.3 → J_B2_NODE16.4	39,22 → 39,18
W56 / NODE16	J_B2_NODE16.4 → J_B2_NODE16.5	39,18 → 39,14
W57 / NODE16	J_B2_NODE16.5 → J_B2_NODE16.6	39,14 → 39,10
W58 / NODE16	J_B2_NODE16.6 → J_B2_NODE.3	39,10 → 33,4
W59 / NODE16	J_B2_NODE.3 → J_B2_SMART.3	33,4 → 47,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP30 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

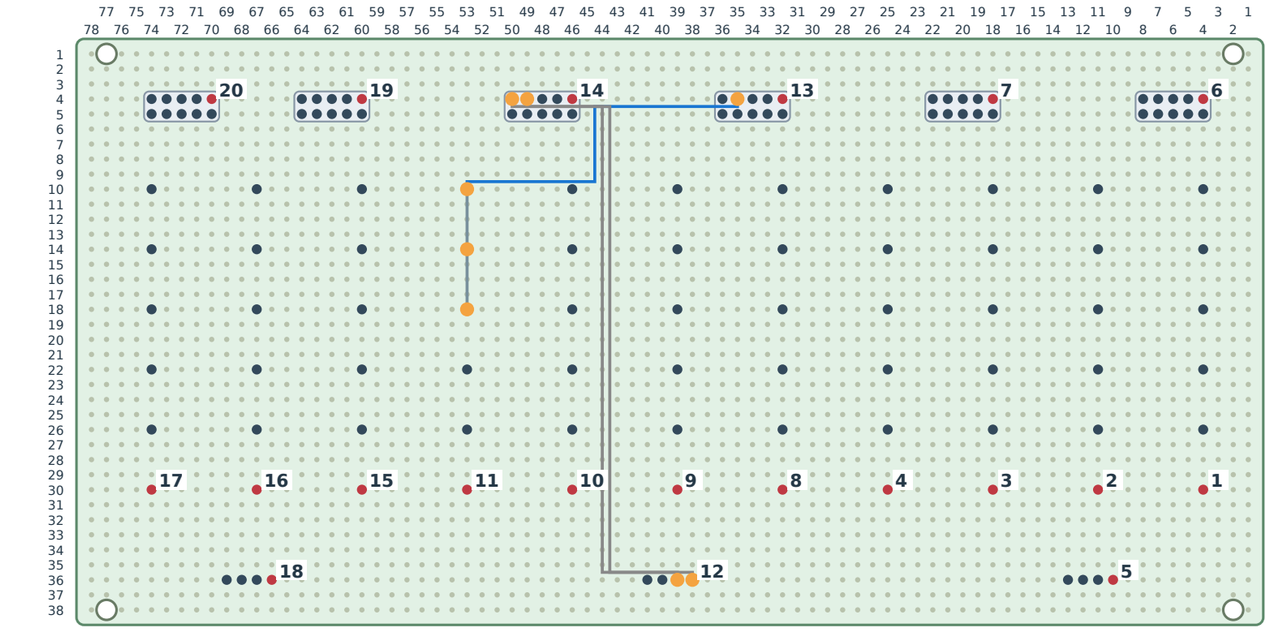
Wire / net	From → to	TOP holes
W60 / NODE17	J_B2_NODE17.1 → J_B2_NODE17.2	46,30 → 46,26
W61 / NODE17	J_B2_NODE17.2 → J_B2_NODE17.3	46,26 → 46,22
W62 / NODE17	J_B2_NODE17.3 → J_B2_NODE17.4	46,22 → 46,18
W63 / NODE17	J_B2_NODE17.4 → J_B2_NODE17.5	46,18 → 46,14
W64 / NODE17	J_B2_NODE17.5 → J_B2_NODE17.6	46,14 → 46,10
W65 / NODE17	J_B2_NODE17.6 → J_B2_NODE.5	46,10 → 34,4
W66 / NODE17	J_B2_NODE.5 → J_B2_SMART.5	34,4 → 48,4
W67 / NODE18	J_B2_NODE18.1 → J_B2_NODE18.2	53,30 → 53,26
W68 / NODE18	J_B2_NODE18.2 → J_B2_NODE18.3	53,26 → 53,22
W69 / NODE18	J_B2_NODE18.3 → J_B2_NODE18.4	53,22 → 53,18

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP30 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

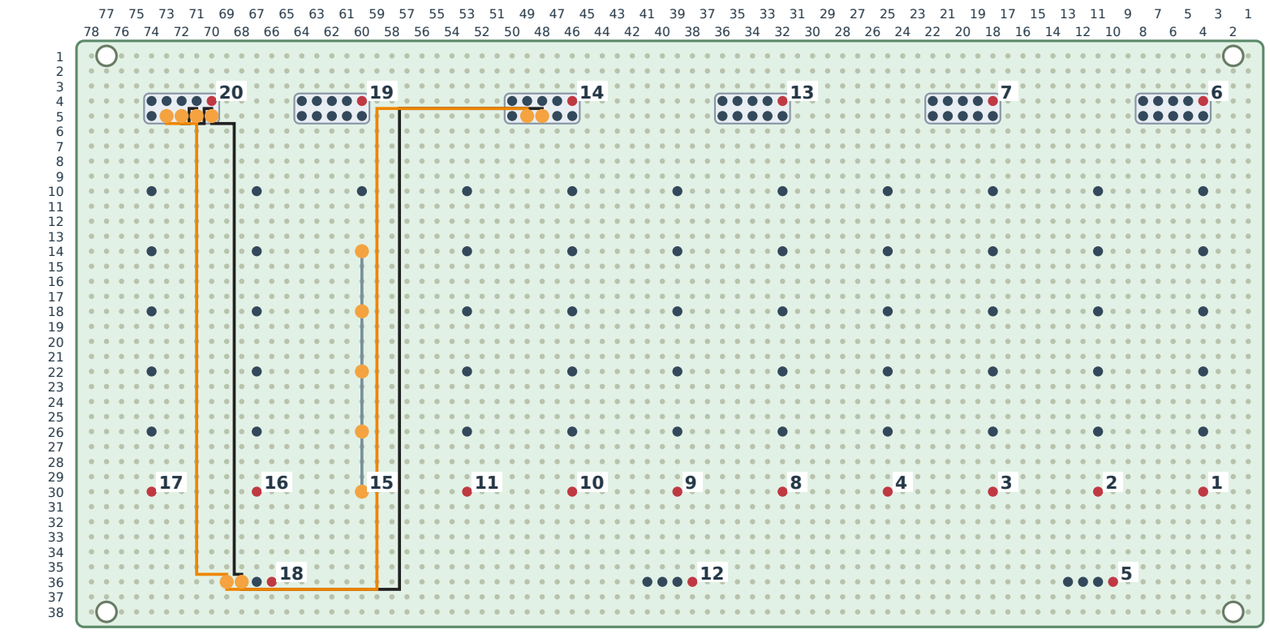
Wire / net	From → to	TOP holes
W70 / NODE18	J_B2_NODE18.4 → J_B2_NODE18.5	53,18 → 53,14
W71 / NODE18	J_B2_NODE18.5 → J_B2_NODE18.6	53,14 → 53,10
W72 / NODE18	J_B2_NODE18.6 → J_B2_NODE.7	53,10 → 35,4
W73 / SCL_POSITION_4	J_B2_SERVICE.1 → J_B2_SMART.9	38,36 → 50,4
W74 / SDA_POSITION_4	J_B2_SERVICE.2 → J_B2_SMART.7	39,36 → 49,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP31 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

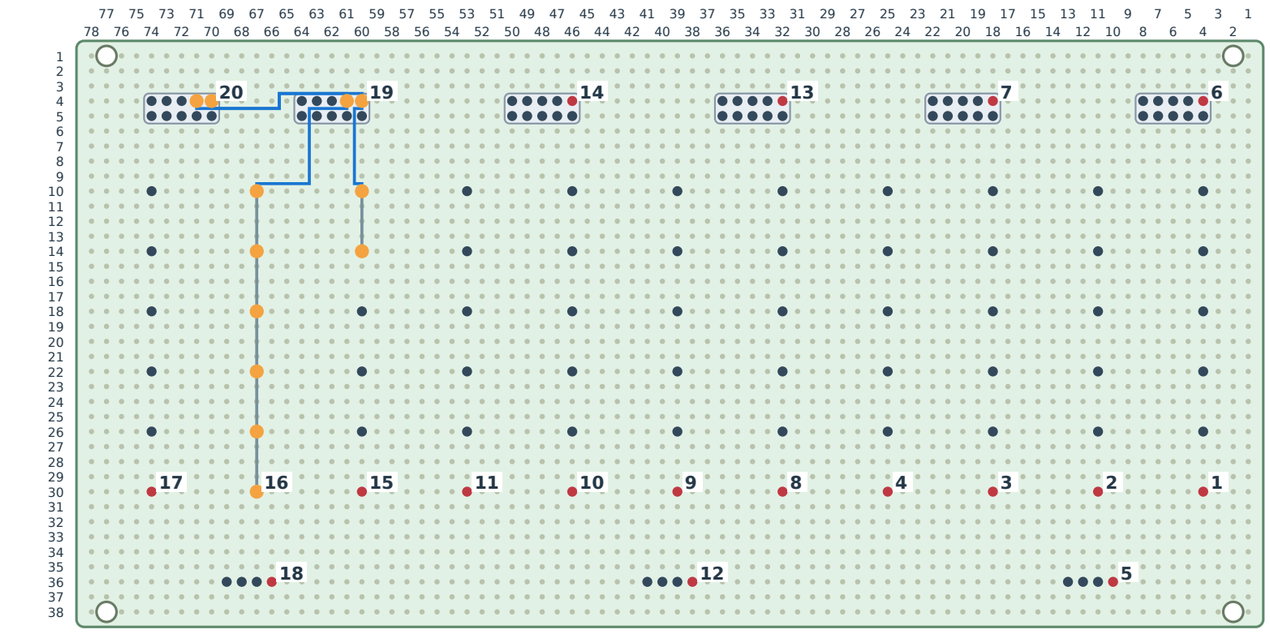
Wire / net	From → to	TOP holes
W37 / GND	J_B2_SMART.6 → J_B3_SERVICE.3	48,5 → 68,36
W38 / GND	J_B3_SERVICE.3 → J_B3_SMART.2	68,36 → 70,5
W39 / GND	J_B3_SMART.2 → J_B3_SMART.4	70,5 → 71,5
W40 / GND	J_B3_SMART.4 → J_B3_SMART.6	71,5 → 72,5
W44 / 3V3_SYS	J_B2_SMART.8 → J_B3_SERVICE.4	49,5 → 69,36
W45 / 3V3_SYS	J_B3_SERVICE.4 → J_B3_SMART.8	69,36 → 73,5
W75 / NODE19	J_B3_NODE19.1 → J_B3_NODE19.2	60,30 → 60,26
W76 / NODE19	J_B3_NODE19.2 → J_B3_NODE19.3	60,26 → 60,22
W77 / NODE19	J_B3_NODE19.3 → J_B3_NODE19.4	60,22 → 60,18
W78 / NODE19	J_B3_NODE19.4 → J_B3_NODE19.5	60,18 → 60,14

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP31 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

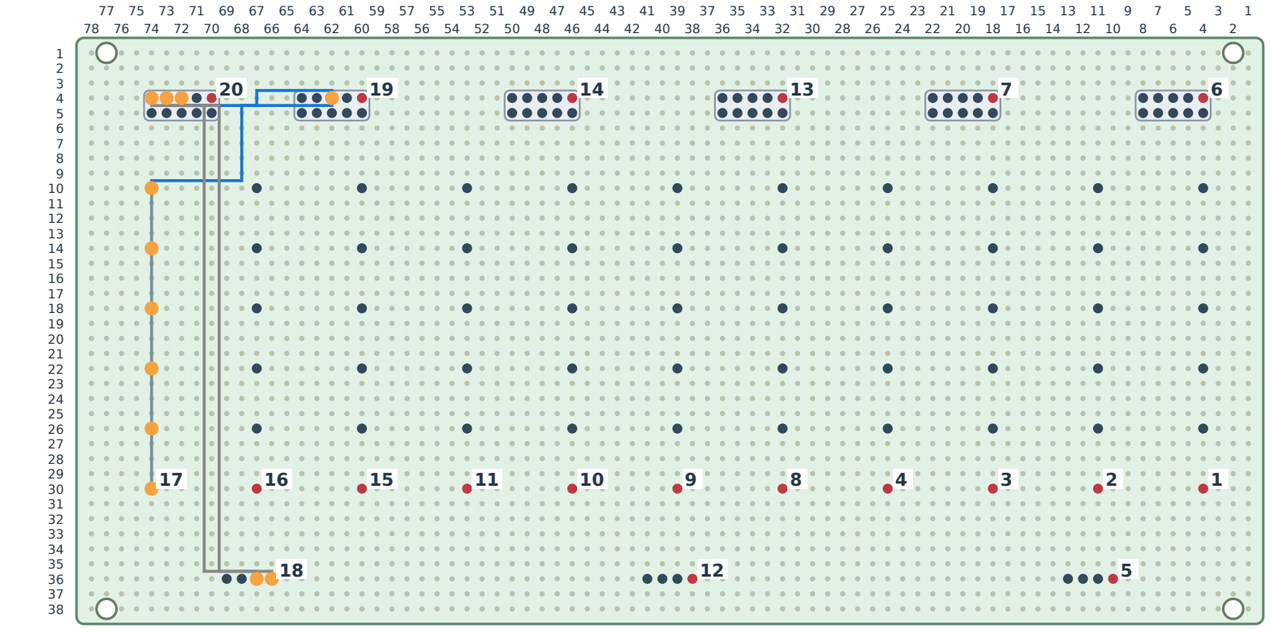
Wire / net	From → to	TOP holes
W79 / NODE19	J_B3_NODE19.5 → J_B3_NODE19.6	60,14 → 60,10
W80 / NODE19	J_B3_NODE19.6 → J_B3_NODE.1	60,10 → 60,4
W81 / NODE19	J_B3_NODE.1 → J_B3_SMART.1	60,4 → 70,4
W82 / NODE20	J_B3_NODE20.1 → J_B3_NODE20.2	67,30 → 67,26
W83 / NODE20	J_B3_NODE20.2 → J_B3_NODE20.3	67,26 → 67,22
W84 / NODE20	J_B3_NODE20.3 → J_B3_NODE20.4	67,22 → 67,18
W85 / NODE20	J_B3_NODE20.4 → J_B3_NODE20.5	67,18 → 67,14
W86 / NODE20	J_B3_NODE20.5 → J_B3_NODE20.6	67,14 → 67,10
W87 / NODE20	J_B3_NODE20.6 → J_B3_NODE.3	67,10 → 61,4
W88 / NODE20	J_B3_NODE.3 → J_B3_SMART.3	61,4 → 71,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP31 — contact-b

At most ten added wires per view. Inspect each net with power isolated.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

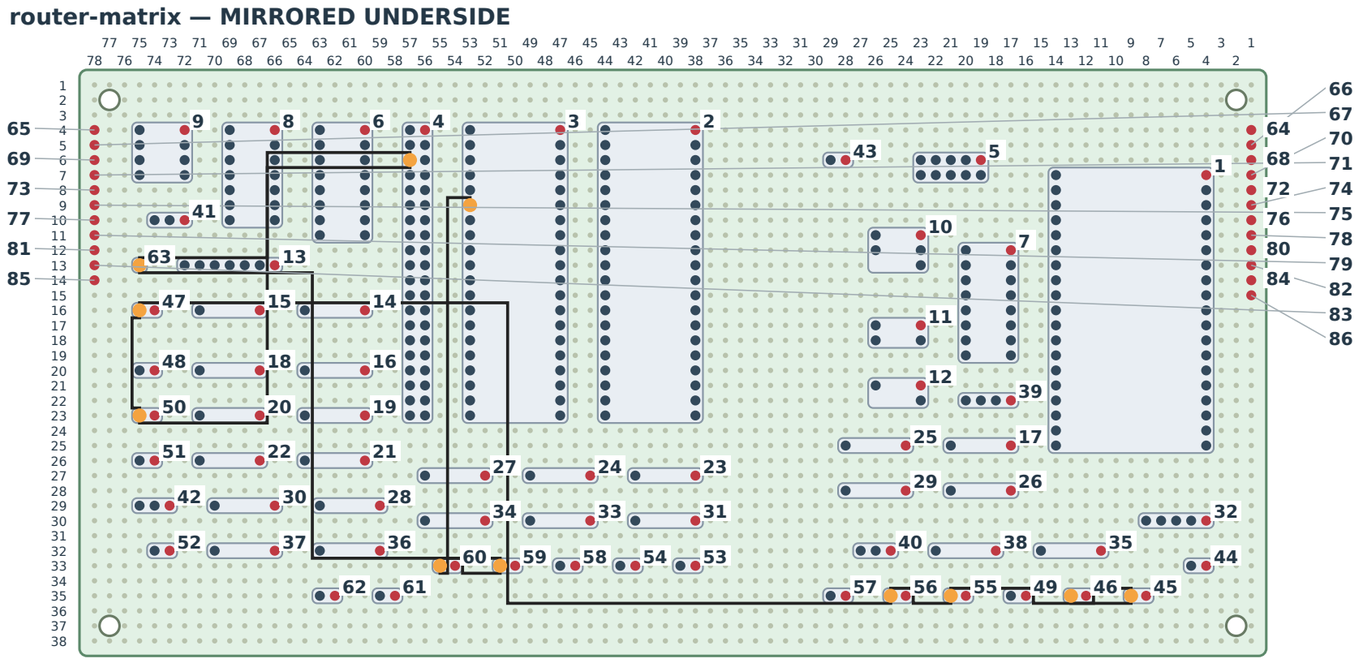
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W89 / NODE21	J_B3_NODE21.1 → J_B3_NODE21.2	74,30 → 74,26
W90 / NODE21	J_B3_NODE21.2 → J_B3_NODE21.3	74,26 → 74,22
W91 / NODE21	J_B3_NODE21.3 → J_B3_NODE21.4	74,22 → 74,18
W92 / NODE21	J_B3_NODE21.4 → J_B3_NODE21.5	74,18 → 74,14
W93 / NODE21	J_B3_NODE21.5 → J_B3_NODE21.6	74,14 → 74,10
W94 / NODE21	J_B3_NODE21.6 → J_B3_NODE.5	74,10 → 62,4
W95 / NODE21	J_B3_NODE.5 → J_B3_SMART.5	62,4 → 72,4
W96 / SCL_POSITION_5	J_B3_SERVICE.1 → J_B3_SMART.9	66,36 → 74,4
W97 / SDA_POSITION_5	J_B3_SERVICE.2 → J_B3_SMART.7	67,36 → 73,4

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP32 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

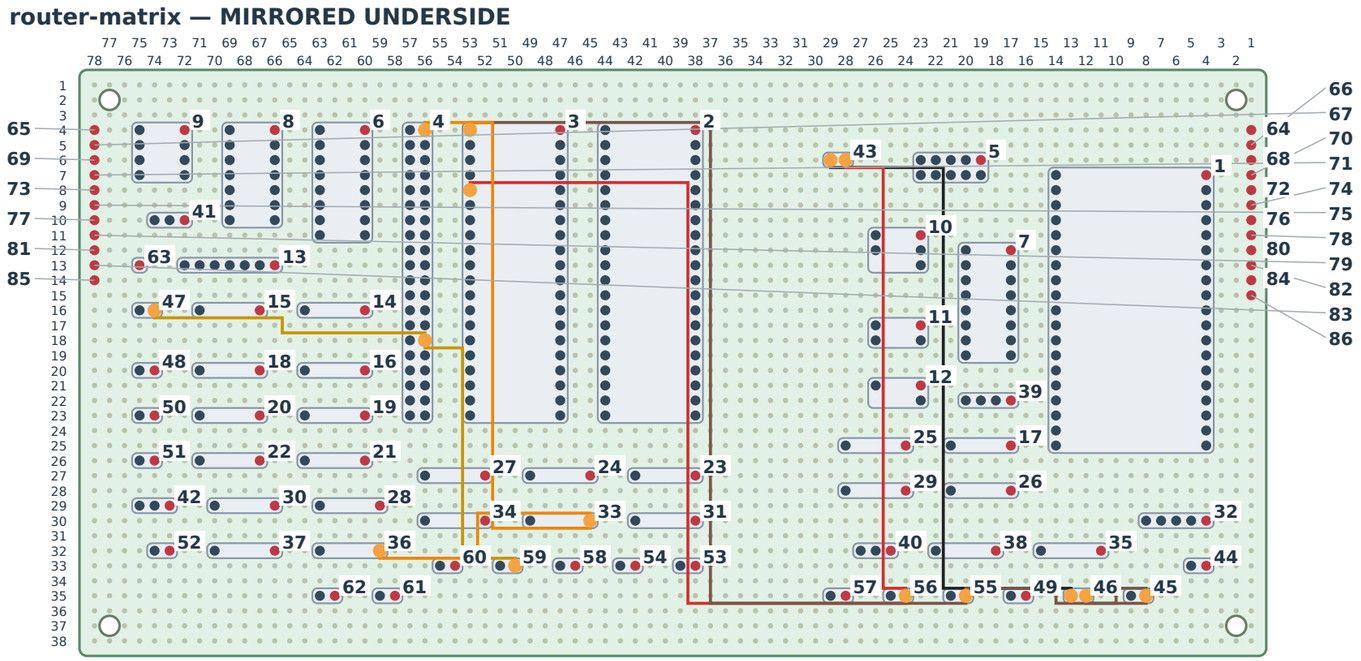
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W75 / GND	U_MATRIX_SWITCHES_MT_B.35 → C_INTEGRATION_BACK.2	53,9 → 55,33
W76 / GND	C_INTEGRATION_BACK.2 → C_INTEGRATION_ENERGIZE.2	55,33 → 51,33
W77 / GND	C_INTEGRATION_ENERGIZE.2 → J_INTEGRATION_GROUND.1	51,33 → 75,13
W78 / GND	J_INTEGRATION_GROUND.1 → J_INTEGRATION_PI.6	75,13 → 57,6
W79 / GND	J_INTEGRATION_PI.6 → J_INTEGRATION_SW_BACK.2	57,6 → 75,23
W80 / GND	J_INTEGRATION_SW_BACK.2 → J_INTEGRATION_SW_ENERGIZE.2	75,23 → 75,16
W81 / GND	J_INTEGRATION_SW_ENERGIZE.2 → J_ROUTER_CORE_5V.2	75,16 → 25,35
W82 / GND	J_ROUTER_CORE_5V.2 → J_ROUTER_CORE_12V.2	25,35 → 21,35
W83 / GND	J_ROUTER_CORE_12V.2 → J_ROUTER_CORE_FAN1.2	21,35 → 9,35
W84 / GND	J_ROUTER_CORE_FAN1.2 → J_ROUTER_CORE_FAN2.2	9,35 → 13,35

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP32 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

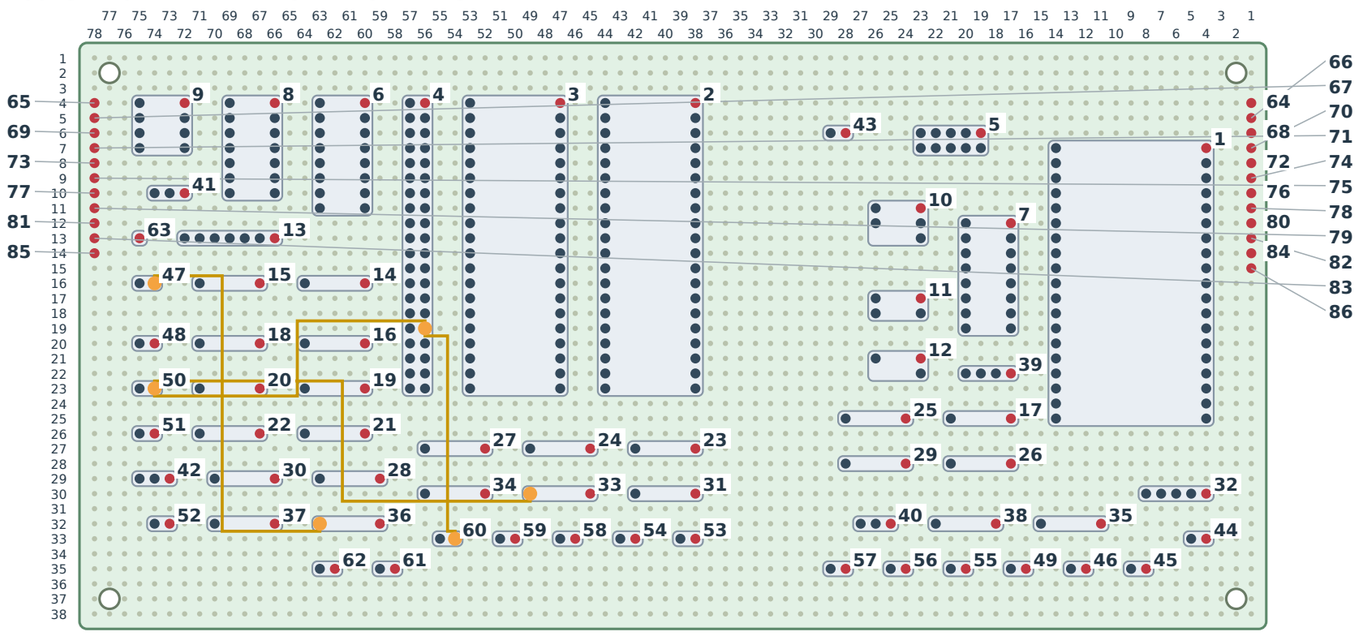
Wire / net	From → to	TOP holes
W85 / GND	J_ROUTER_CORE_FAN2.2 → J_ROUTER_CORE_LIGHT.2	13,35 → 29,6
W152 / 5V_SYS	U_MATRIX_SWITCHES_MT_B.36 → J_ROUTER_CORE_5V.1	53,8 → 24,35
W153 / 5V_SYS	J_ROUTER_CORE_5V.1 → J_ROUTER_CORE_LIGHT.1	24,35 → 28,6
W166 / 12V_SYS	U_MATRIX_SWITCHES_MT_B.40 → J_ROUTER_CORE_12V.1	53,4 → 20,35
W167 / 12V_SYS	J_ROUTER_CORE_12V.1 → J_ROUTER_CORE_FAN1.1	20,35 → 8,35
W168 / 12V_SYS	J_ROUTER_CORE_FAN1.1 → J_ROUTER_CORE_FAN2.1	8,35 → 12,35
W182 / PI_3V3	J_INTEGRATION_PI.1 → R_INTEGRATION_BACK.1	56,4 → 45,30
W183 / PI_3V3	R_INTEGRATION_BACK.1 → R_INTEGRATION_ENERGIZE.1	45,30 → 59,32
W190 / ENERGIZE_N	C_INTEGRATION_ENERGIZE.1 → J_INTEGRATION_PI.29	50,33 → 56,18
W191 / ENERGIZE_N	J_INTEGRATION_PI.29 → J_INTEGRATION_SW_ENERGIZE.1	56,18 → 74,16

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP32 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.

router-matrix — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

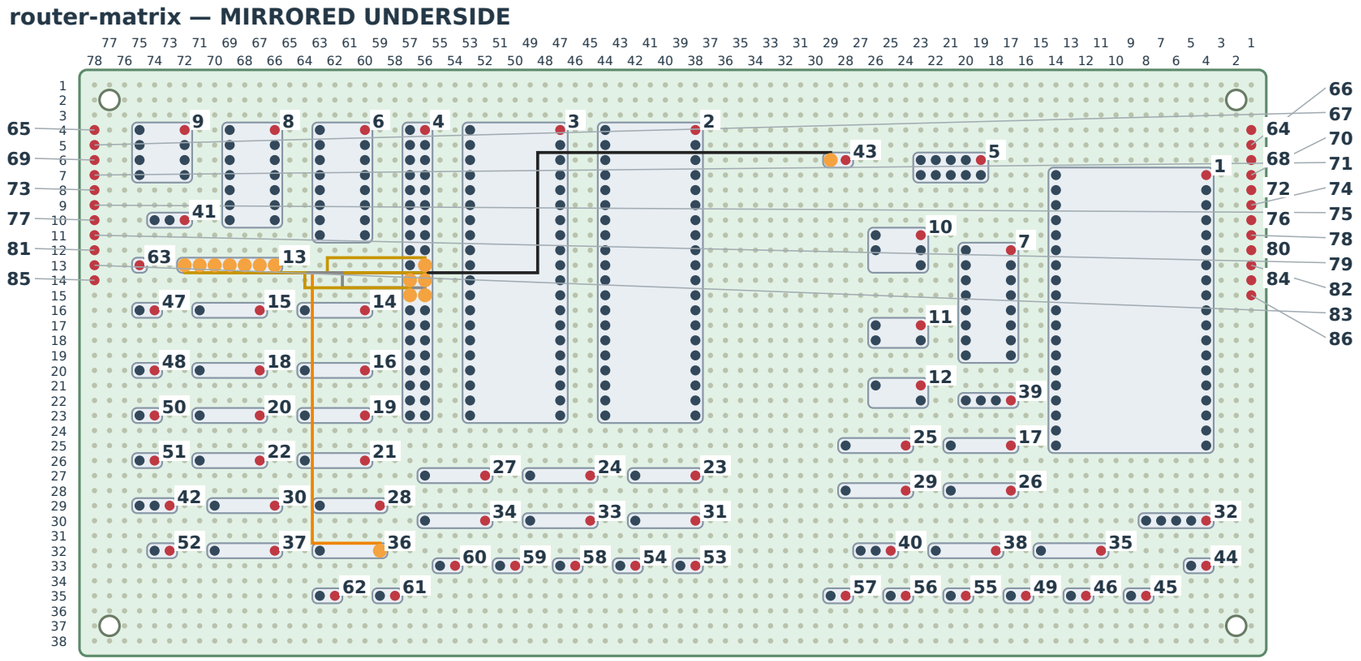
PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W192 / ENERGIZE_N	J_INTEGRATION_SW_ENERGIZE.1 → R_INTEGRATION_ENERGIZE.2	74,16 → 63,32
W193 / BACK_N	C_INTEGRATION_BACK.1 → J_INTEGRATION_PI.31	54,33 → 56,19
W194 / BACK_N	J_INTEGRATION_PI.31 → J_INTEGRATION_SW_BACK.1	56,19 → 74,23
W195 / BACK_N	J_INTEGRATION_SW_BACK.1 → R_INTEGRATION_BACK.2	74,23 → 49,30

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.

CP34 — router-matrix

At most ten added wires per view. Inspect each net with power isolated.



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Wire / net	From → to	TOP holes
W86 / GND	J_ROUTER_CORE_LIGHT.2 → J_INTEGRATION_RFID.2	29,6 → 67,13
W184 / PI_3V3	R_INTEGRATION_ENERGIZE.1 → J_INTEGRATION_RFID.1	59,32 → 66,13
W185 / MOSI	J_INTEGRATION_PI.19 → J_INTEGRATION_RFID.4	56,13 → 69,13
W186 / MISO	J_INTEGRATION_PI.21 → J_INTEGRATION_RFID.5	56,14 → 70,13
W187 / RFID_RESET	J_INTEGRATION_PI.22 → J_INTEGRATION_RFID.7	57,14 → 72,13
W188 / SCLK	J_INTEGRATION_PI.23 → J_INTEGRATION_RFID.3	56,15 → 68,13
W189 / CE0_N	J_INTEGRATION_PI.24 → J_INTEGRATION_RFID.6	57,15 → 71,13

Bare node-column straps only; all other wires insulated underneath. A bend is not a joint. Consult the endpoint CSV; do not infer a connection from a crossing.